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Huang et al.

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(54) **SEMICONDUCTOR DEVICE HAVING NANOSHEET TRANSISTOR AND METHODS OF FABRICATION THEREOF**

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This patent is subject to a terminal disclaimer.

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Related U.S. Application Data

(63) Continuation of application No. 17/883,971, filed on Aug. 9, 2022, now Pat. No. 11,961,840, which is a
(Continued)

(51) **Int. Cl.**
H01L 27/092 (2006.01)
H01L 21/02 (2006.01)
(Continued)

(52) **U.S. Cl.**
CPC **H10D 84/856** (2025.01); **H01L 21/02603** (2013.01); **H10D 30/031** (2025.01);
(Continued)

(58) **Field of Classification Search**
CPC H01L 27/0922; H01L 21/02603; H01L 21/823807; H01L 21/823814;
(Continued)

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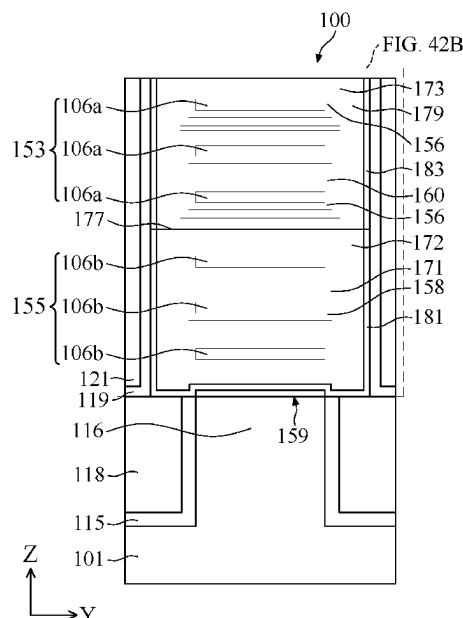
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(57) **ABSTRACT**

A semiconductor device structure is provided. The structure includes a first gate electrode layer having at least three surfaces surrounded by a first intermixed layer, wherein the first intermixed layer comprises a first material and a second material. The structure also includes a second gate electrode layer disposed below and in contact with the first gate electrode layer, the second gate electrode layer having at least three surfaces surrounded by a second intermixed layer, wherein the second intermixed layer comprises the first material and a fifth material, wherein the first gate electrode layer and the second gate electrode layer are disposed between two adjacent dielectric spacers.

18 Claims, 38 Drawing Sheets



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H01L 21/8238 (2006.01)
H01L 29/06 (2006.01)
H01L 29/423 (2006.01)
H01L 29/66 (2006.01)
H01L 29/78 (2006.01)
H01L 29/786 (2006.01)
H10D 30/01 (2025.01)
H10D 30/67 (2025.01)
H10D 30/69 (2025.01)
H10D 62/10 (2025.01)
H10D 64/01 (2025.01)
H10D 84/01 (2025.01)
H10D 84/03 (2025.01)
H10D 84/85 (2025.01)

(52) **U.S. Cl.**

CPC **H10D 30/0415** (2025.01); **H10D 30/6713**
(2025.01); **H10D 30/6735** (2025.01); **H10D**
30/6757 (2025.01); **H10D 30/701** (2025.01);
H10D 62/121 (2025.01); **H10D 64/017**
(2025.01); **H10D 84/0167** (2025.01); **H10D**
84/017 (2025.01); **H10D 84/0181** (2025.01);
H10D 84/038 (2025.01)

(58) **Field of Classification Search**

CPC H01L 21/823857; H01L 29/0673; H01L
29/42392; H01L 29/66545; H01L
29/66742; H01L 29/6684; H01L
29/78391; H01L 29/78618; H01L
29/78696; H01L 29/66439; H01L
21/28185; H01L 27/092; H01L 29/165;
H01L 29/7848; H01L 29/775; B82Y
10/00
USPC 257/295
See application file for complete search history.

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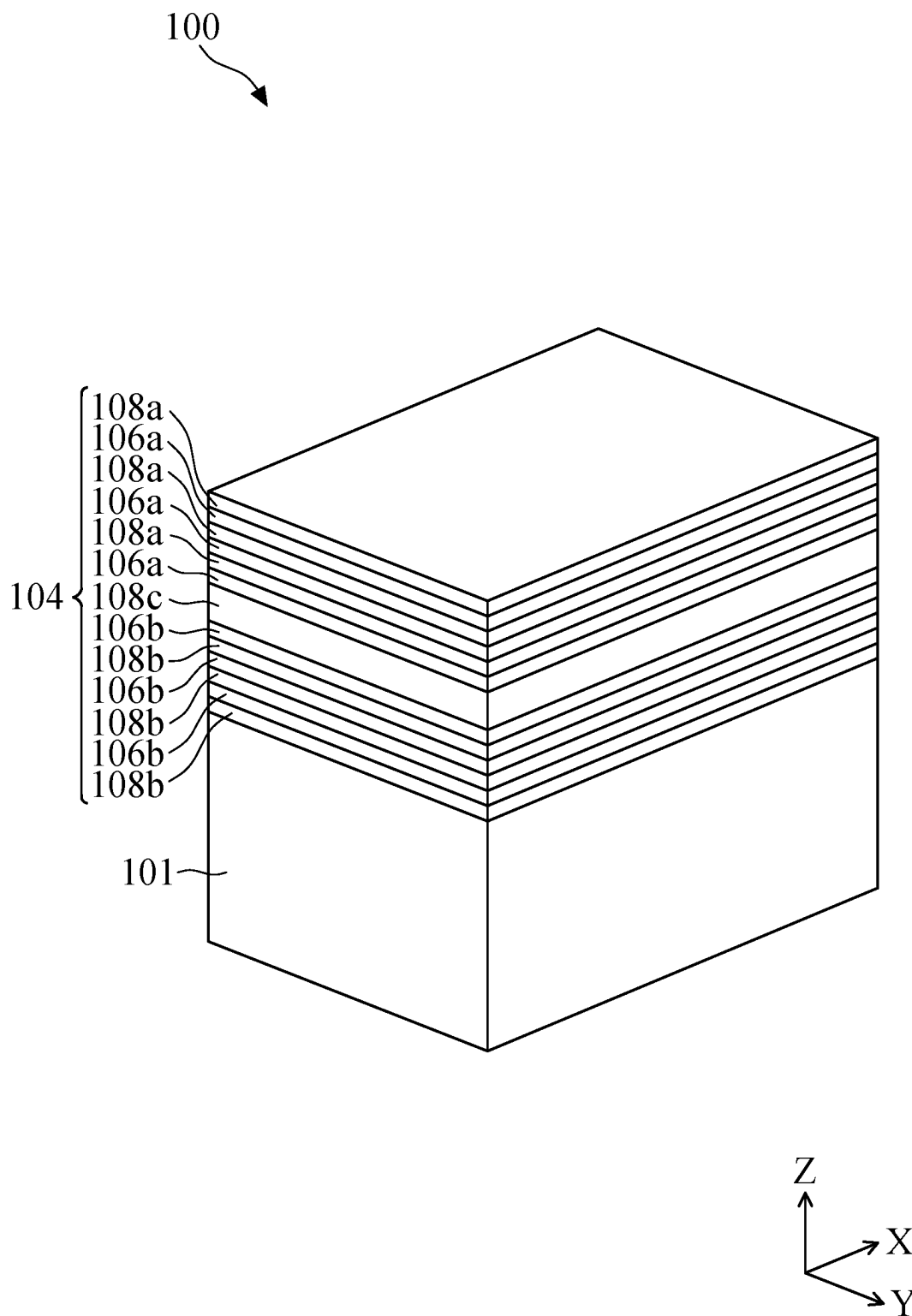


FIG. 1

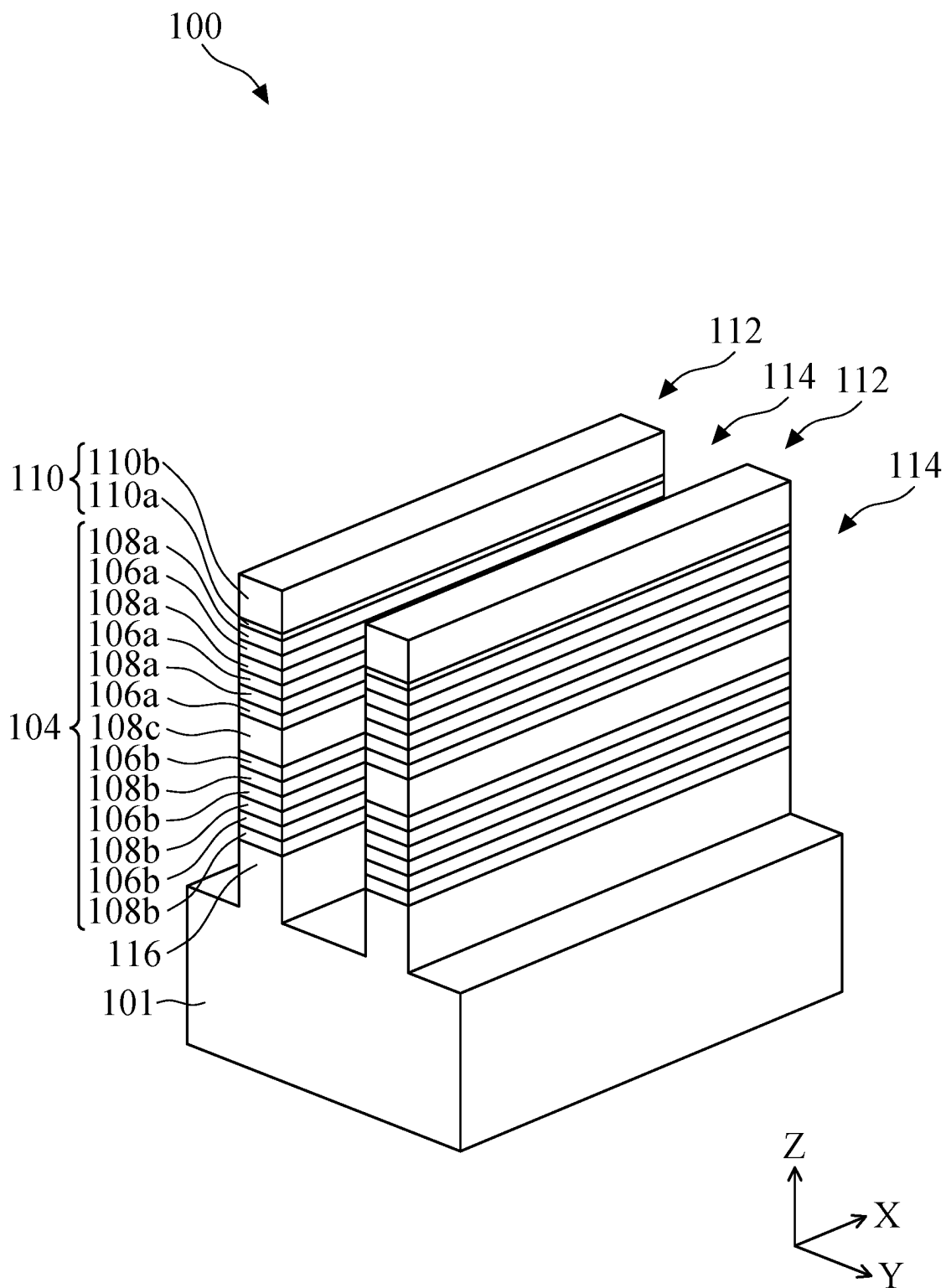


FIG. 2

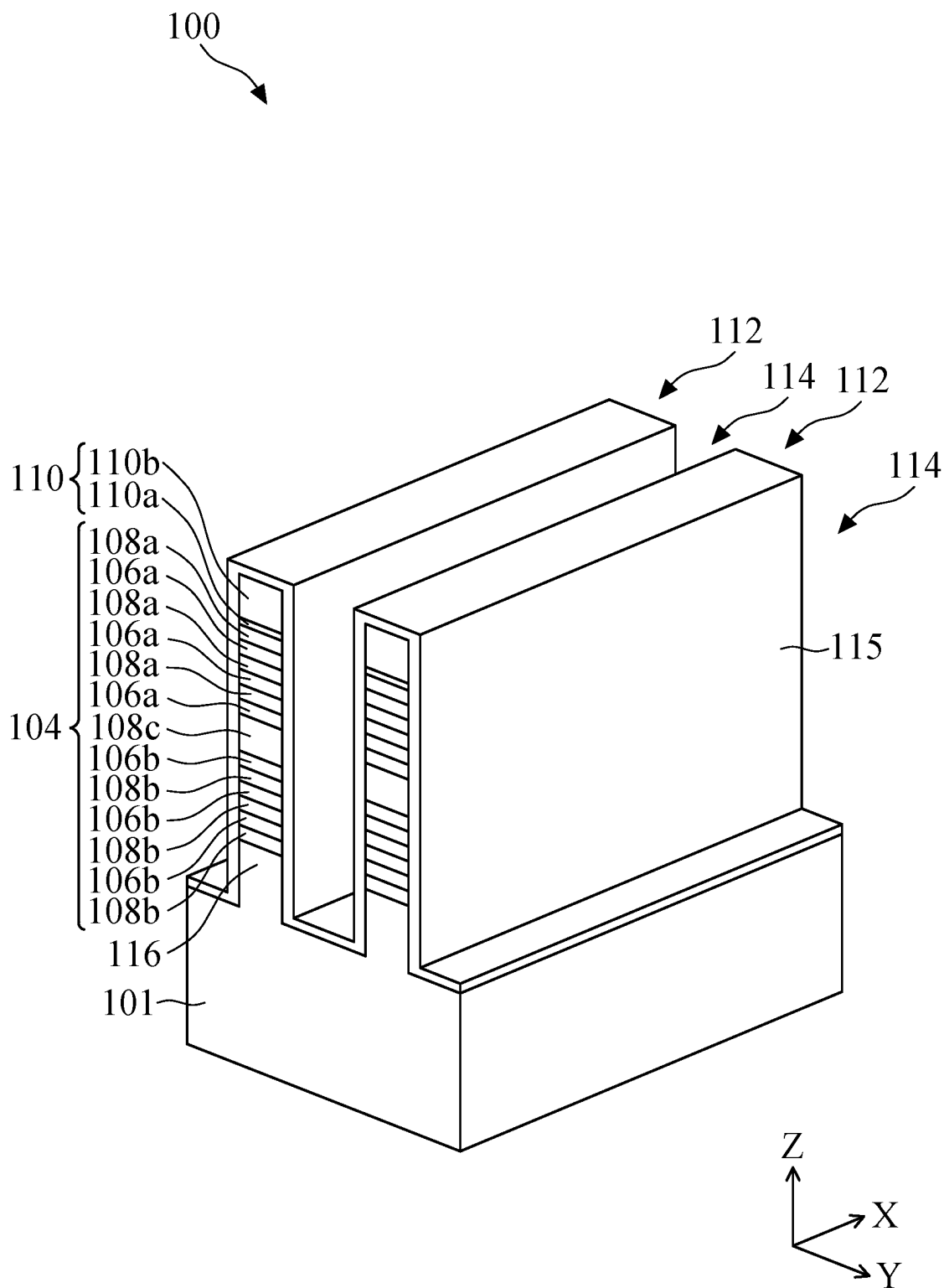


FIG. 3

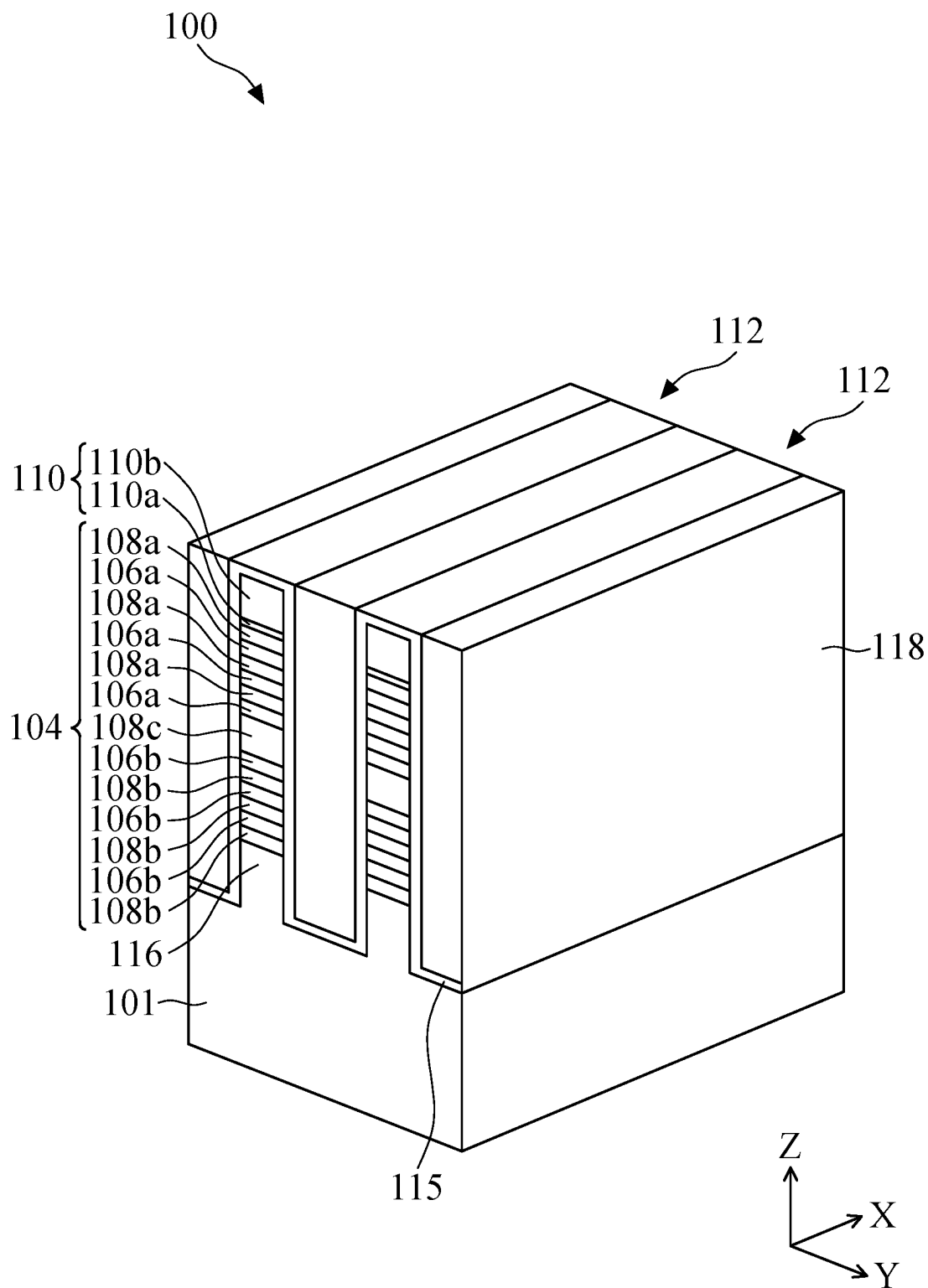


FIG. 4

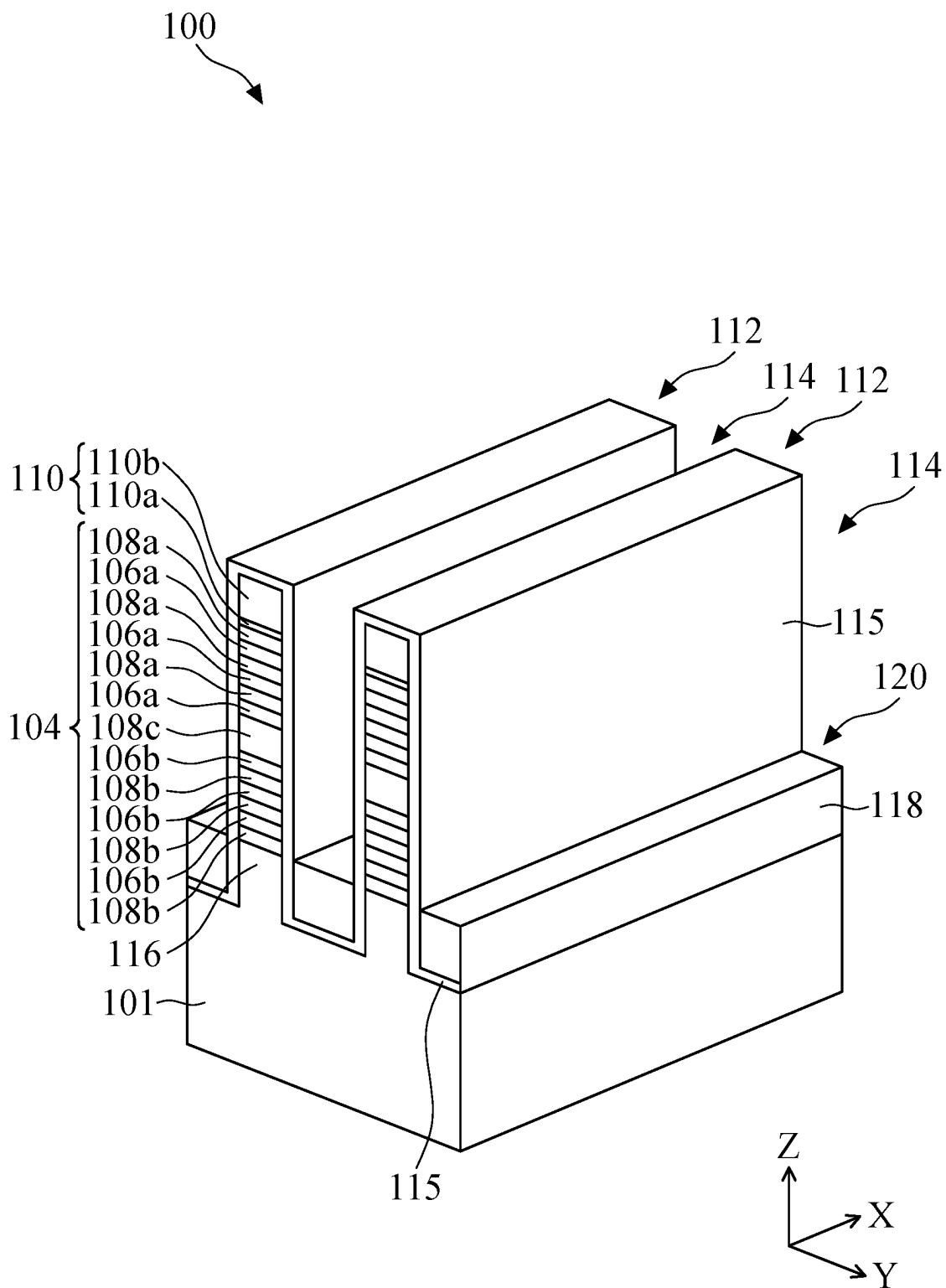


FIG. 5

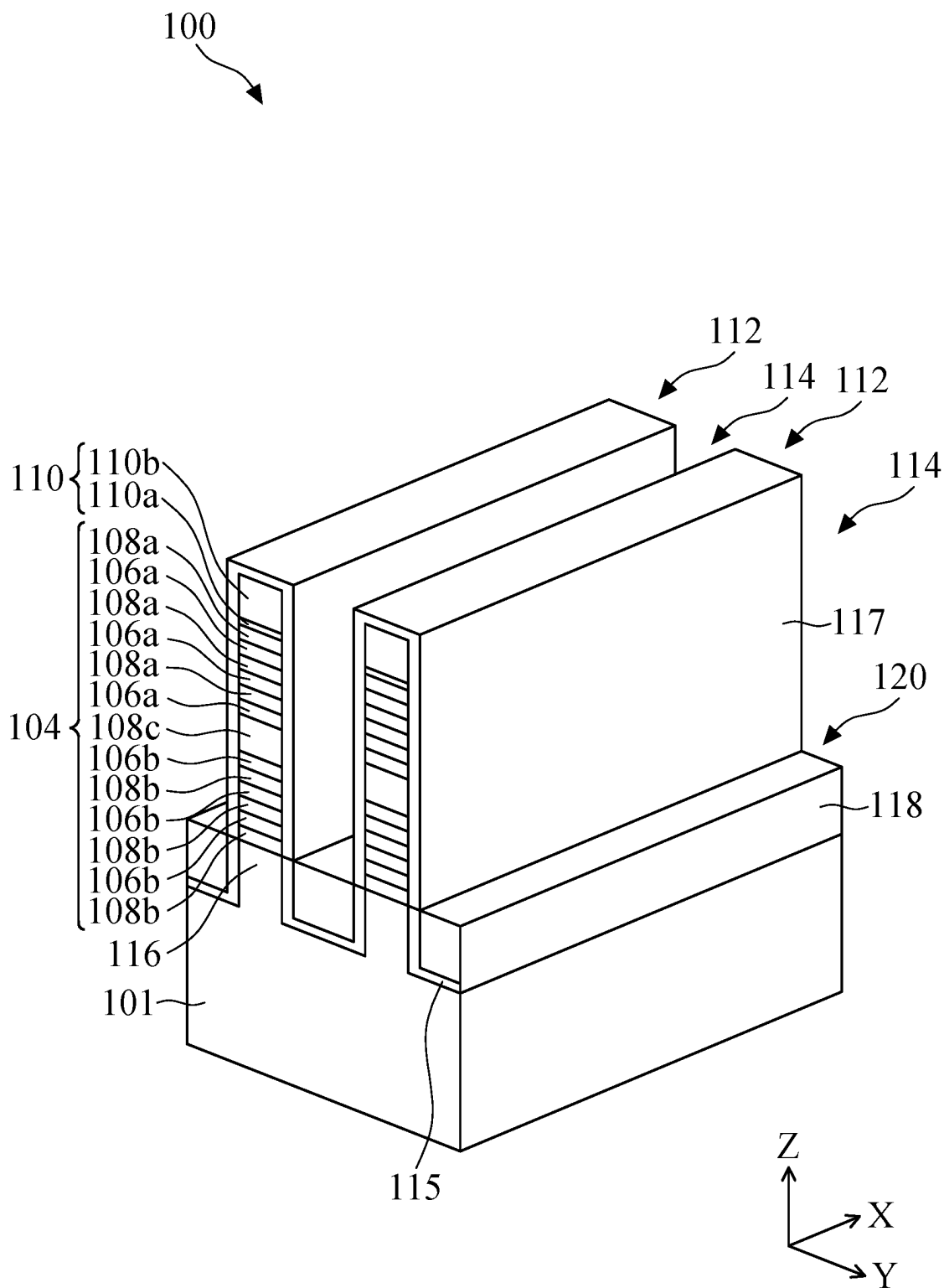


FIG. 6

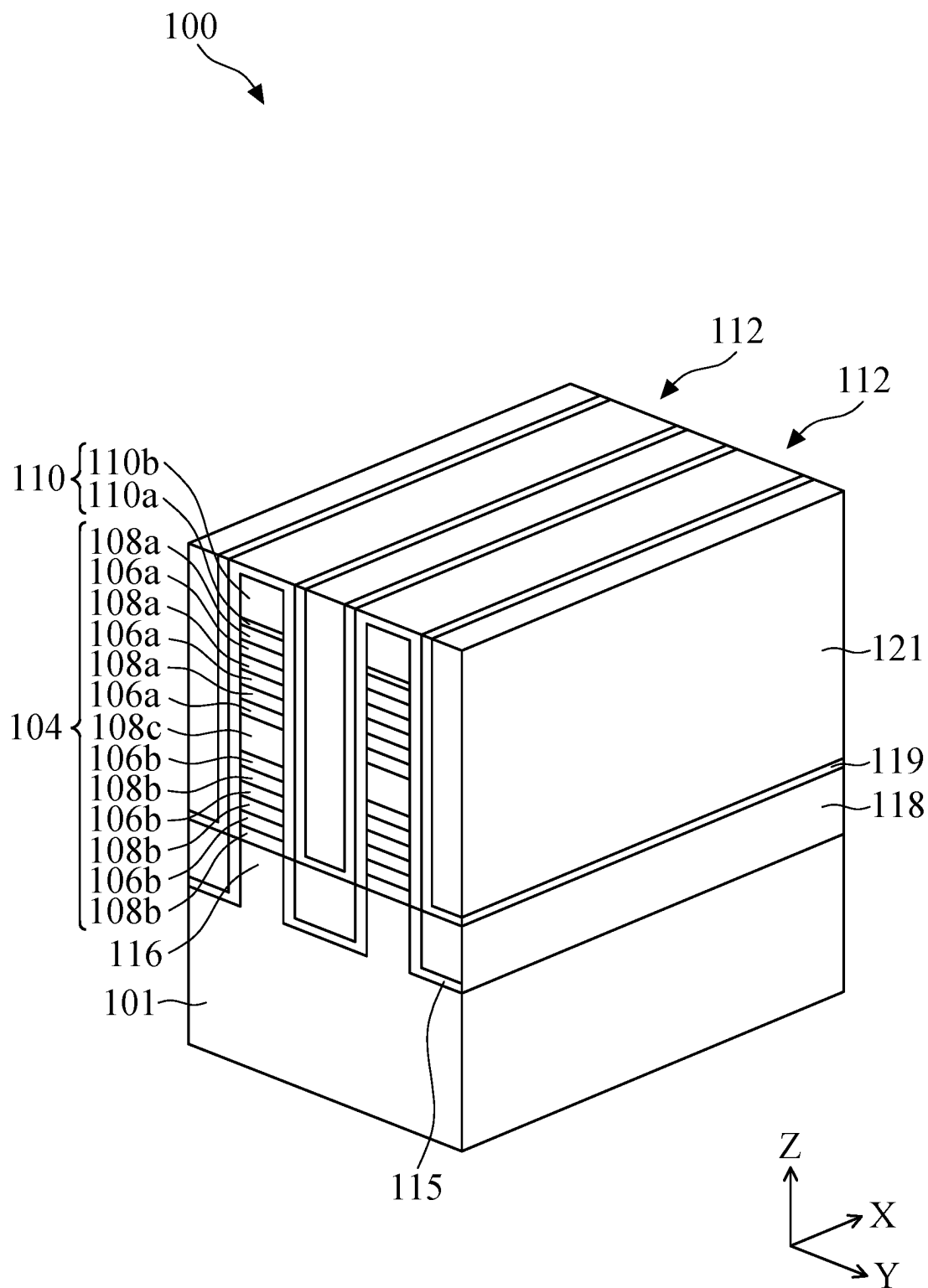


FIG. 7

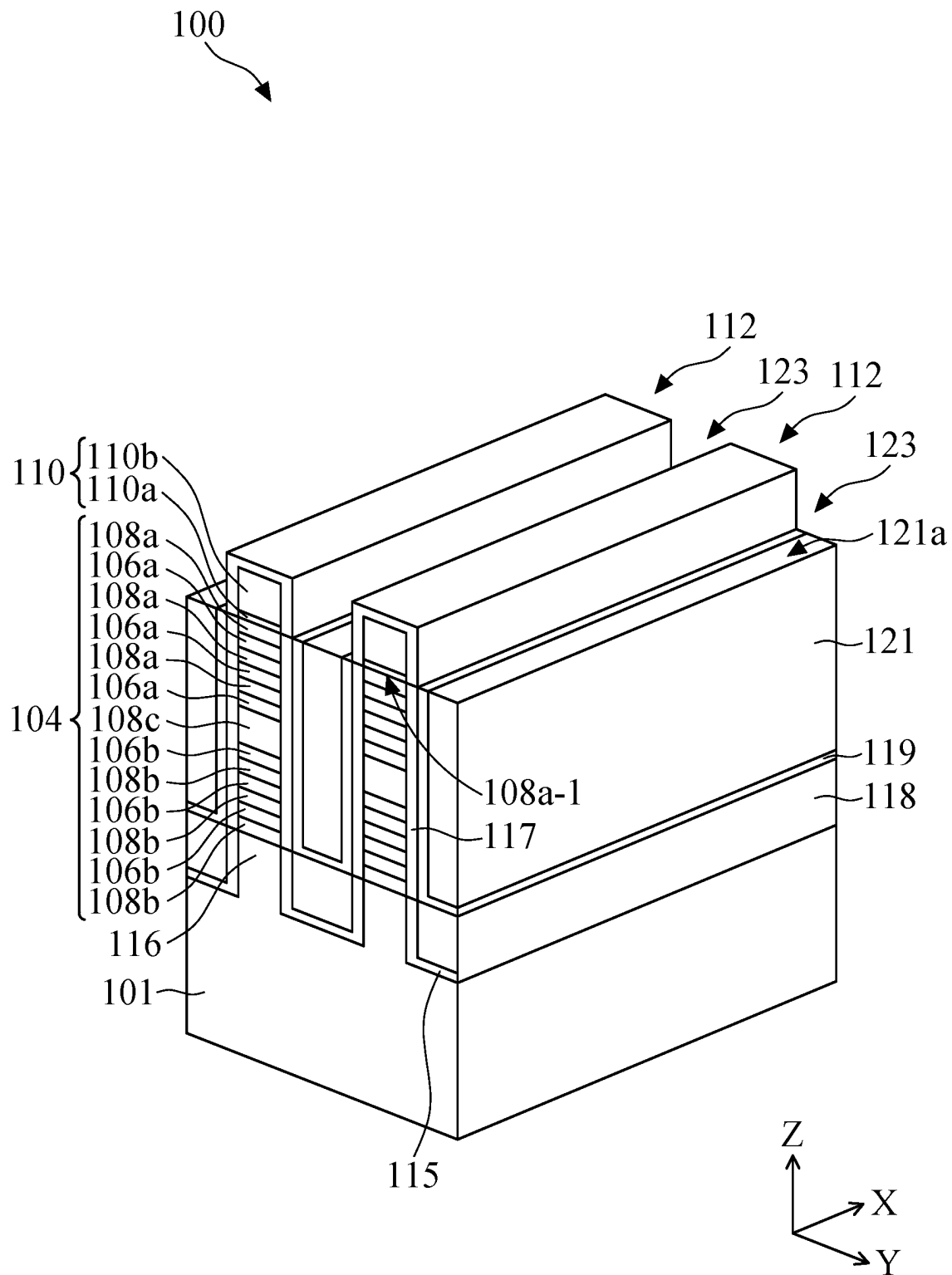


FIG. 8

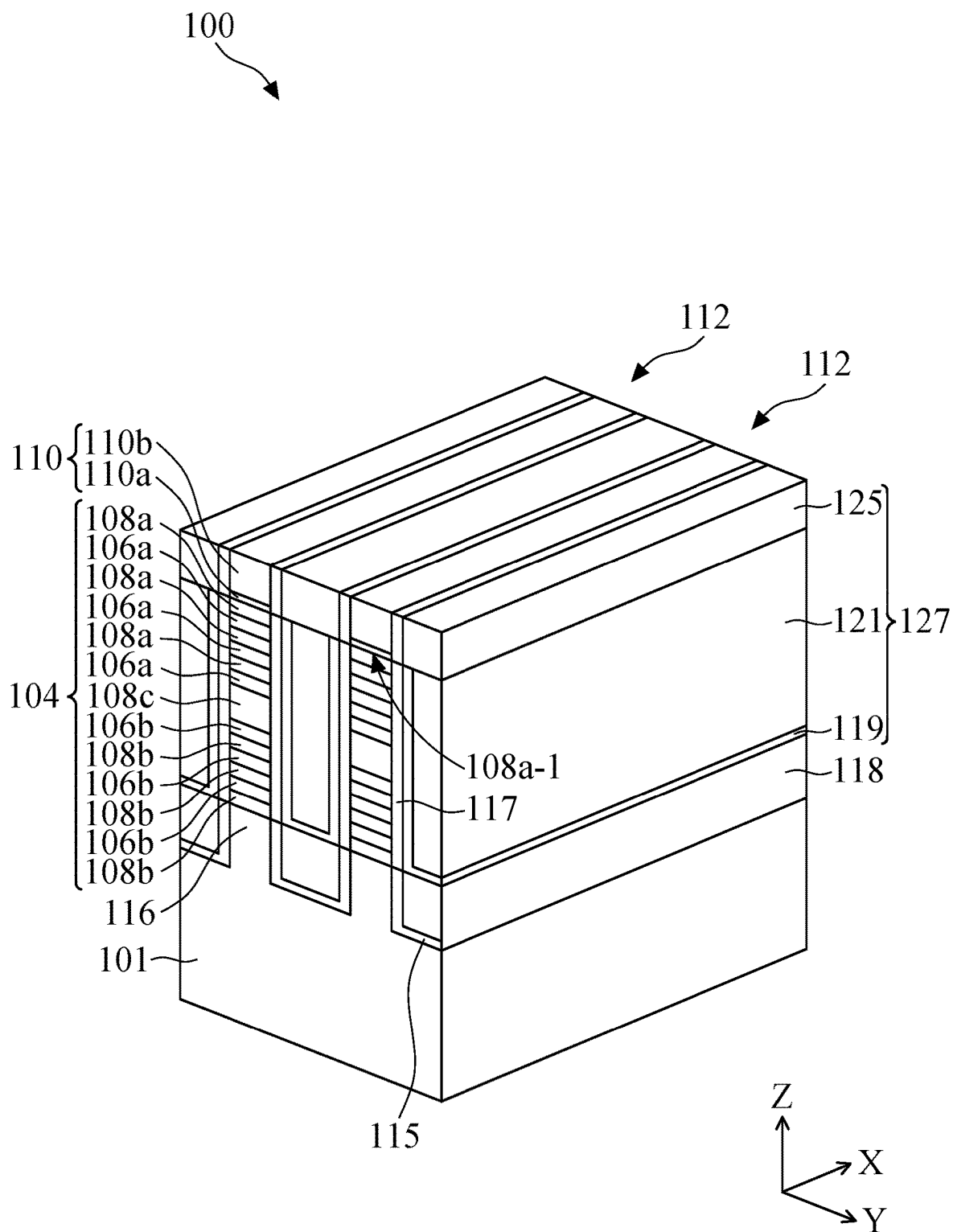


FIG. 9

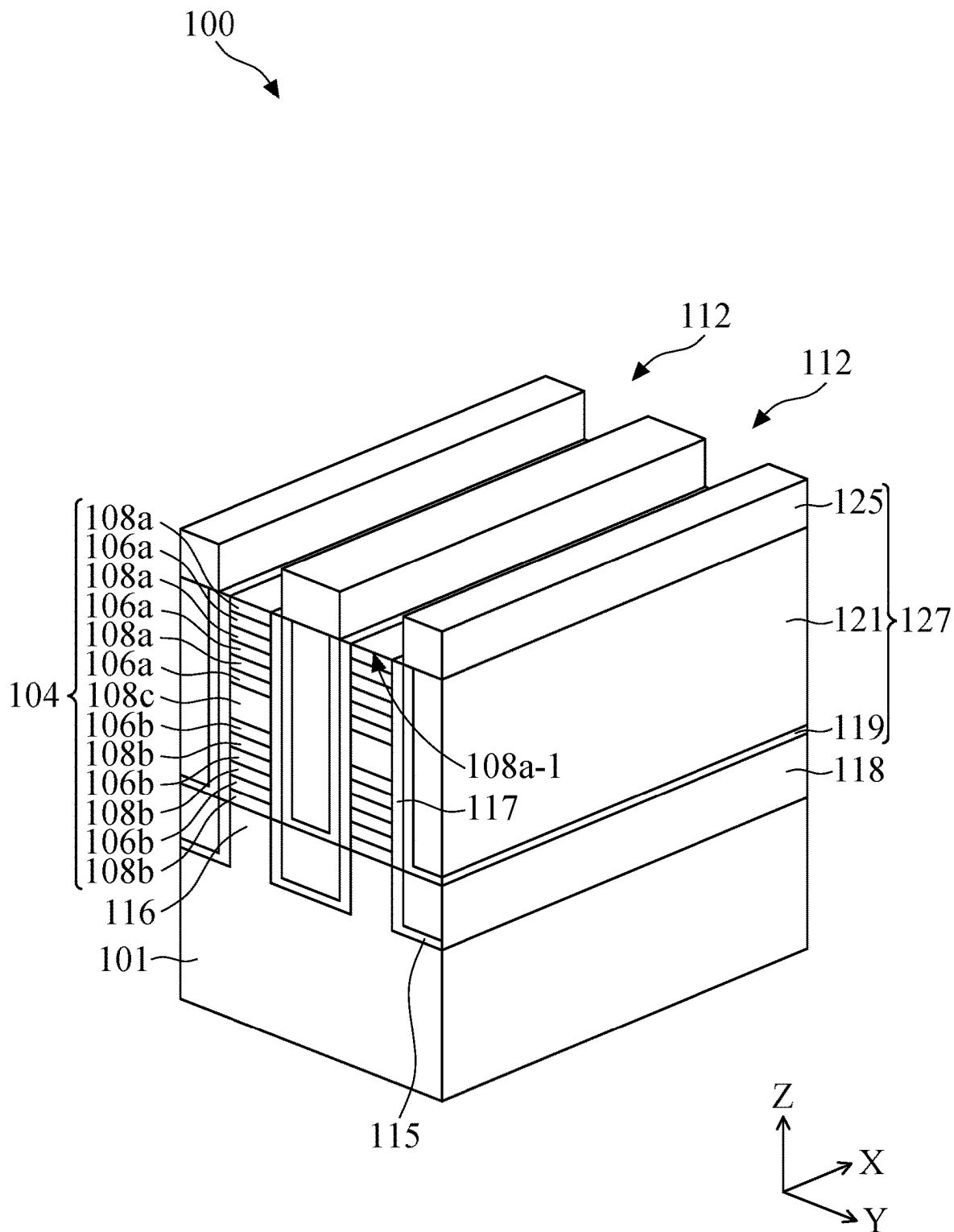


FIG. 10

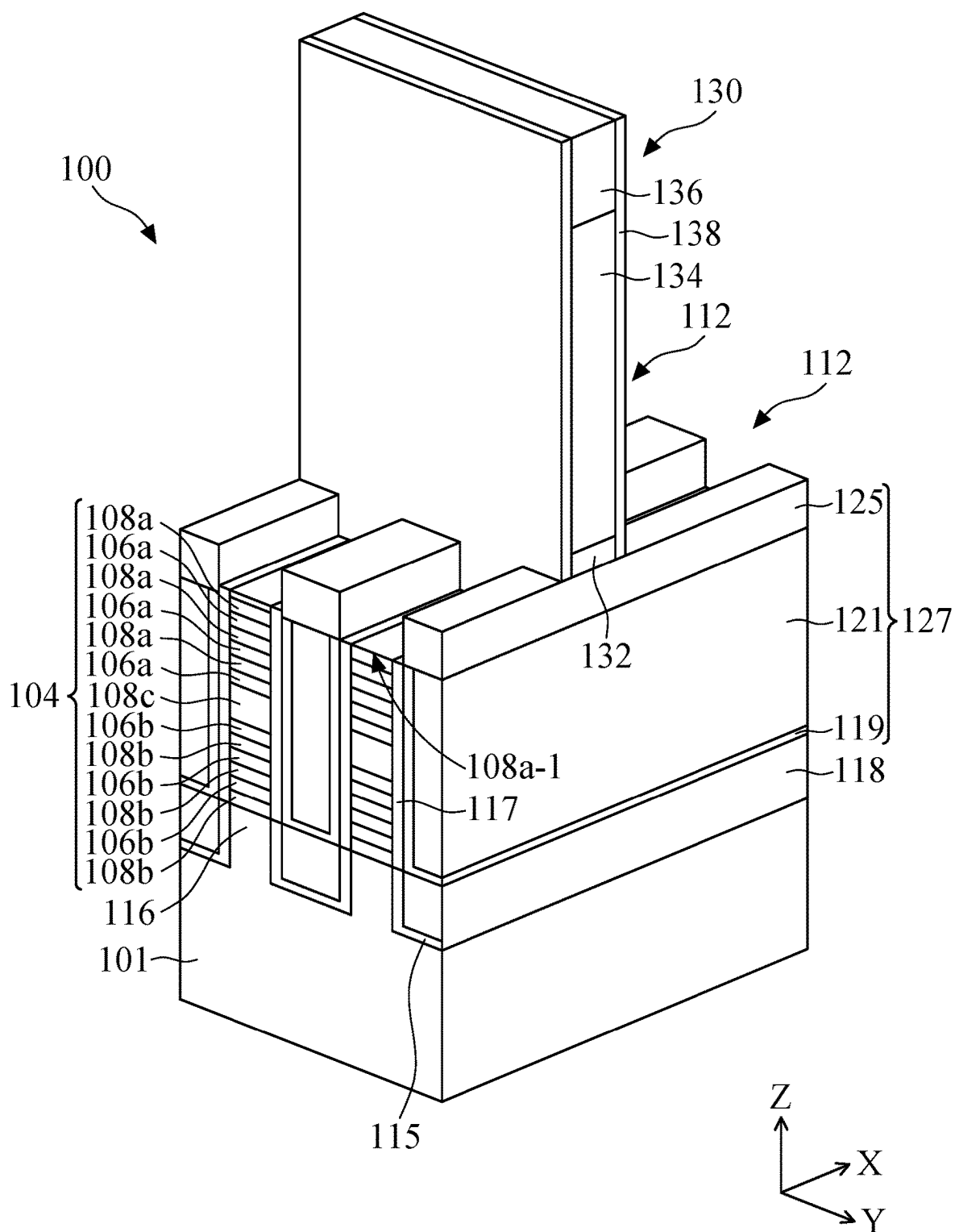


FIG. 11

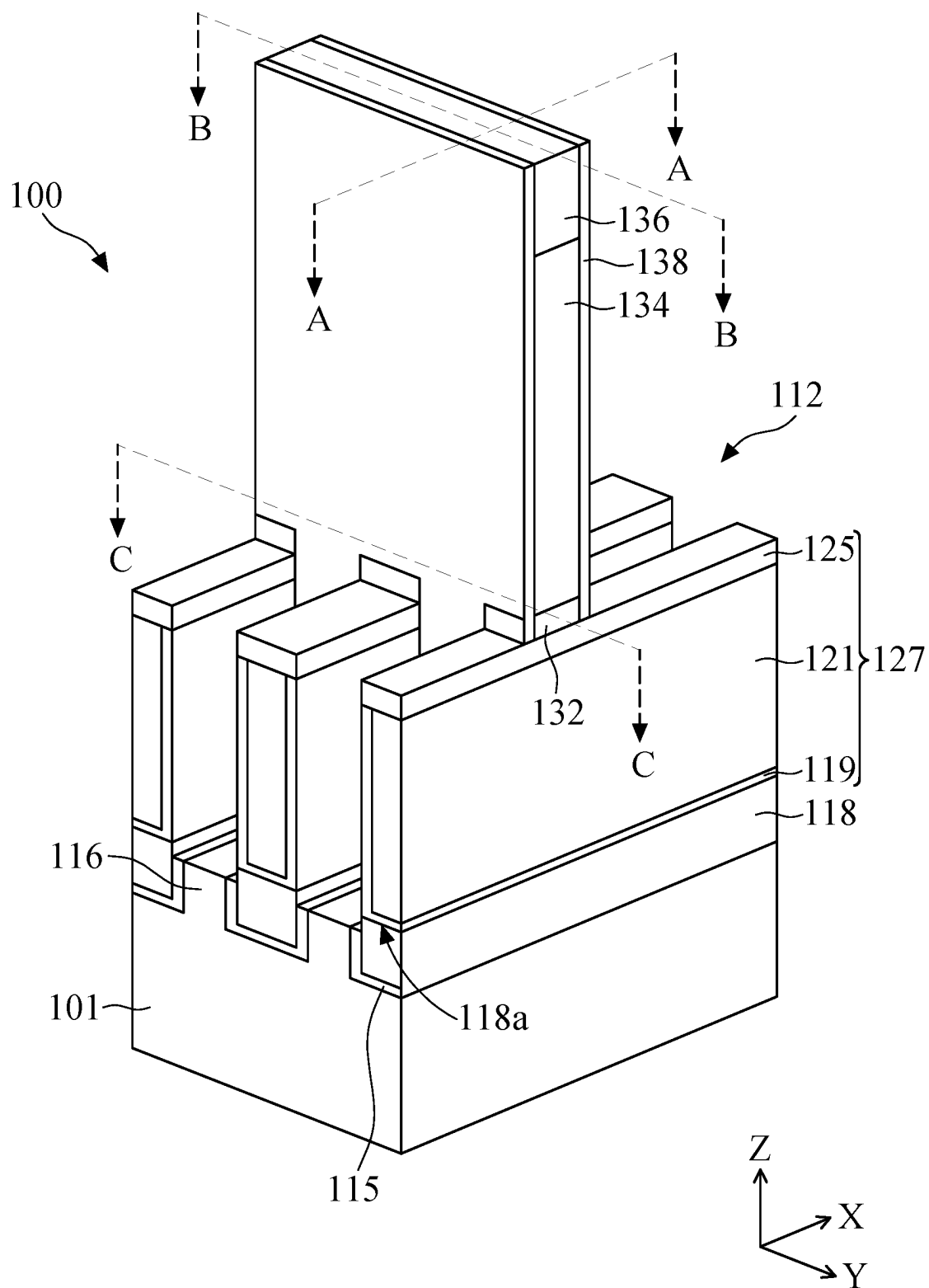
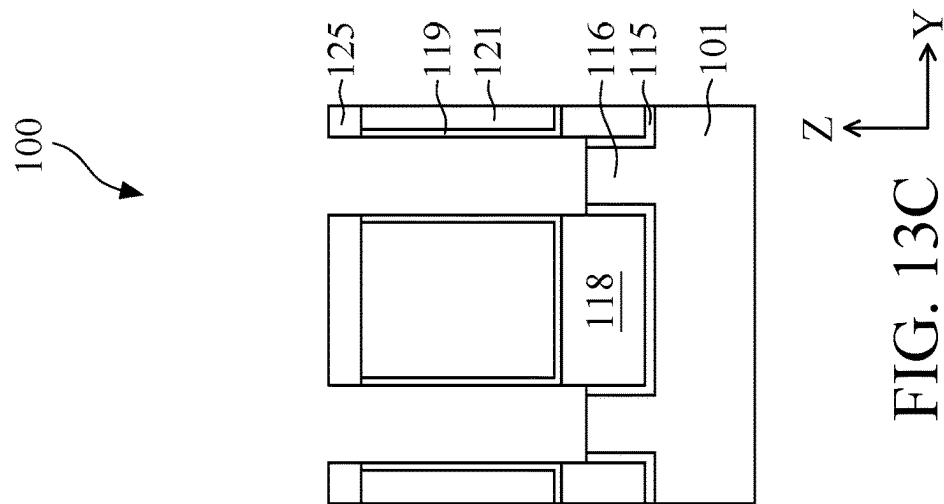
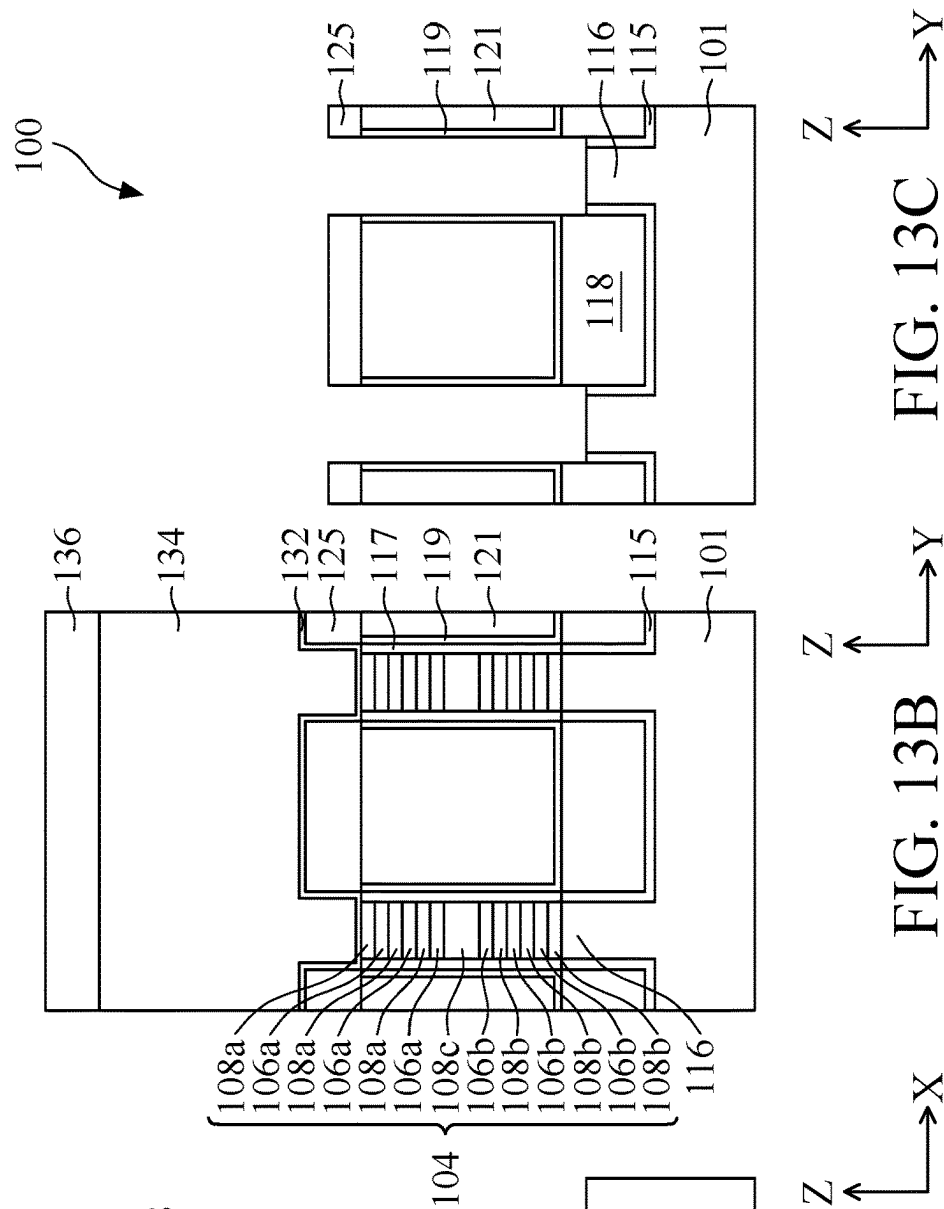
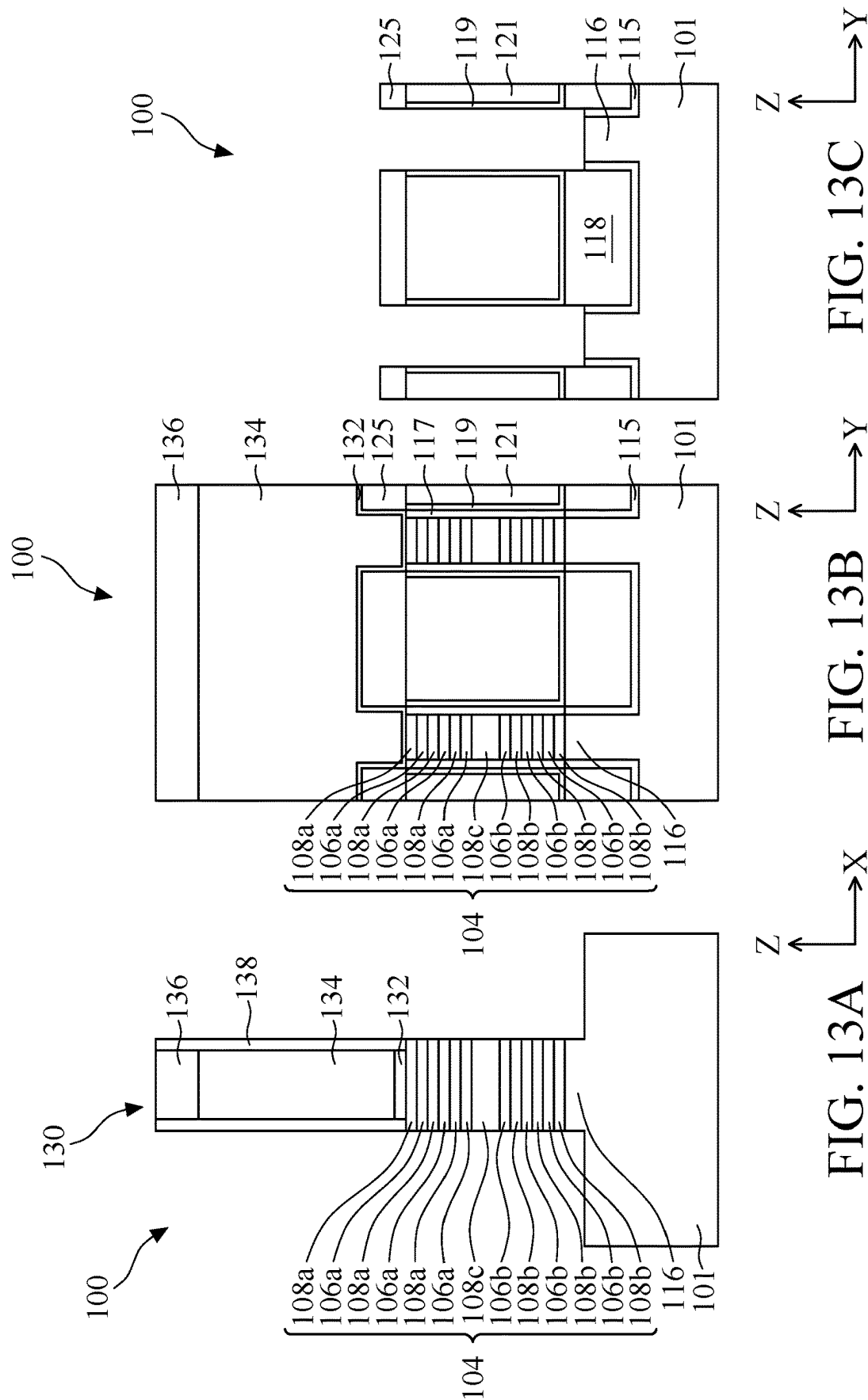
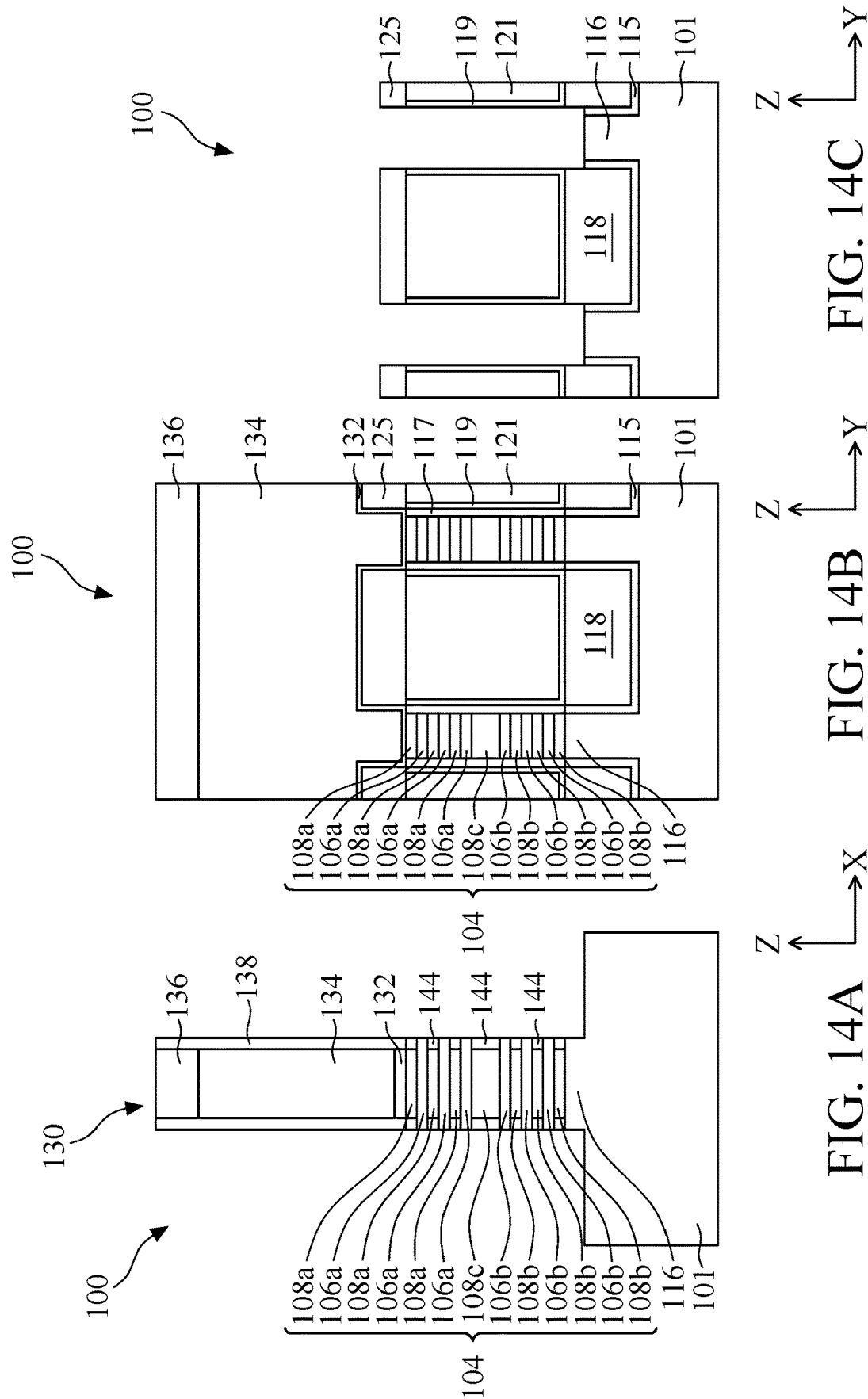
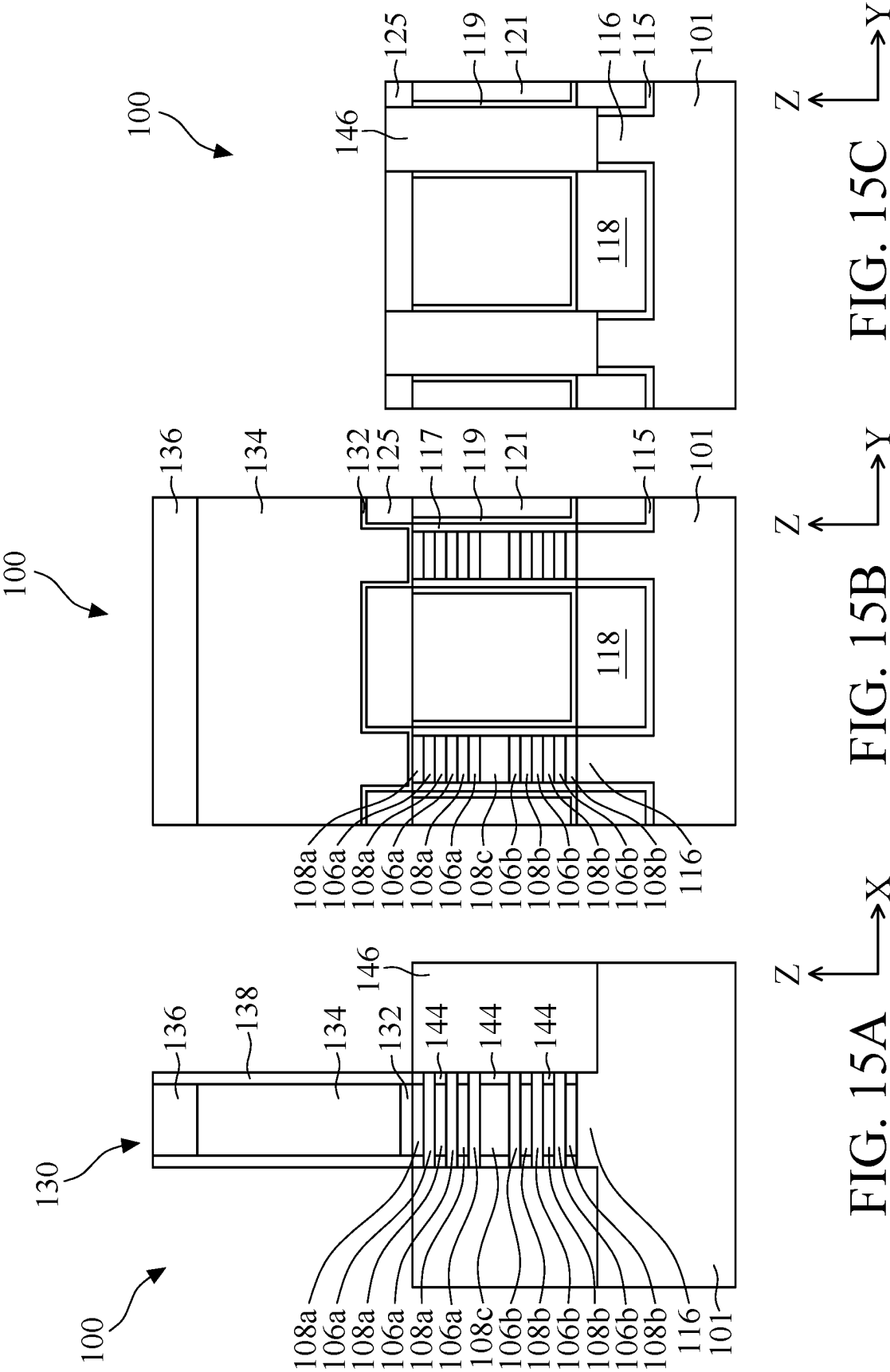
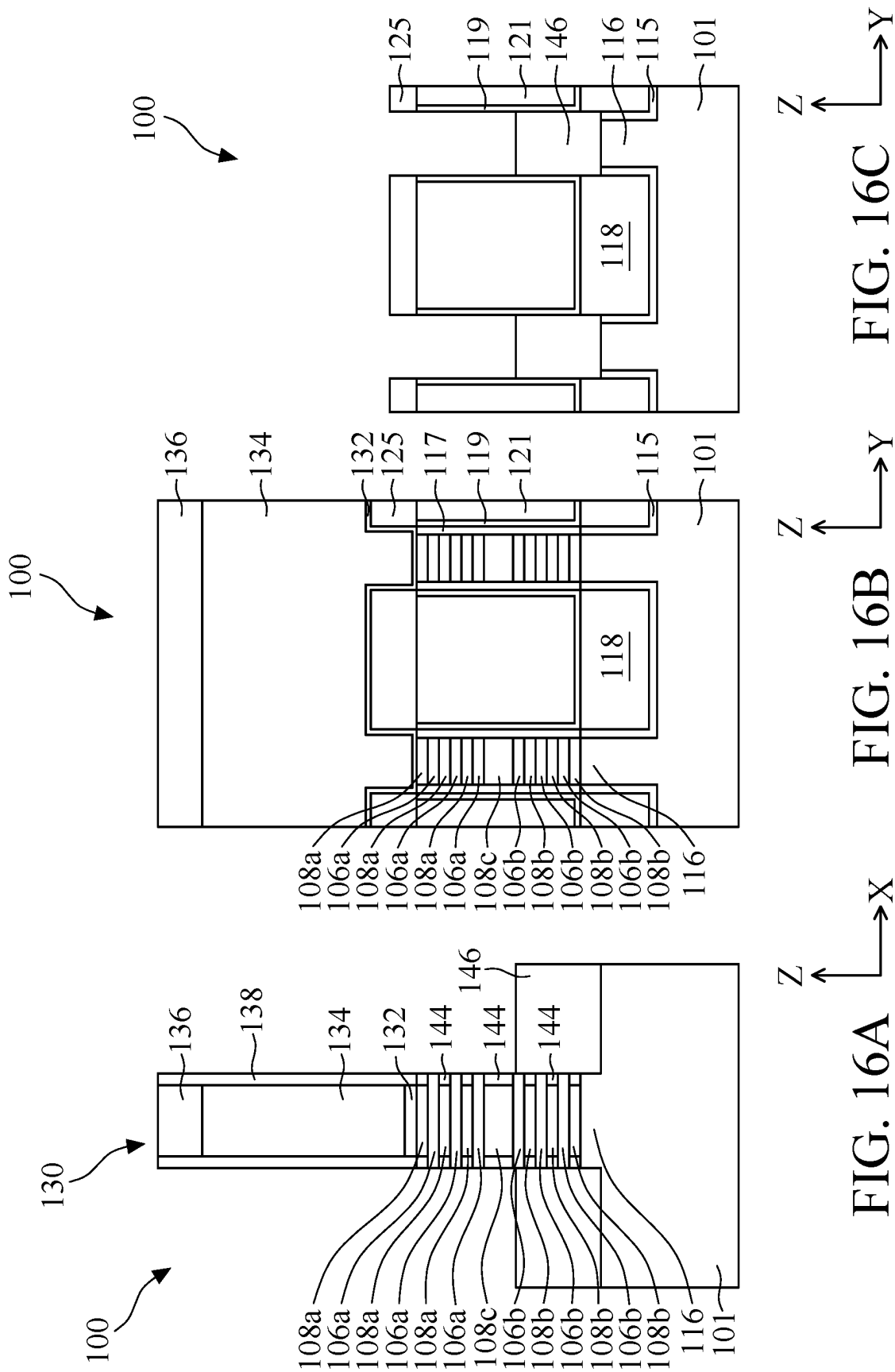


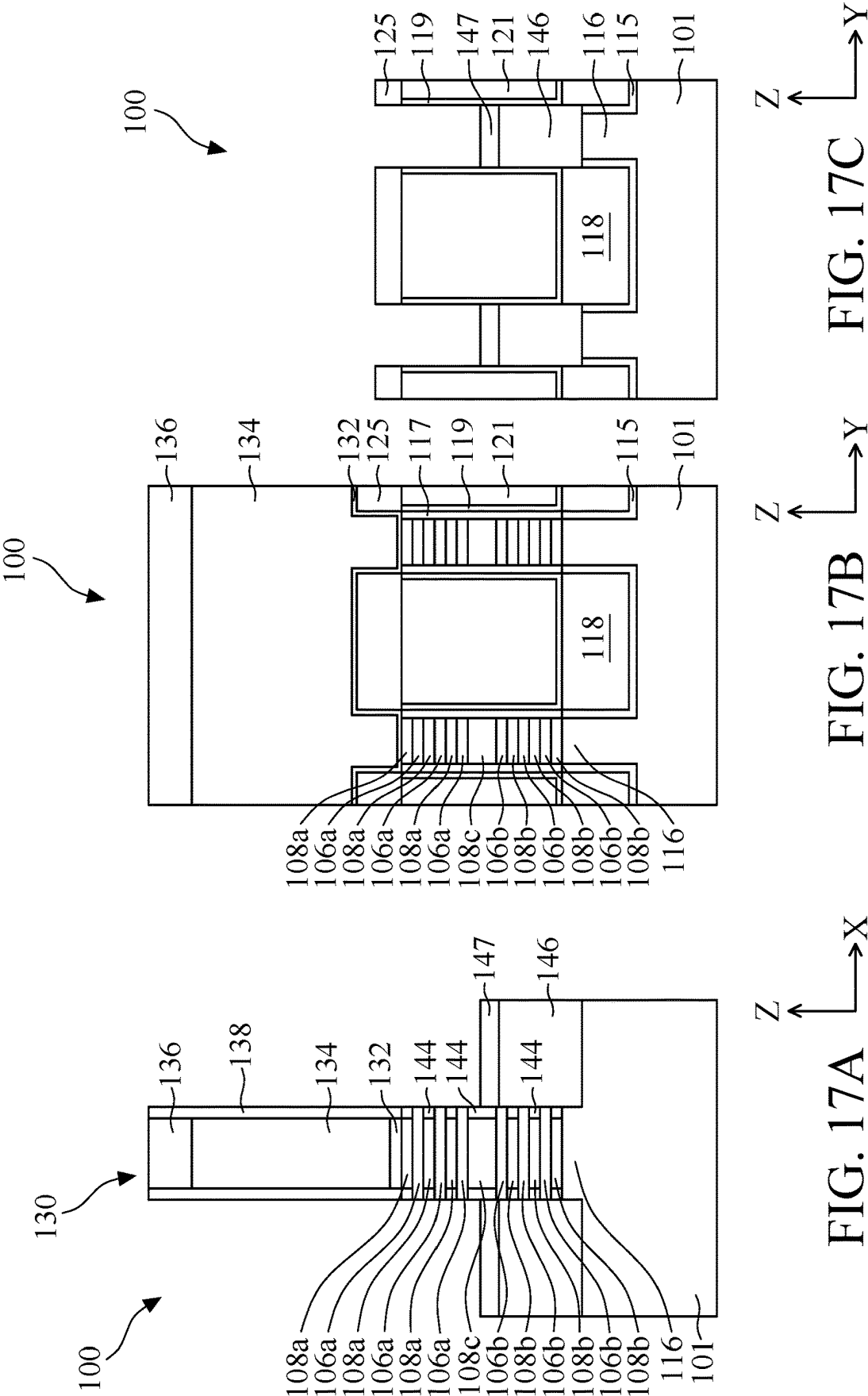
FIG. 12

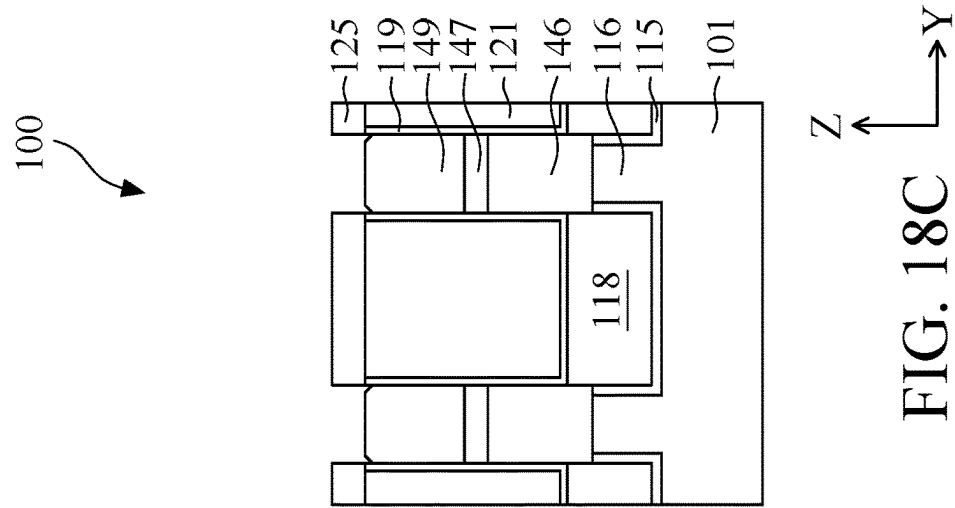
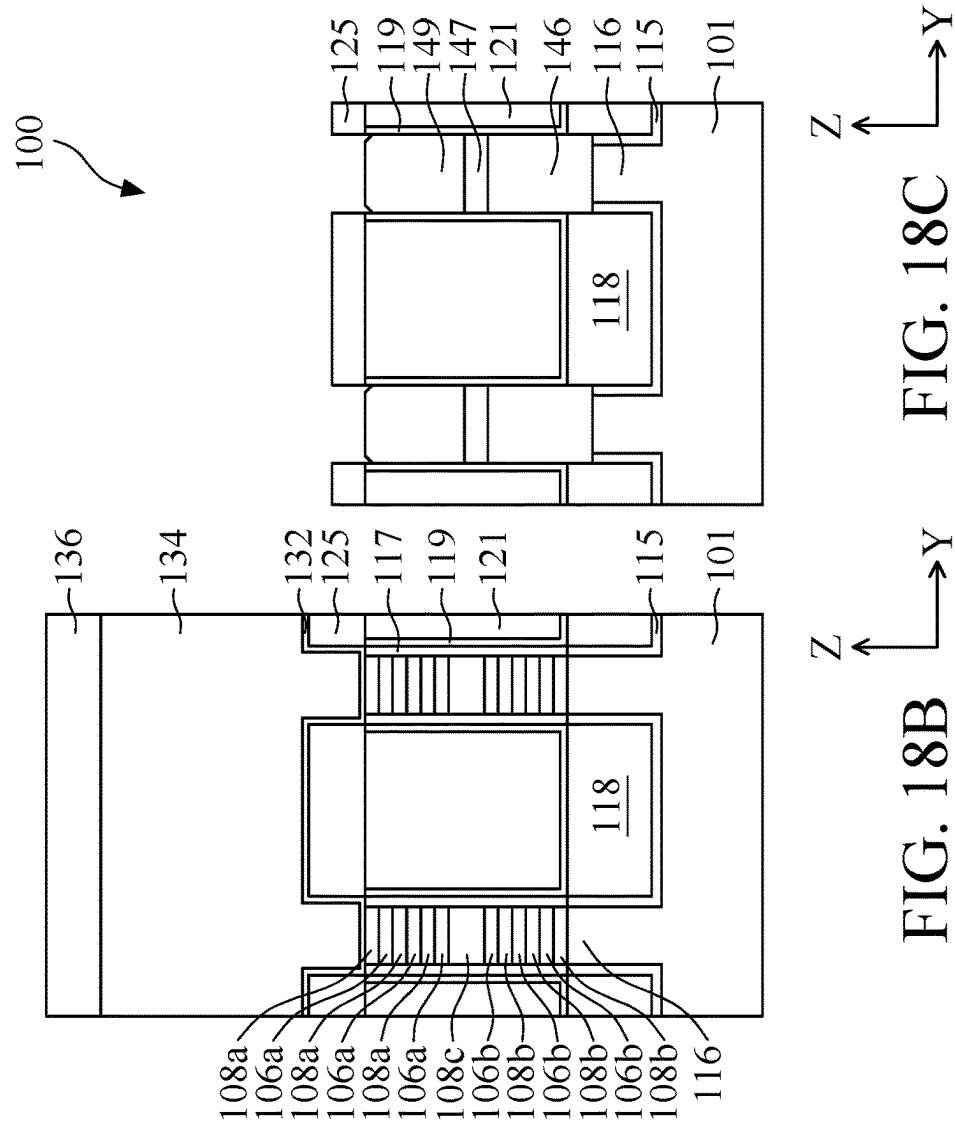
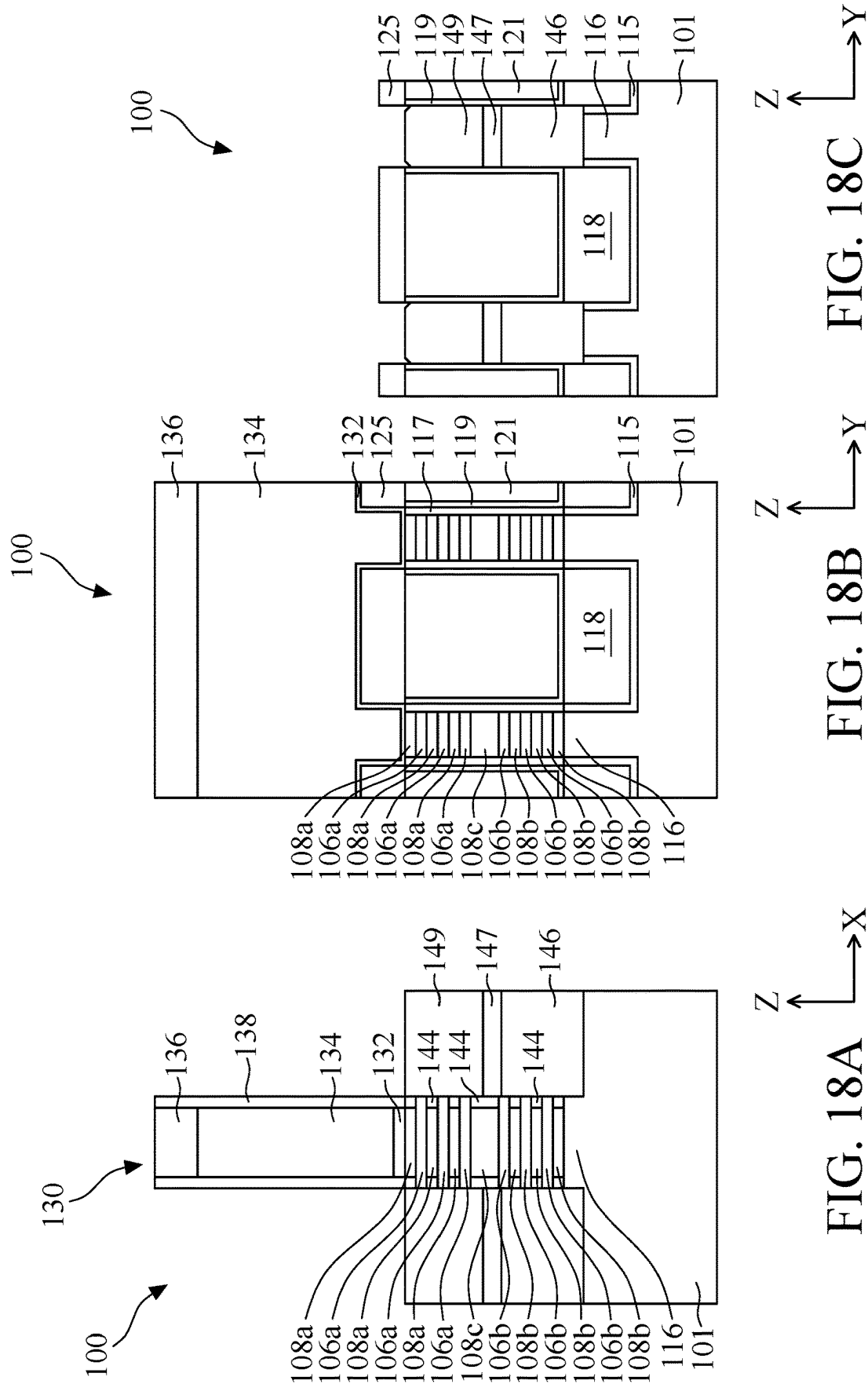


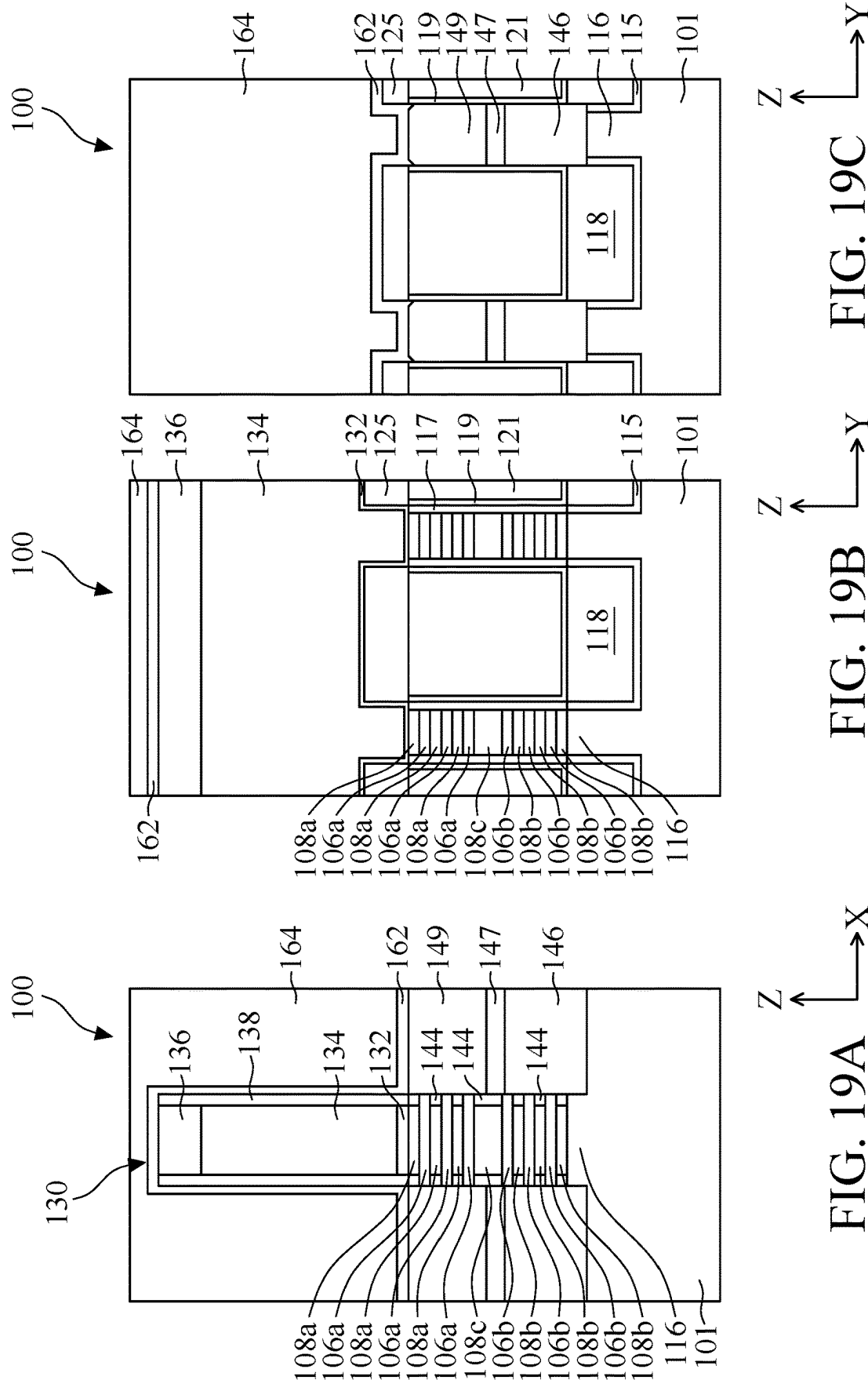


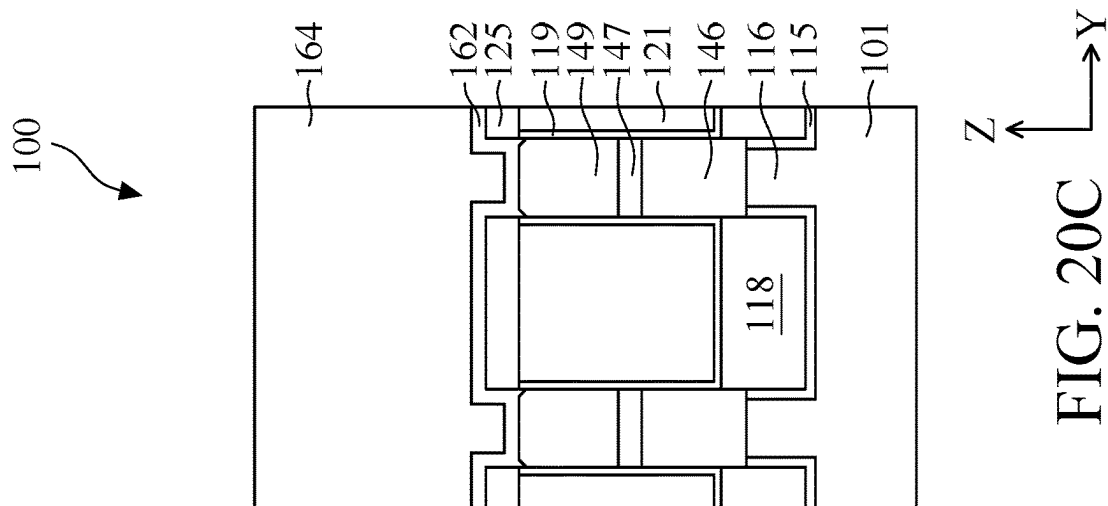
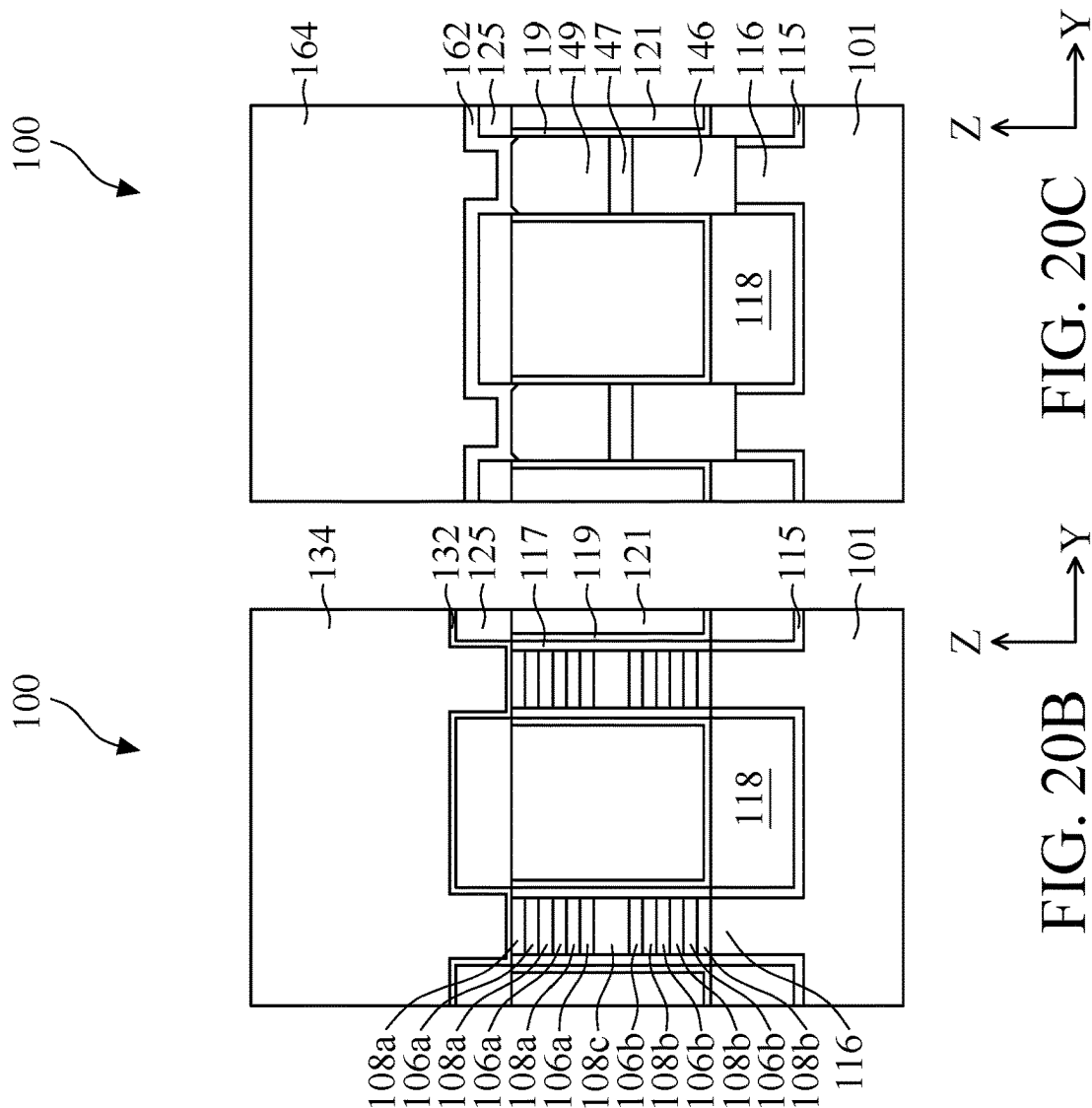
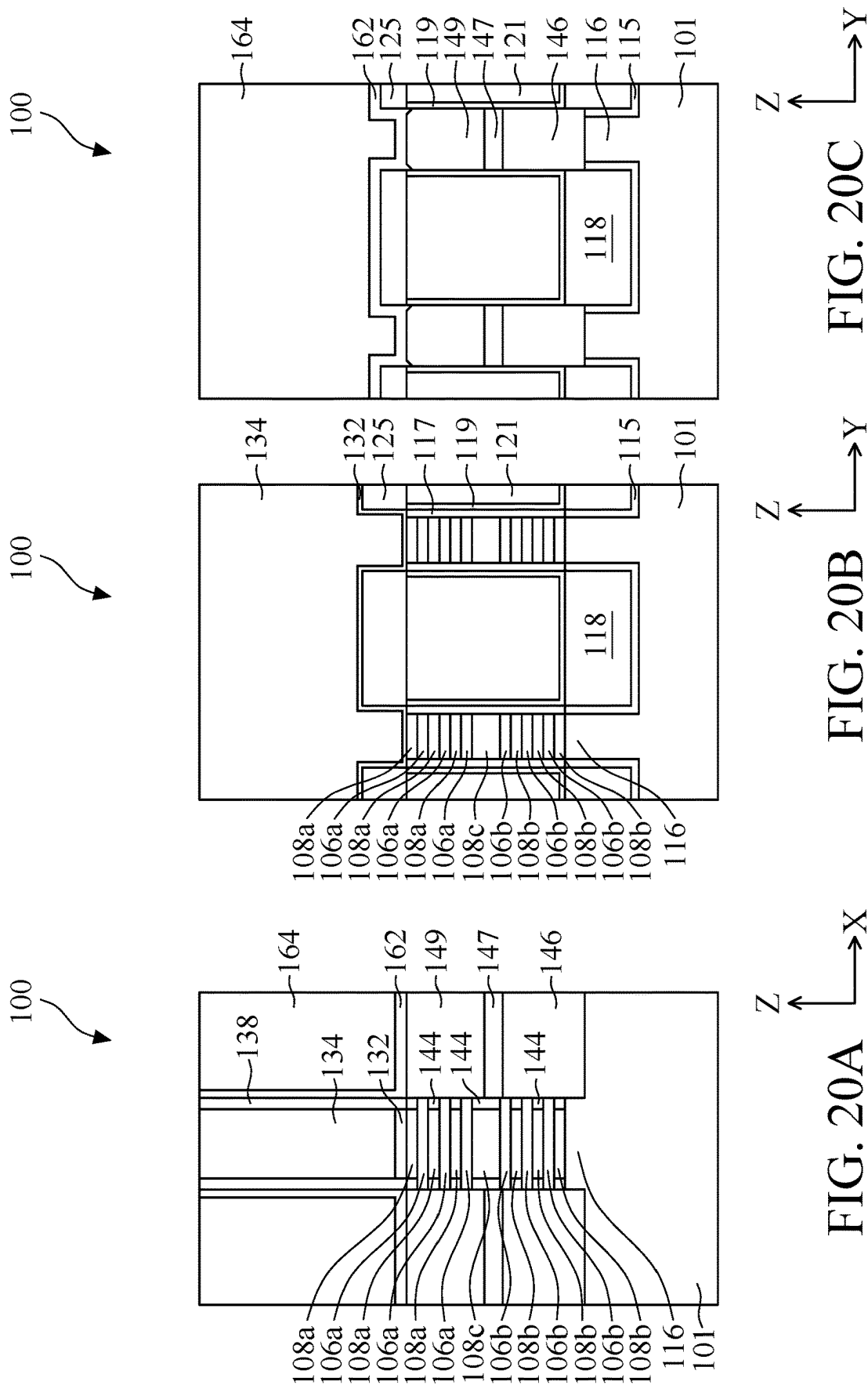


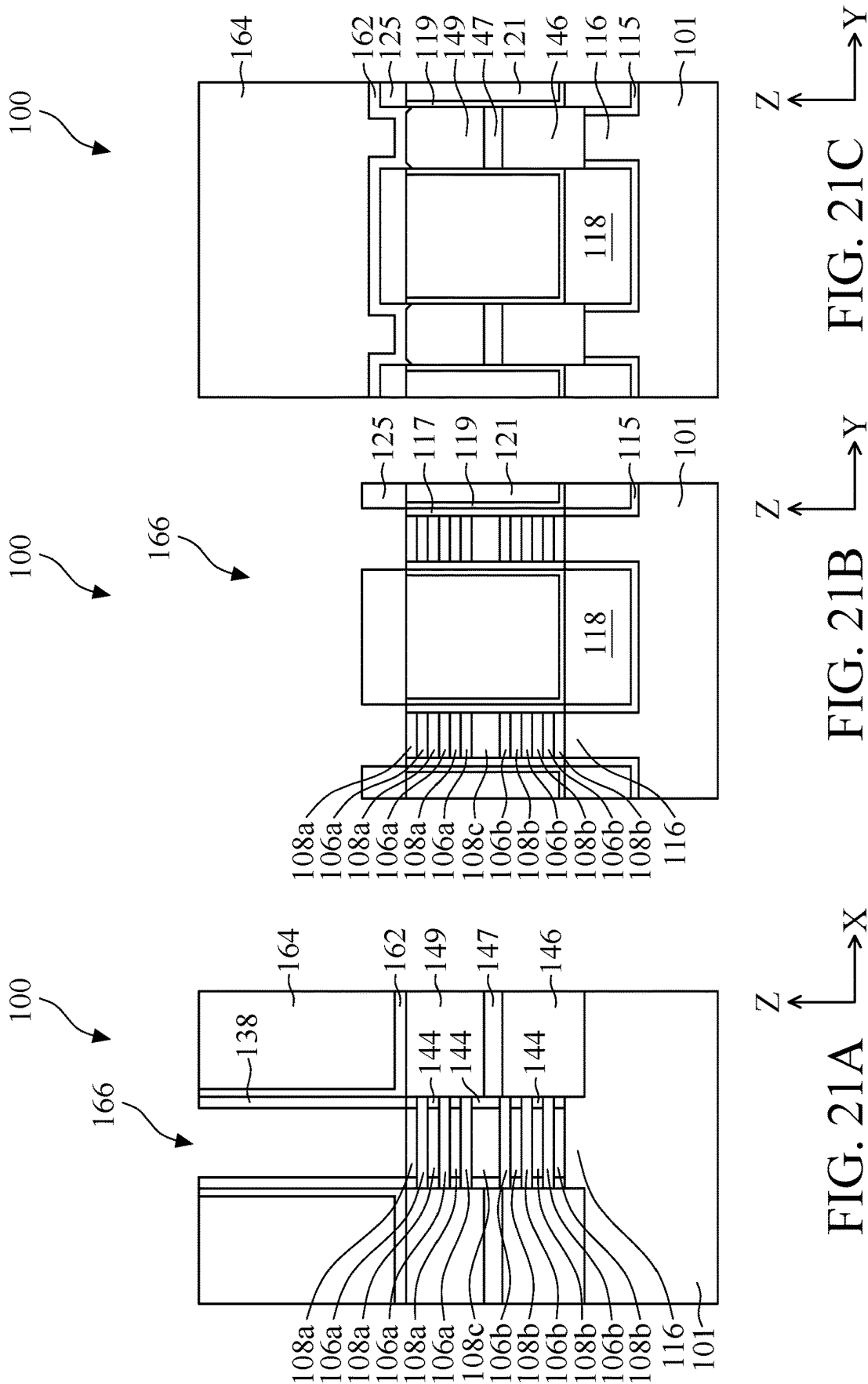


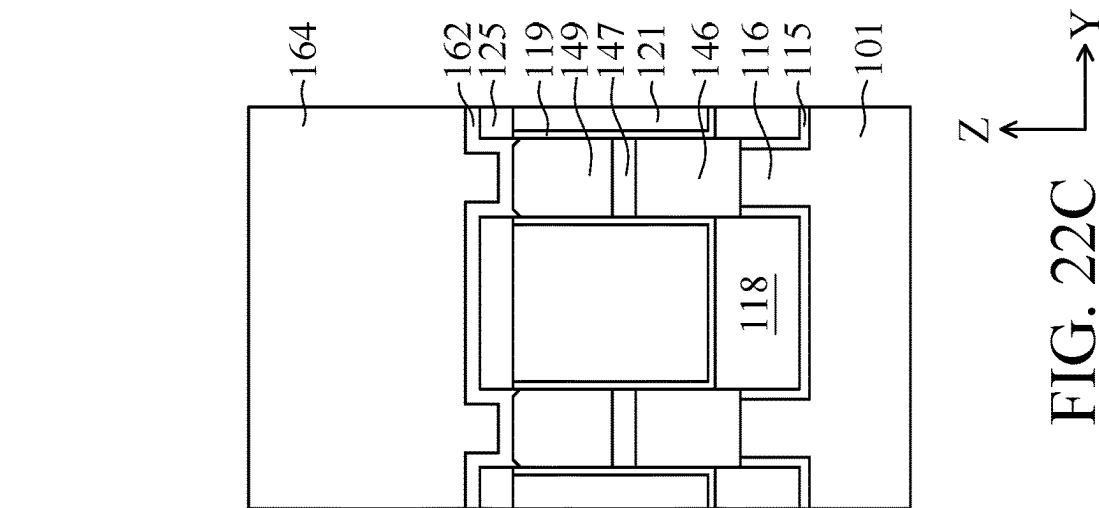
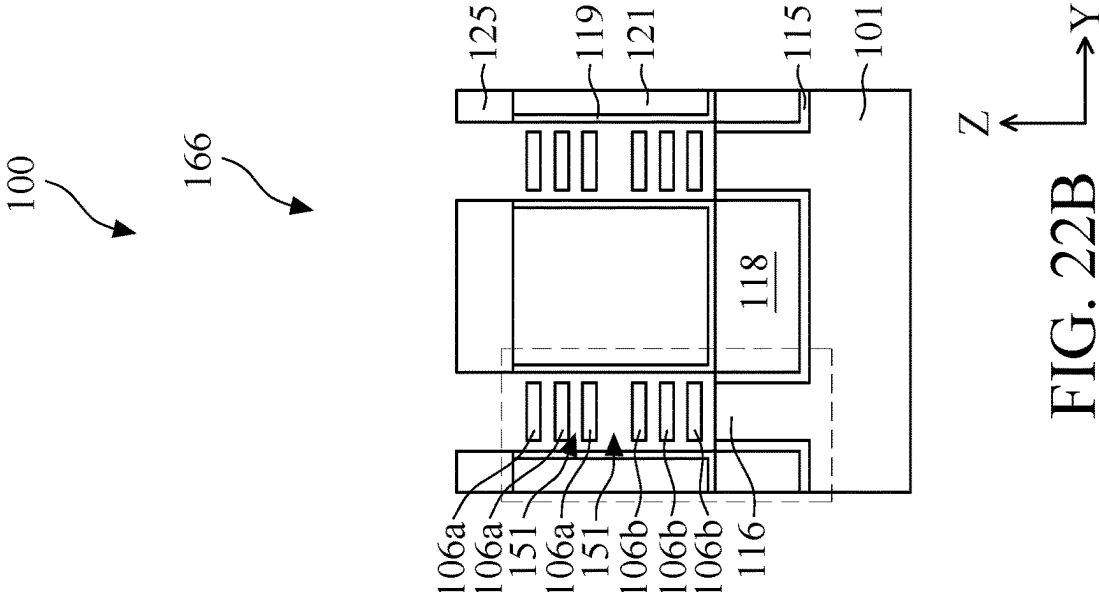
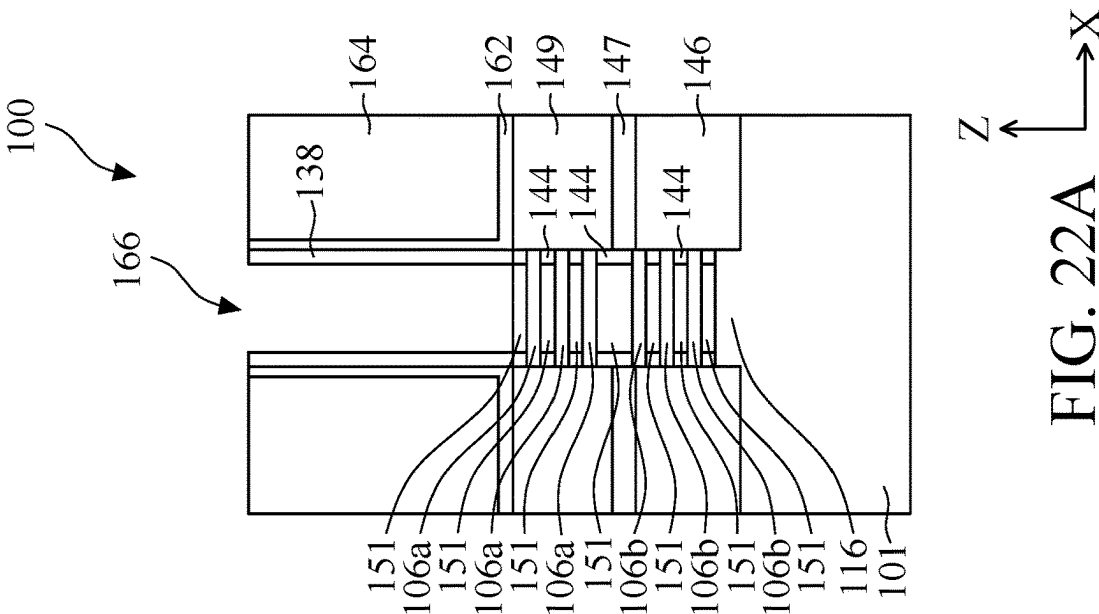


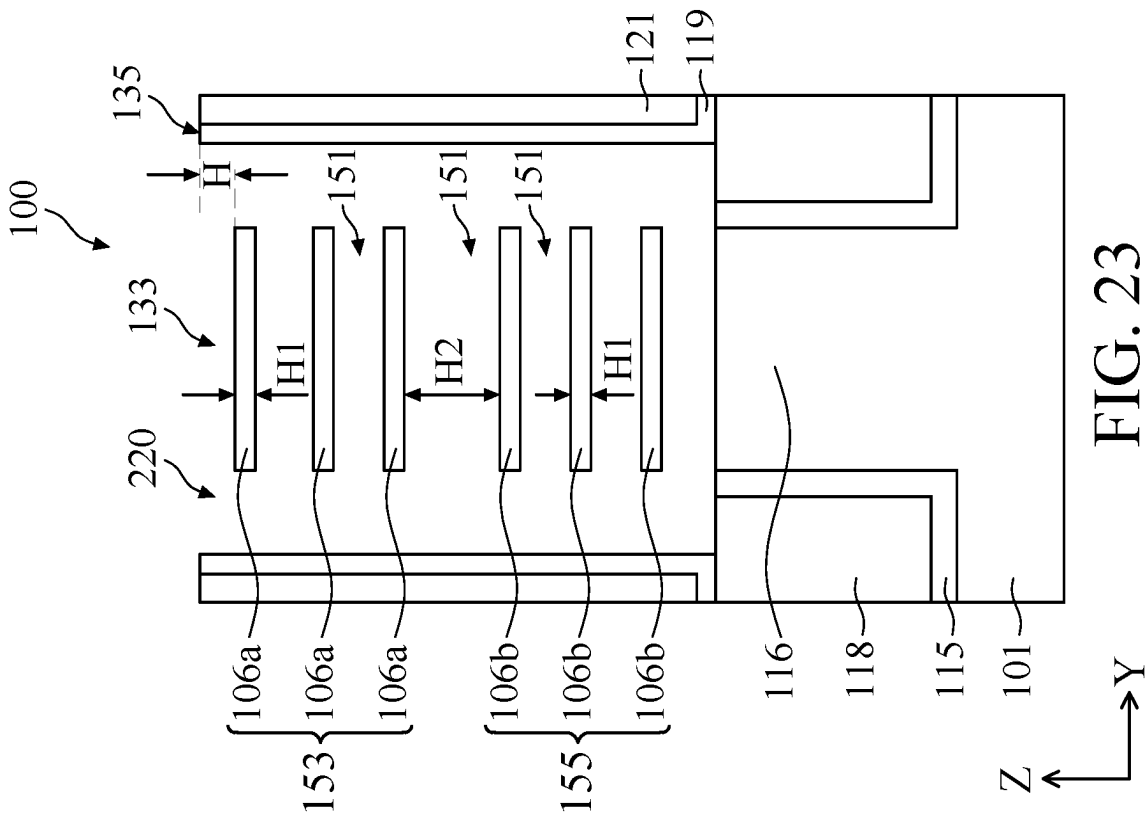
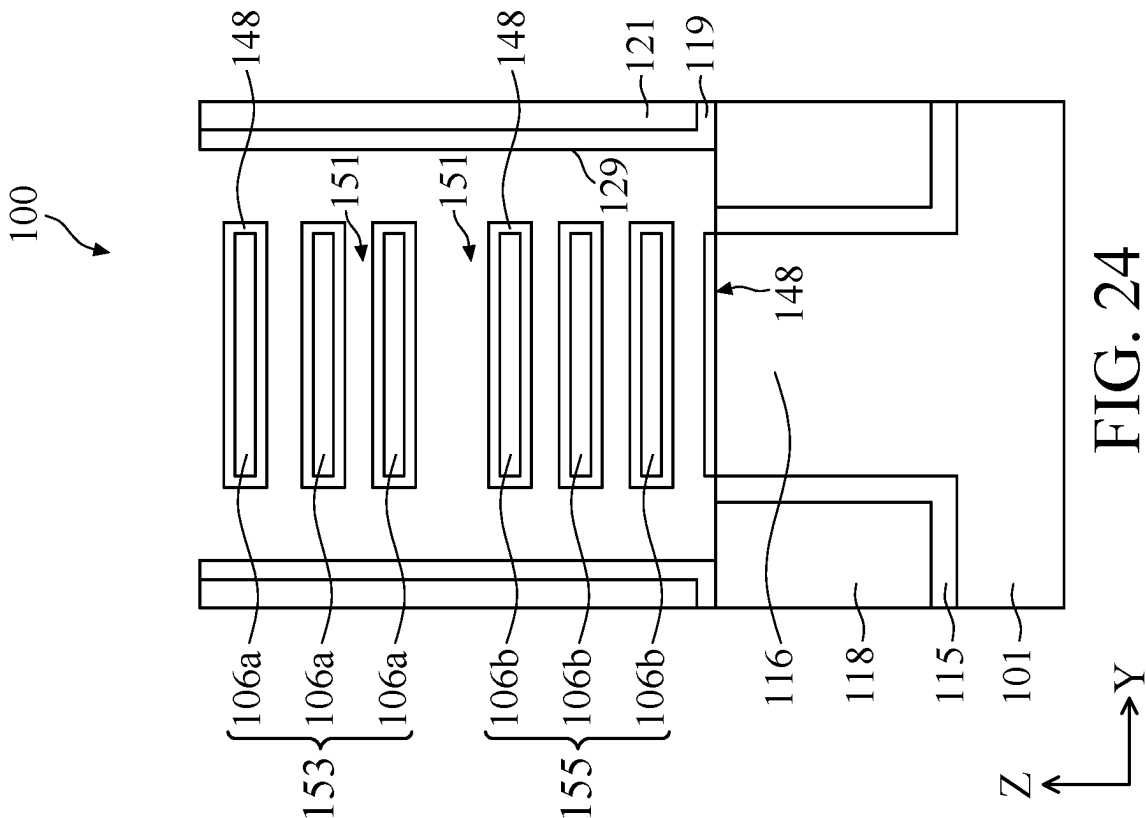


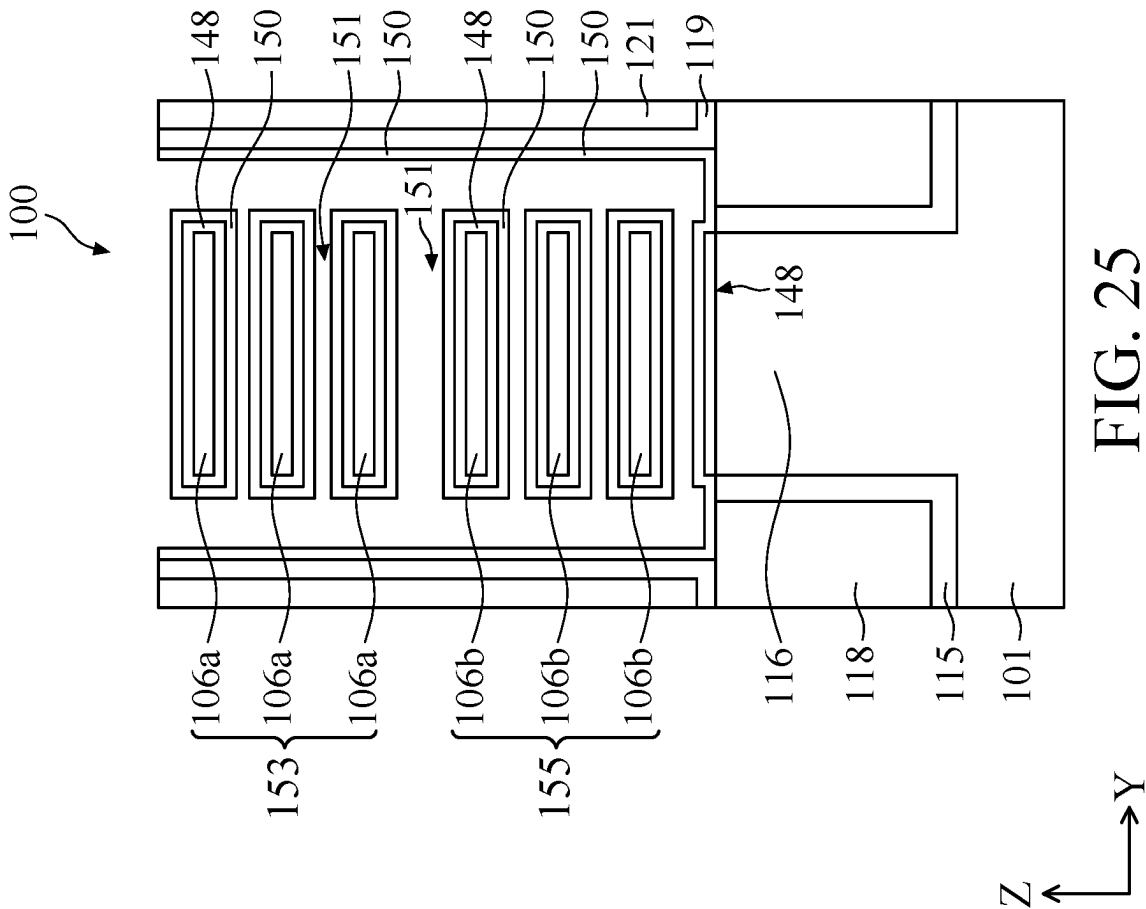












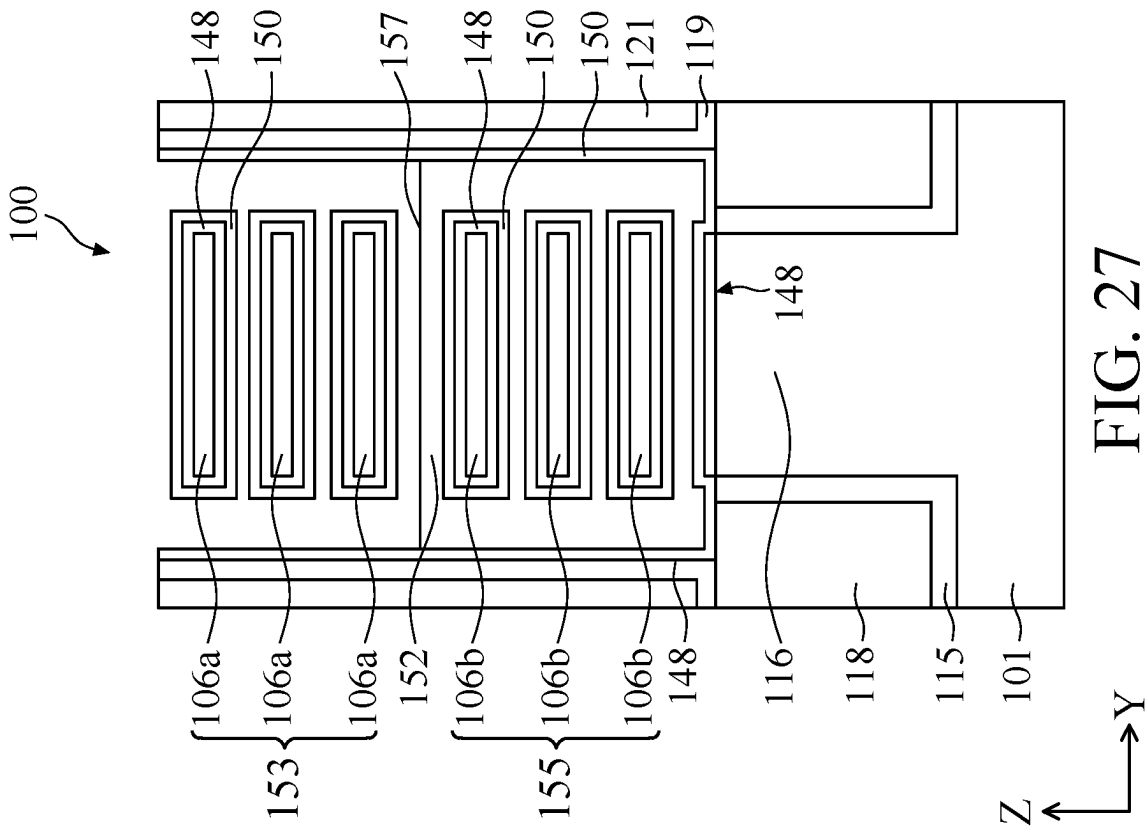


FIG. 27

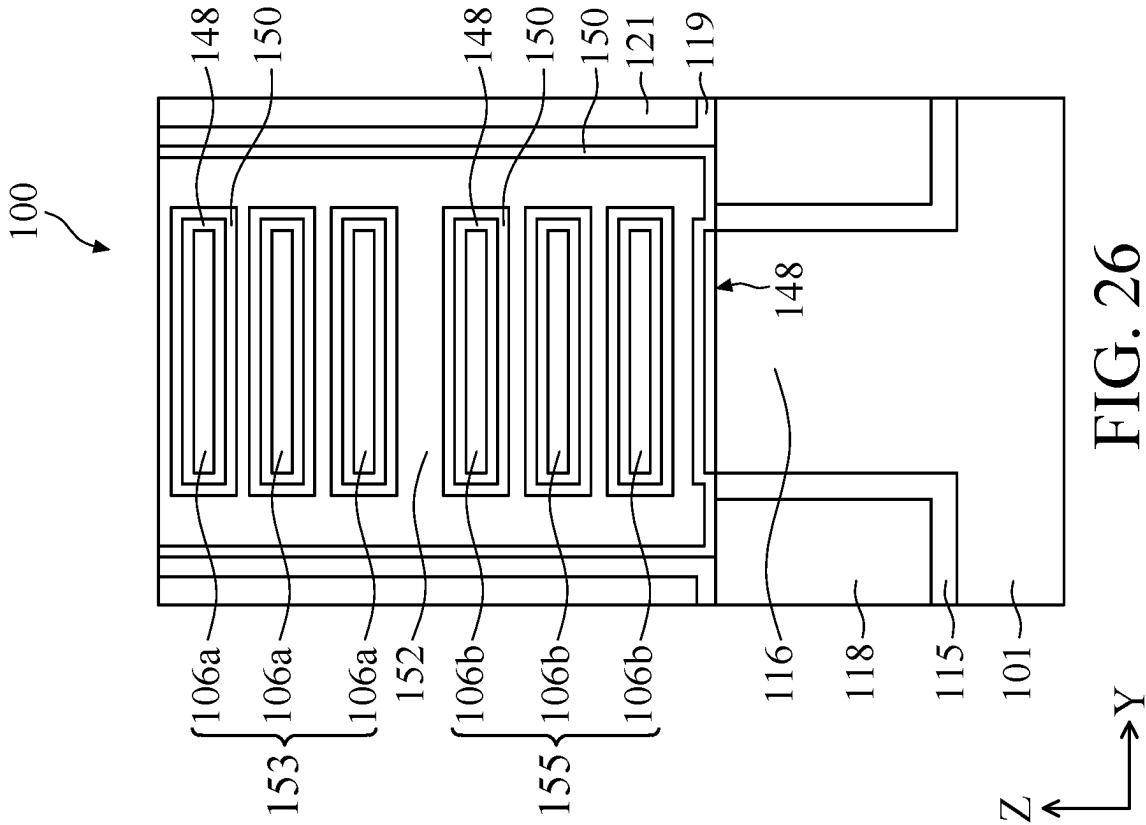
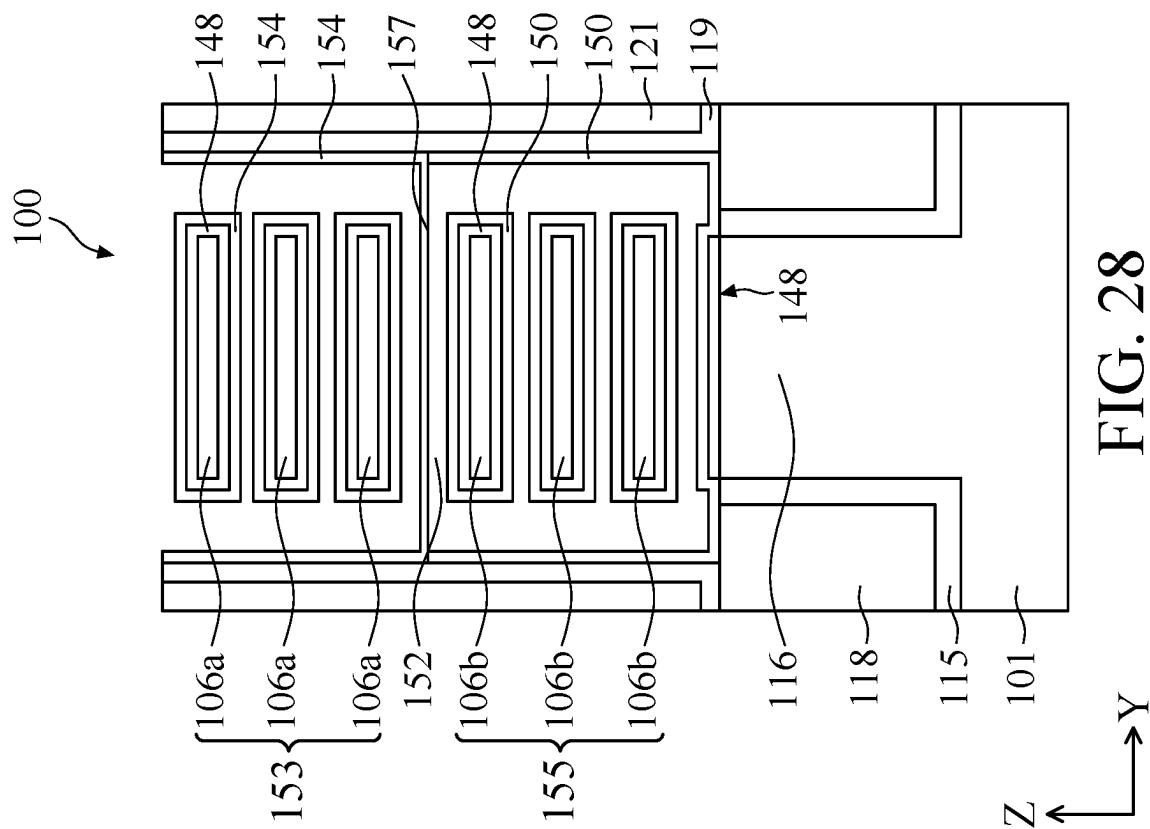
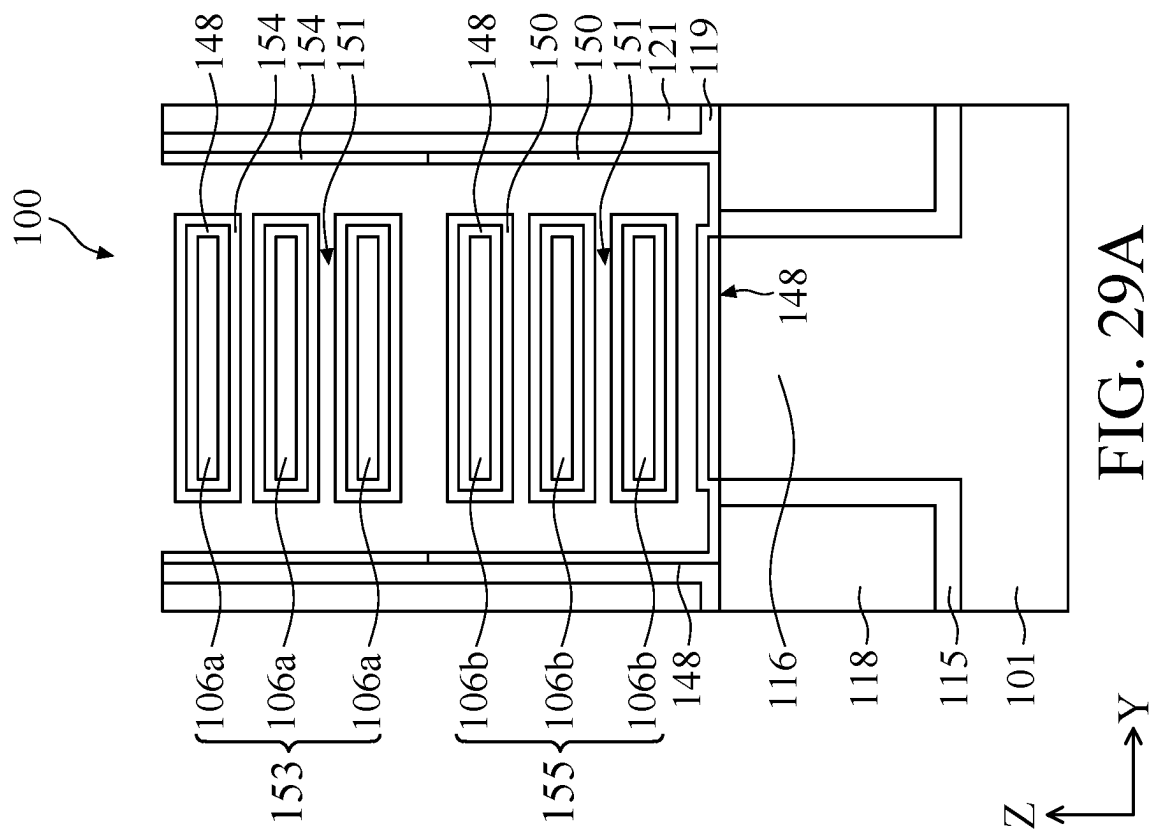


FIG. 26



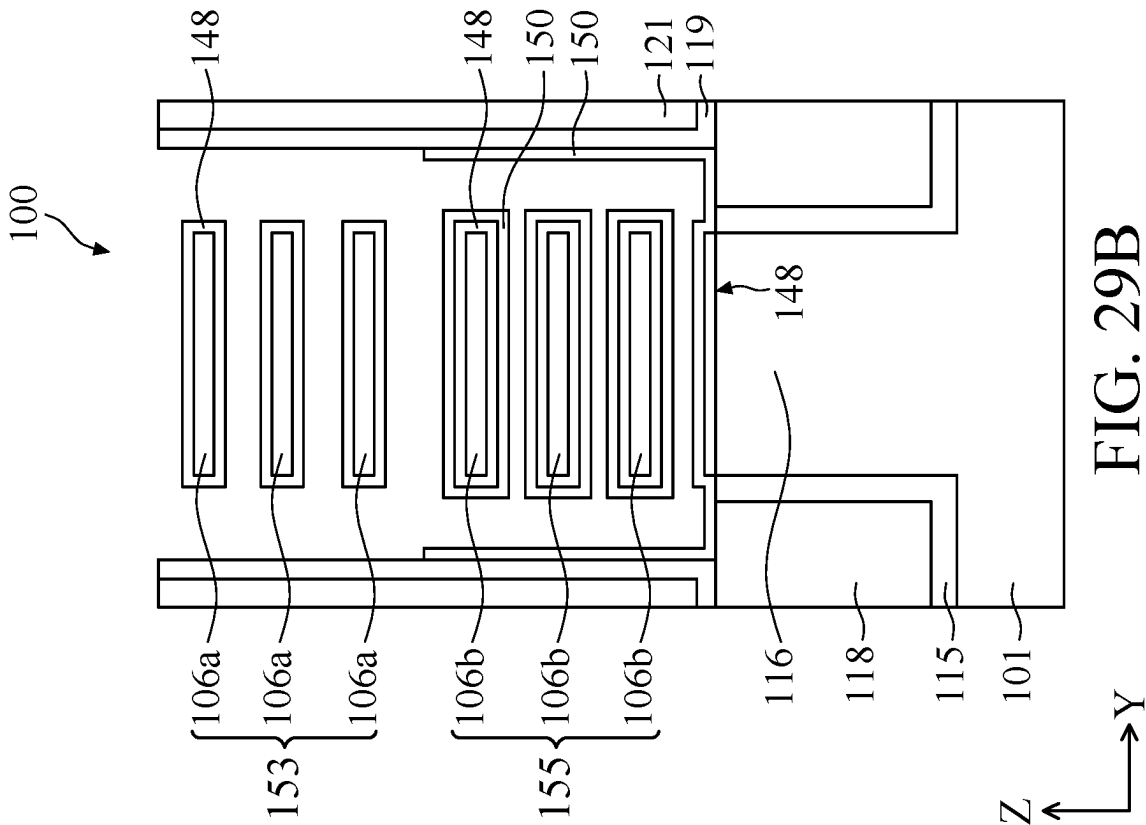


FIG. 29B

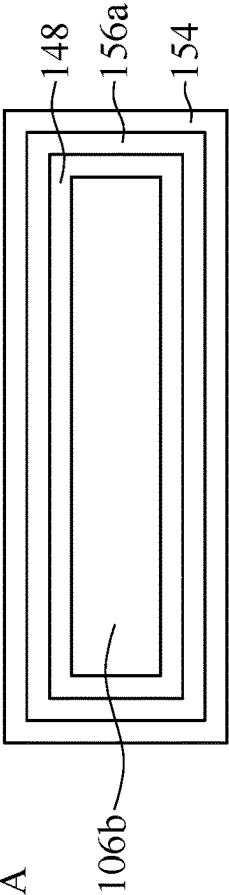
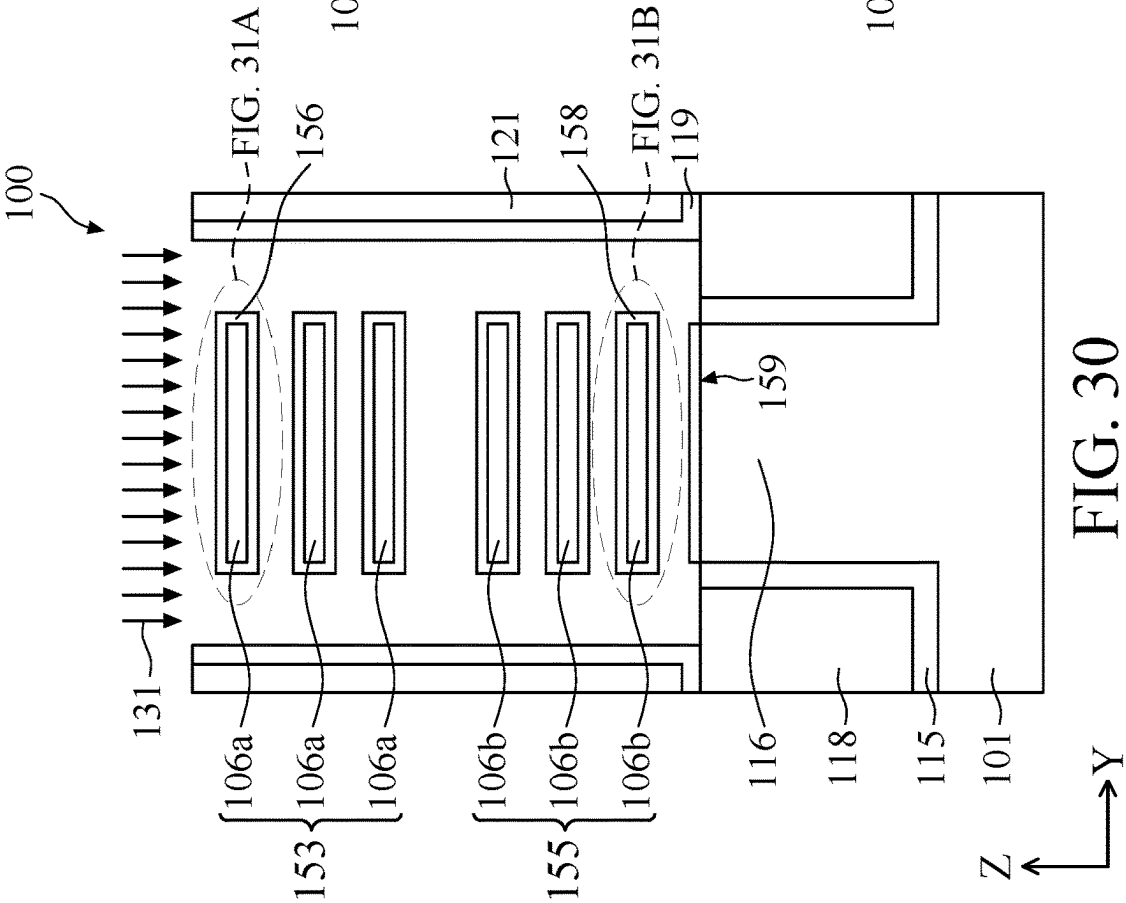


FIG. 31A

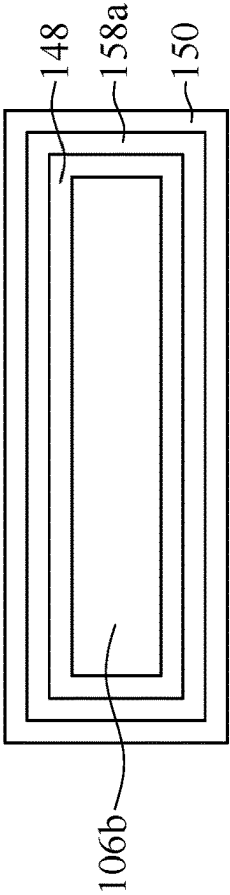


FIG. 31B

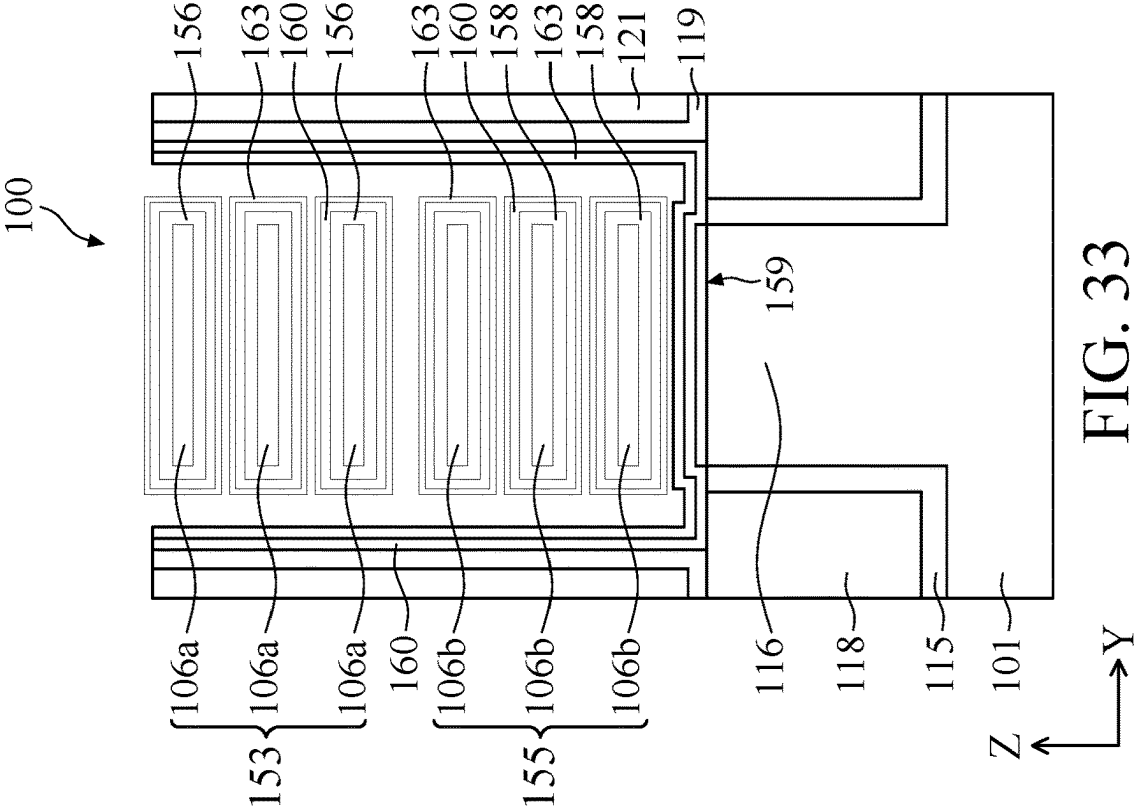


FIG. 32

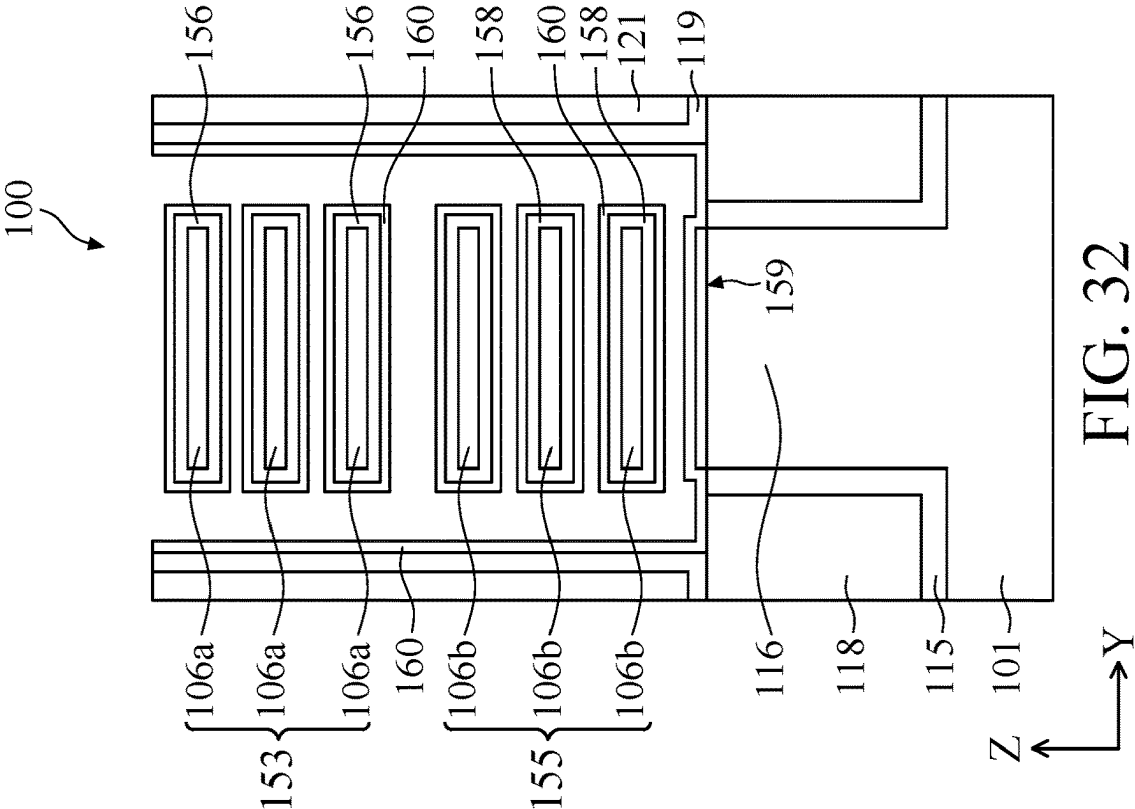


FIG. 33

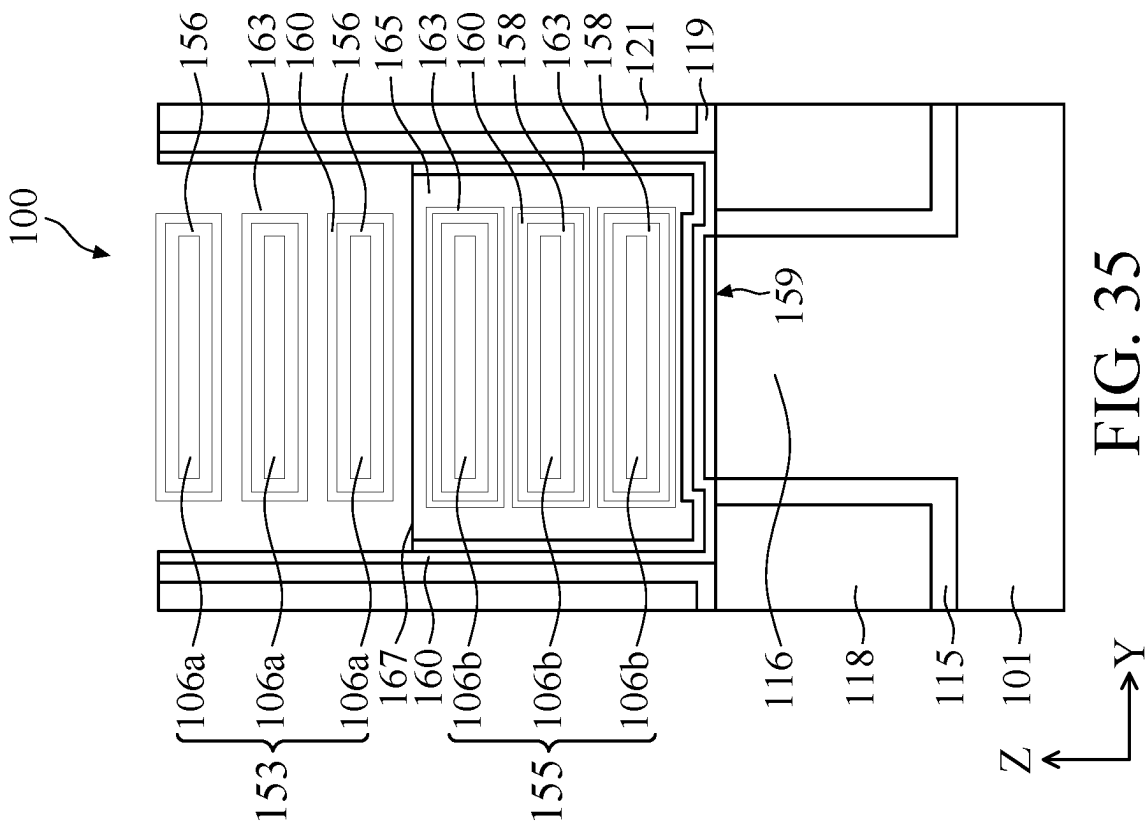


FIG. 34

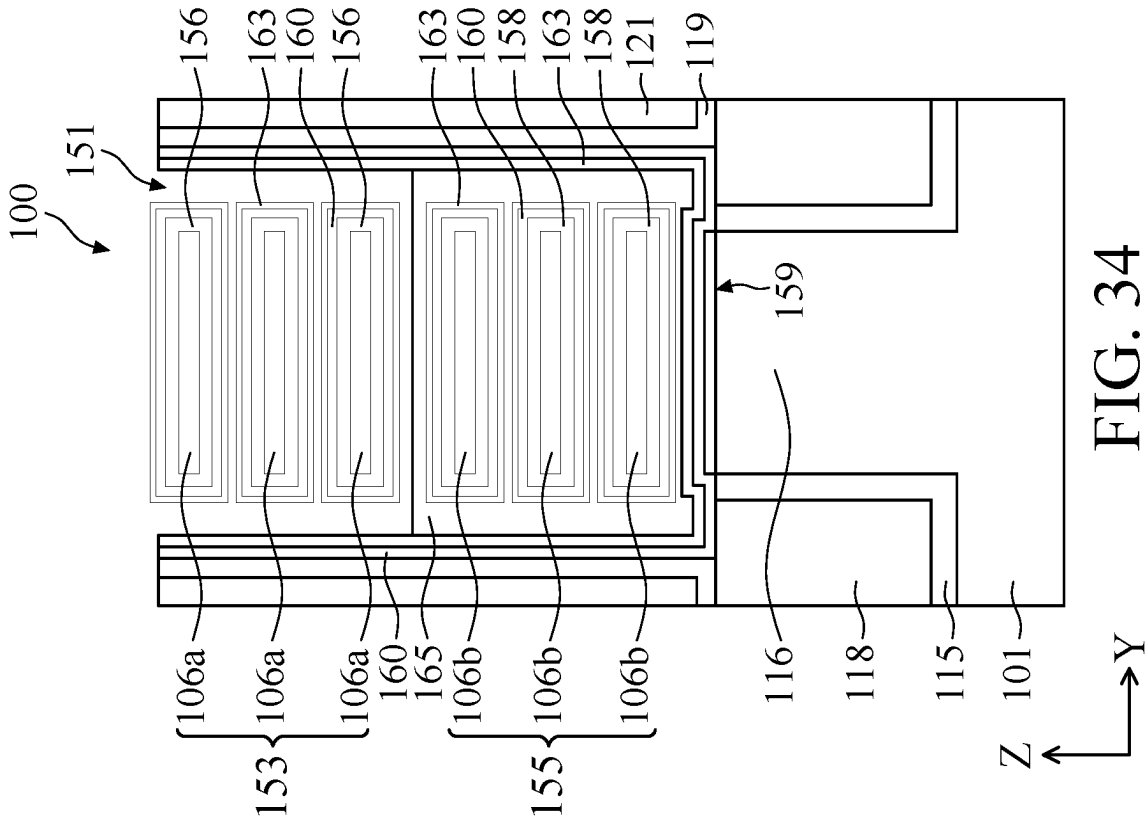
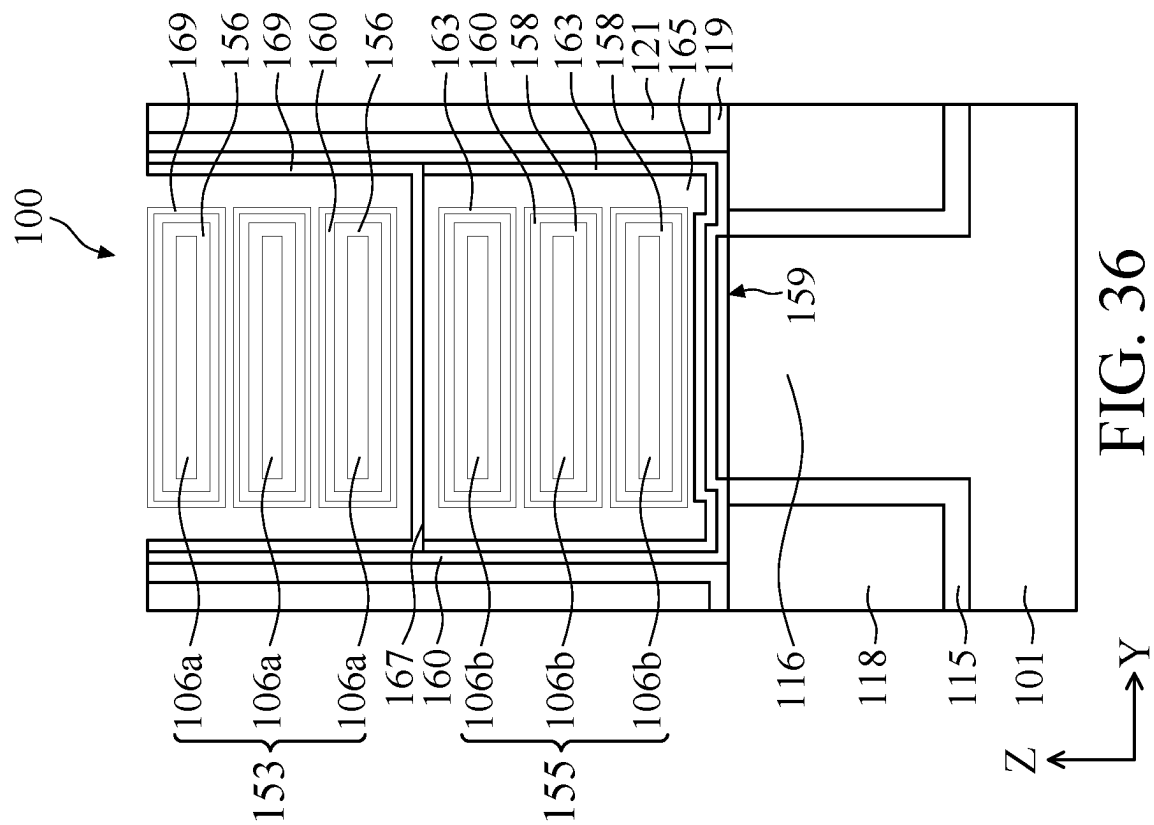
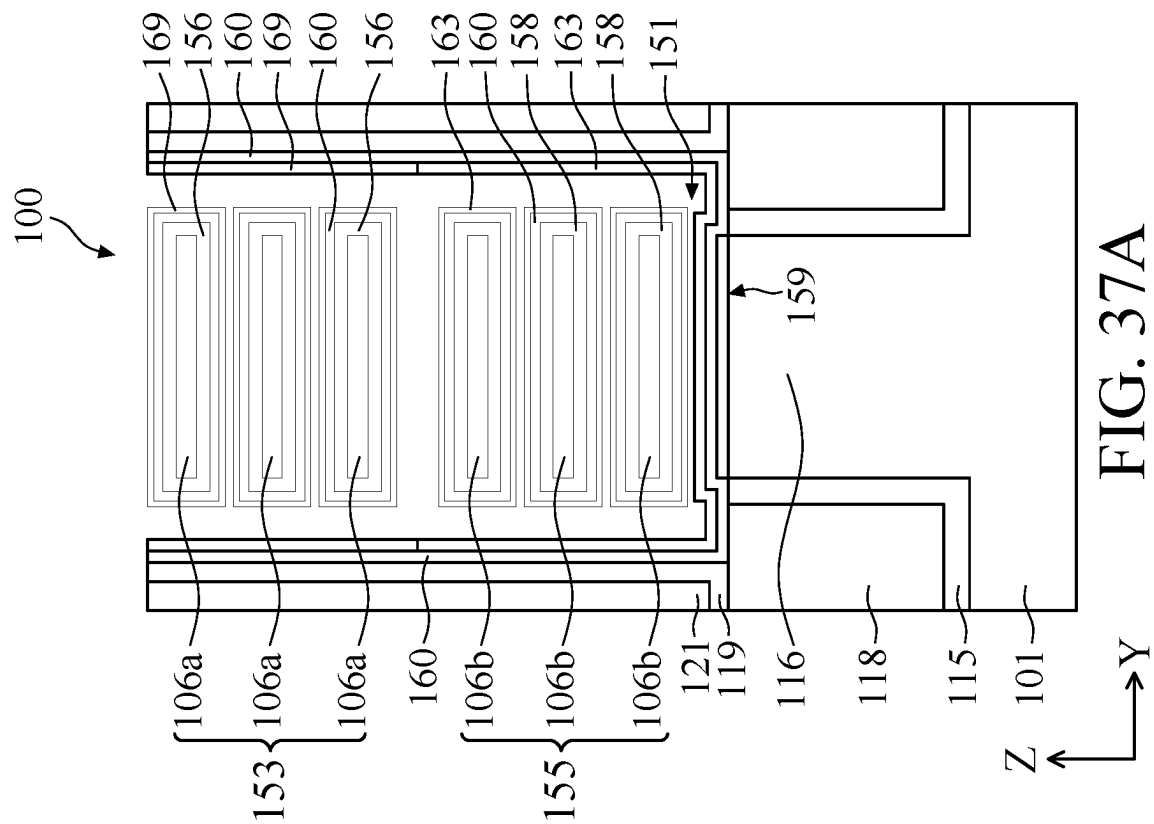
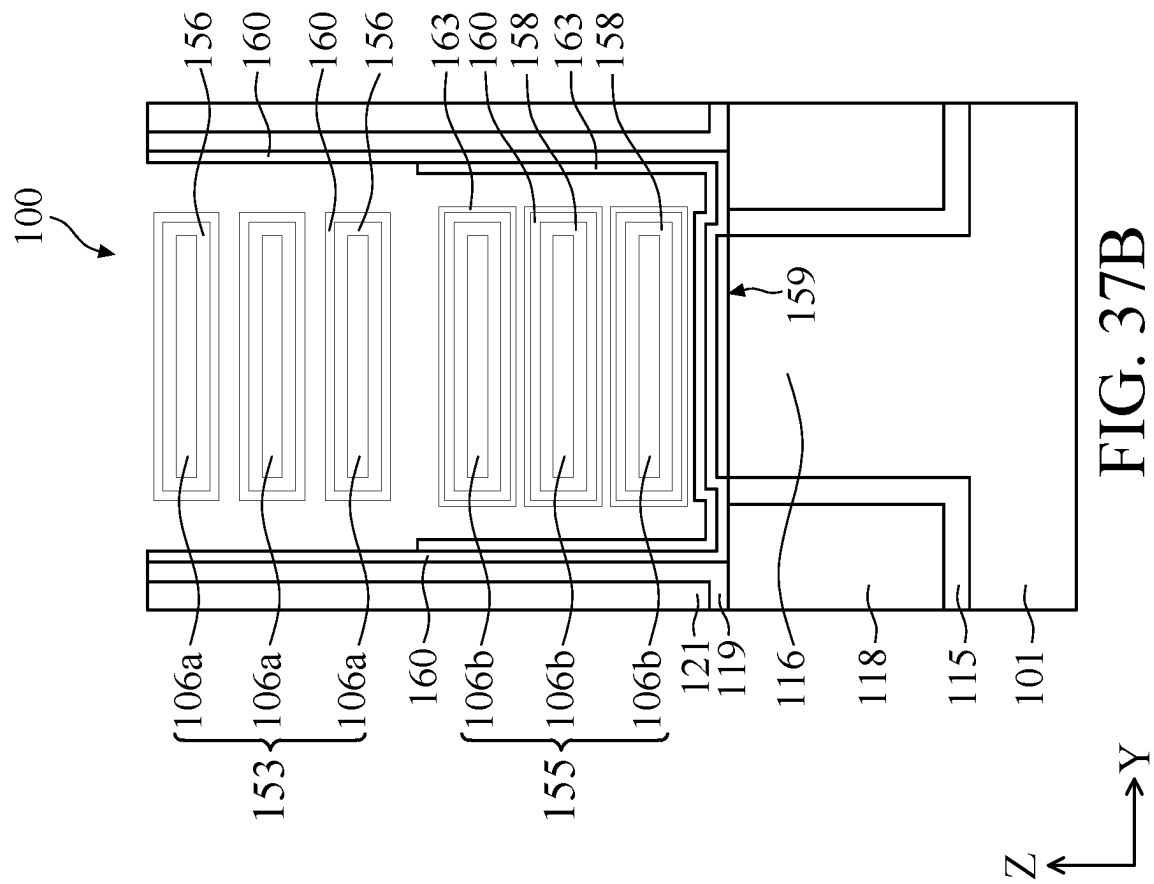
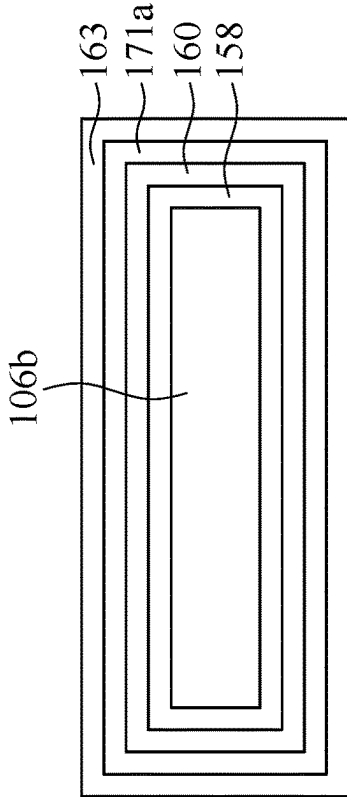
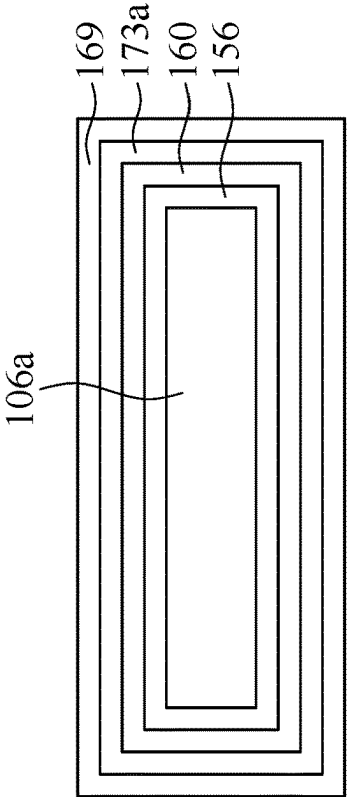
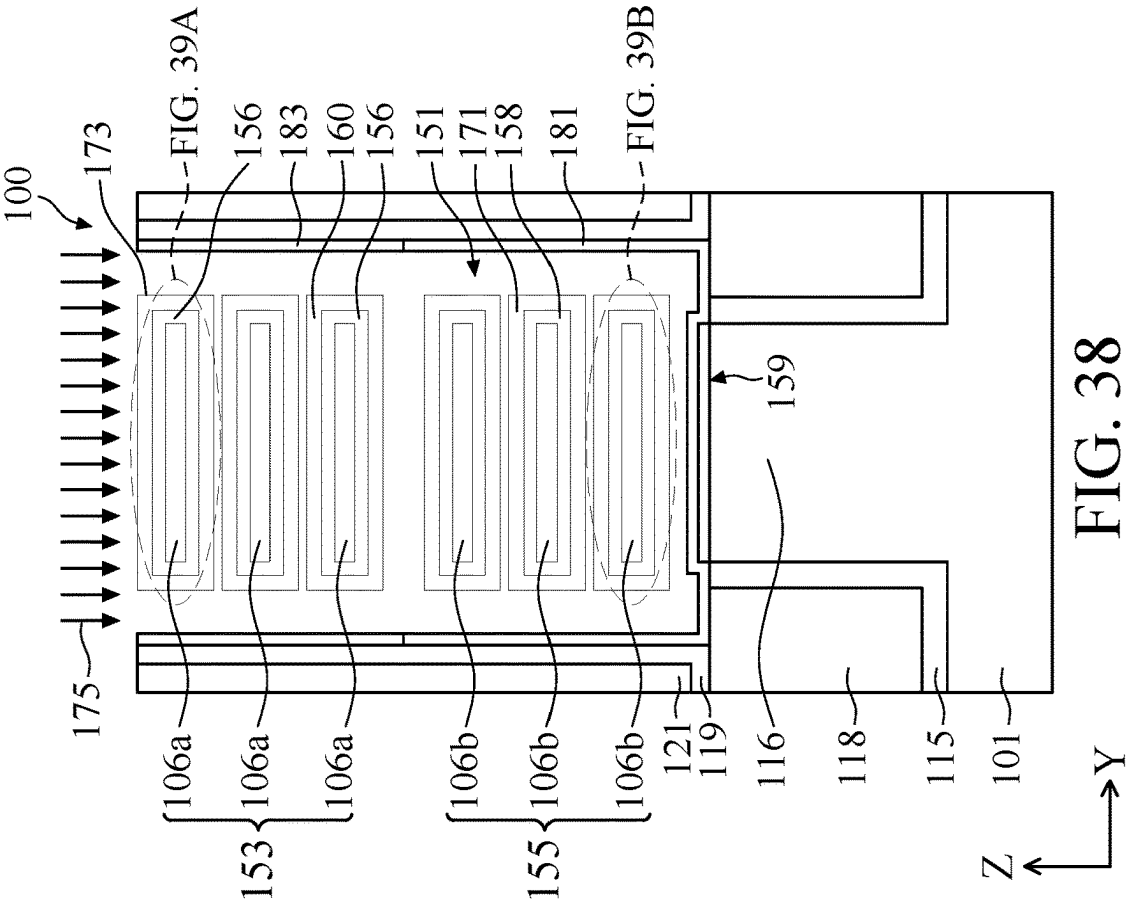


FIG. 35







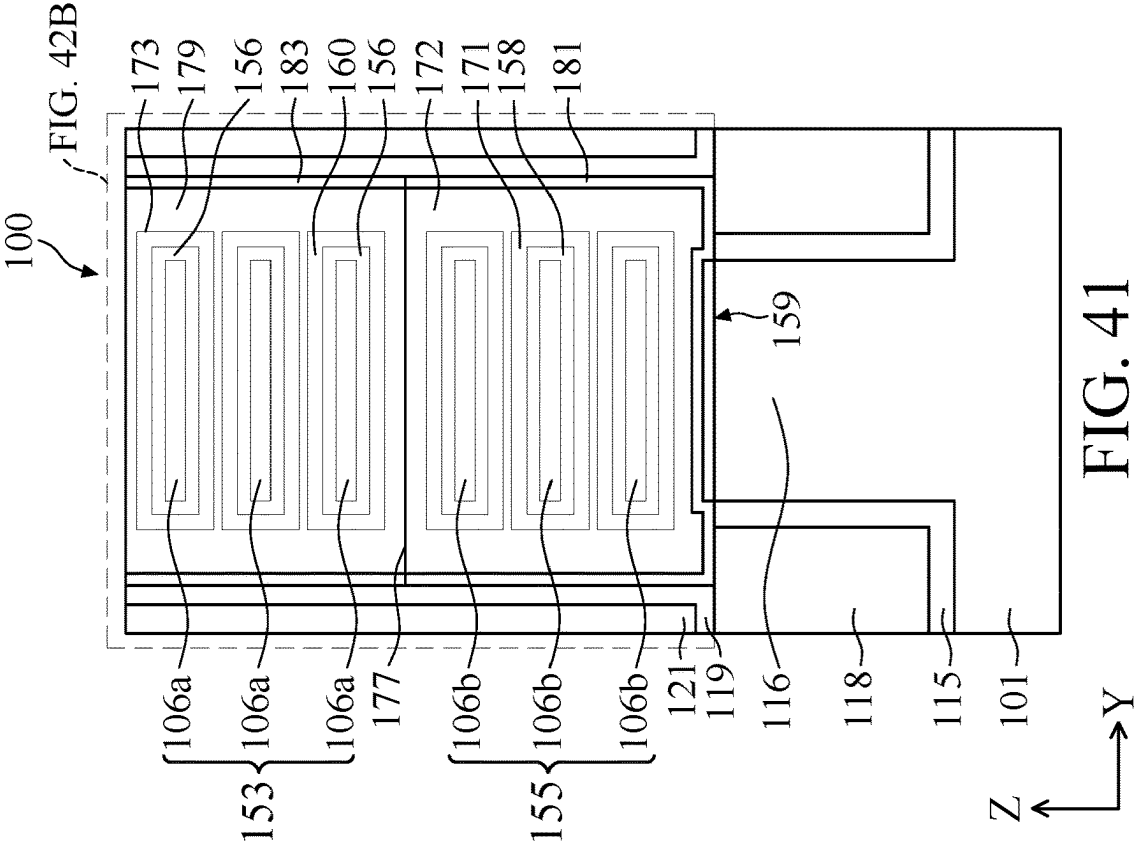


FIG. 40

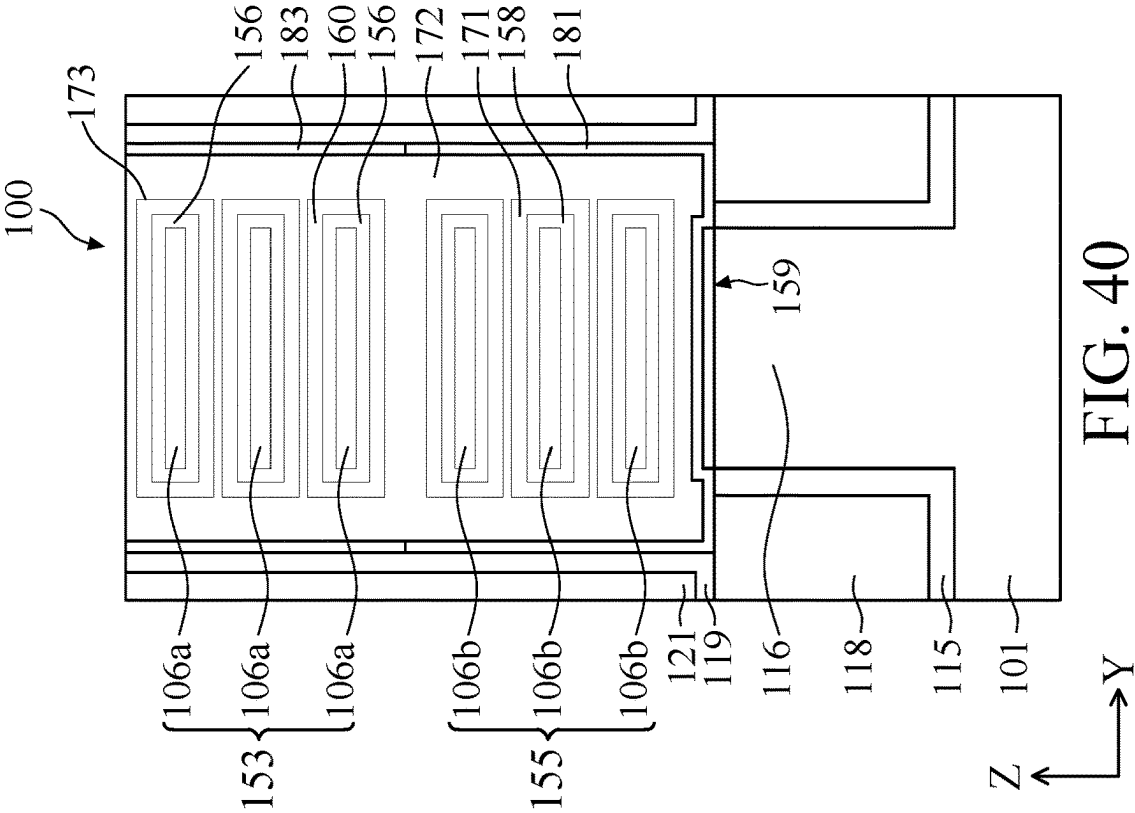


FIG. 41

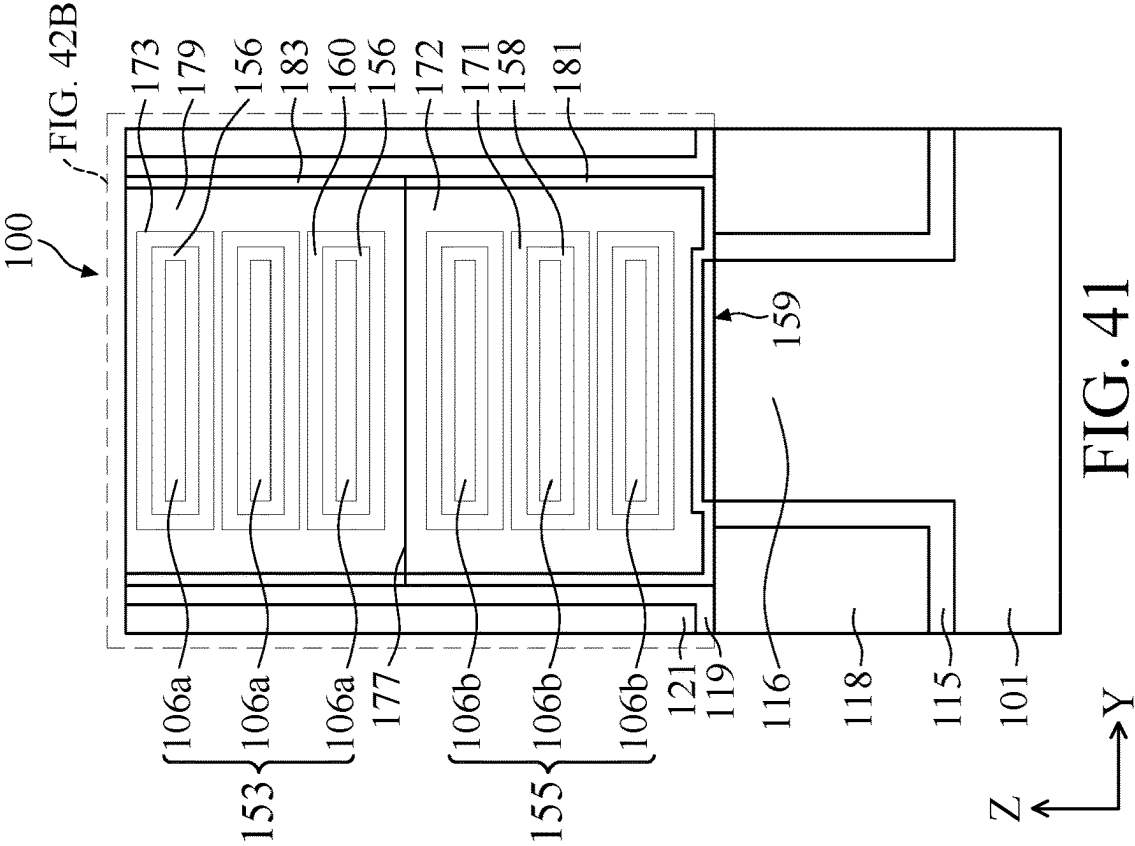


FIG. 42B

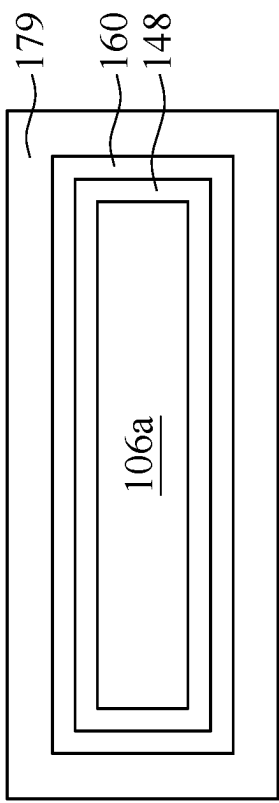
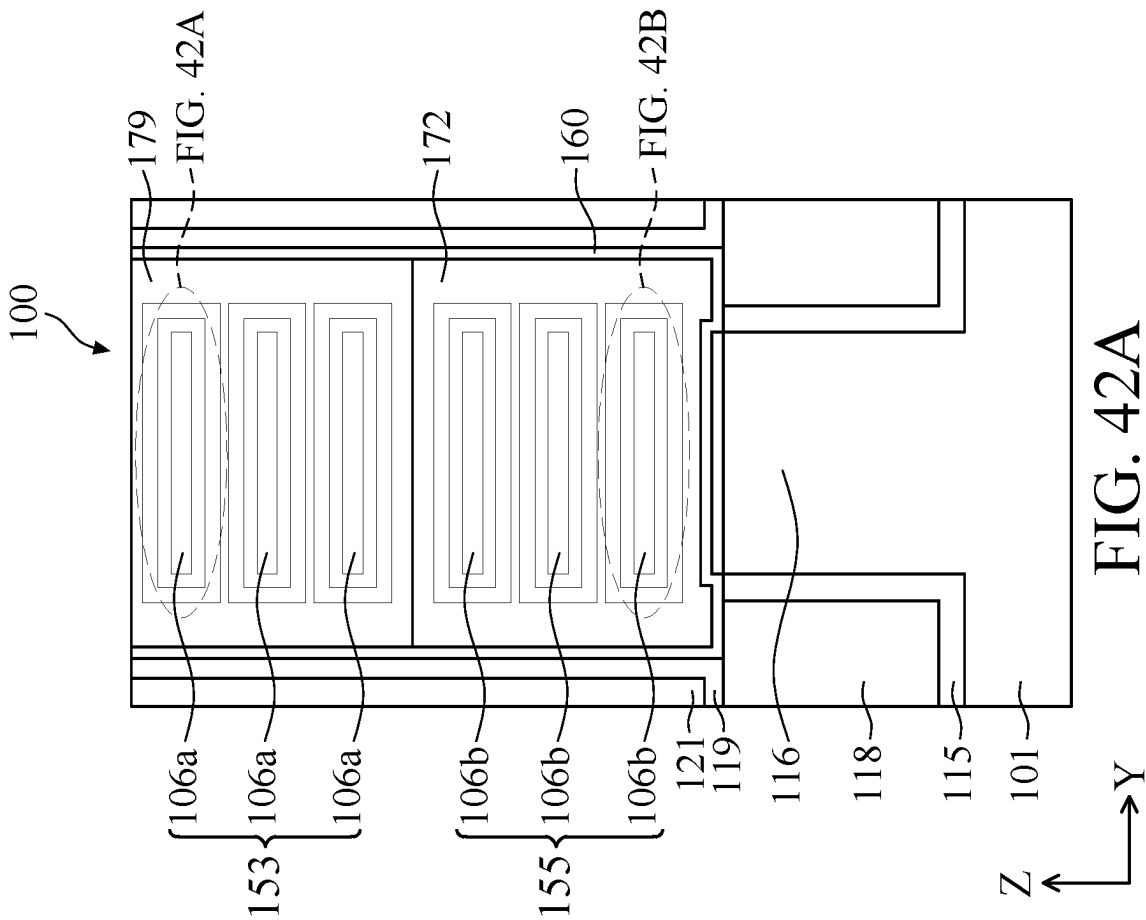


FIG. 42B

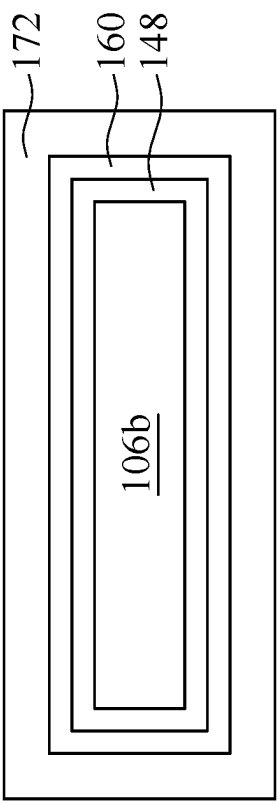
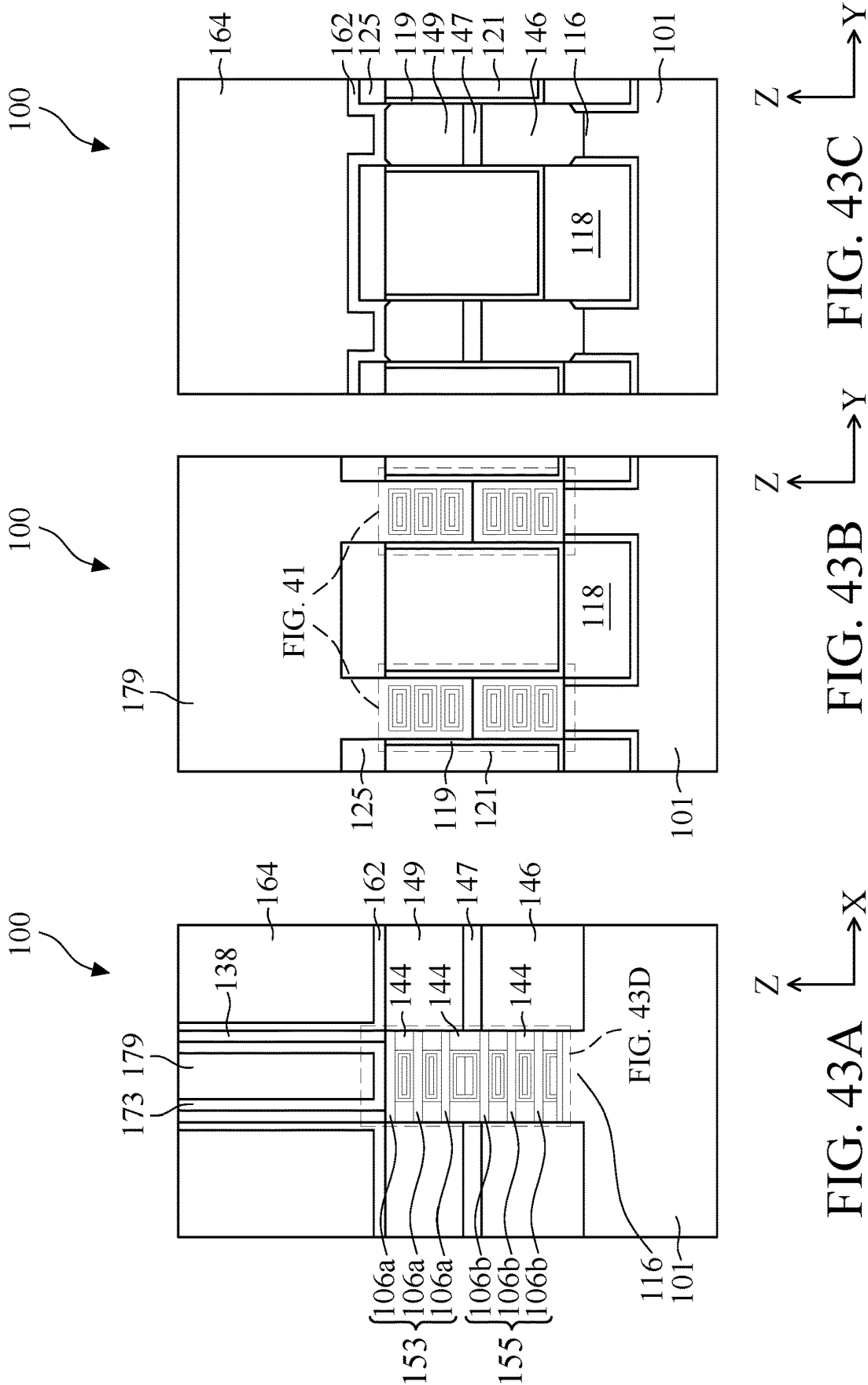
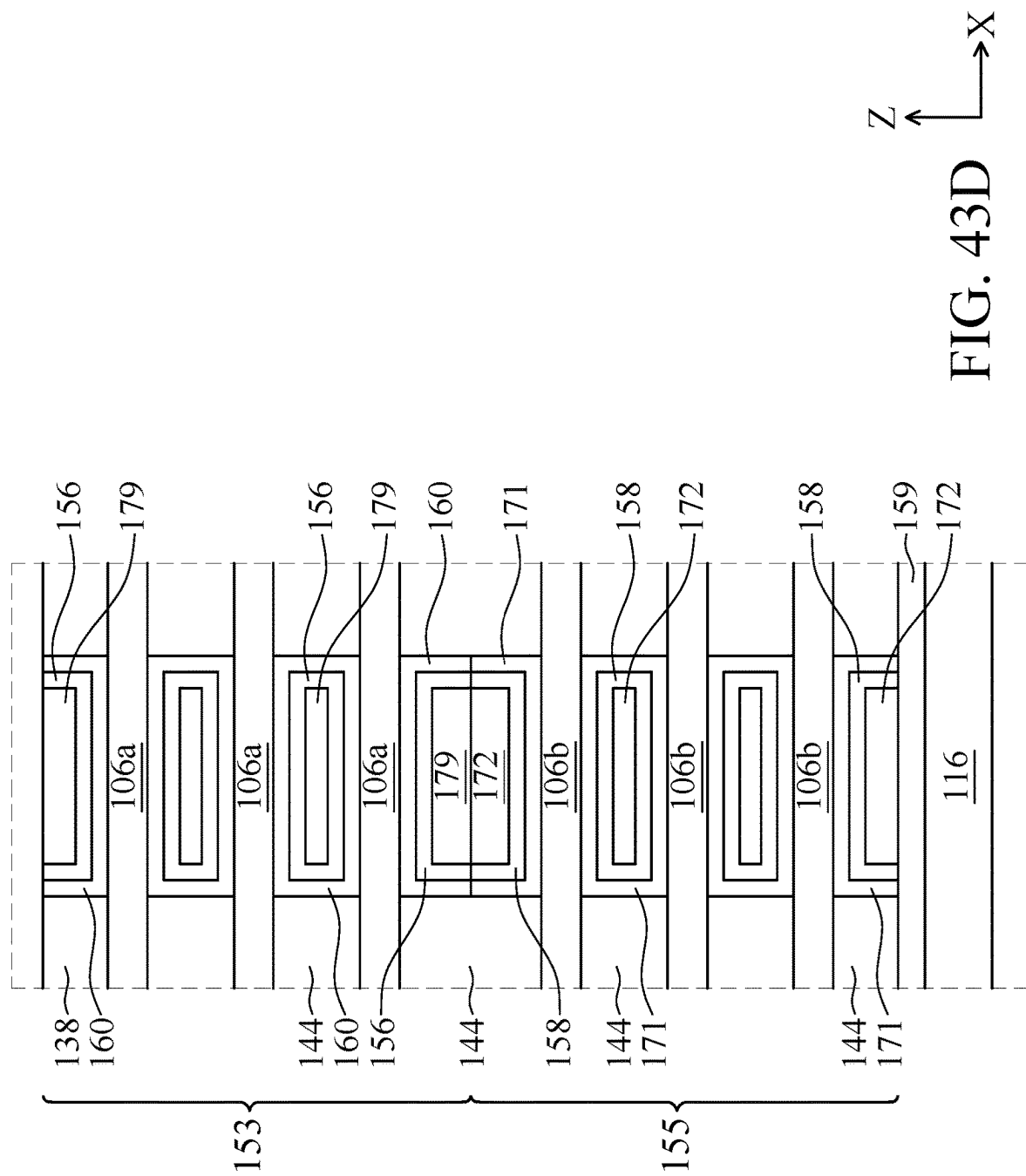


FIG. 42C





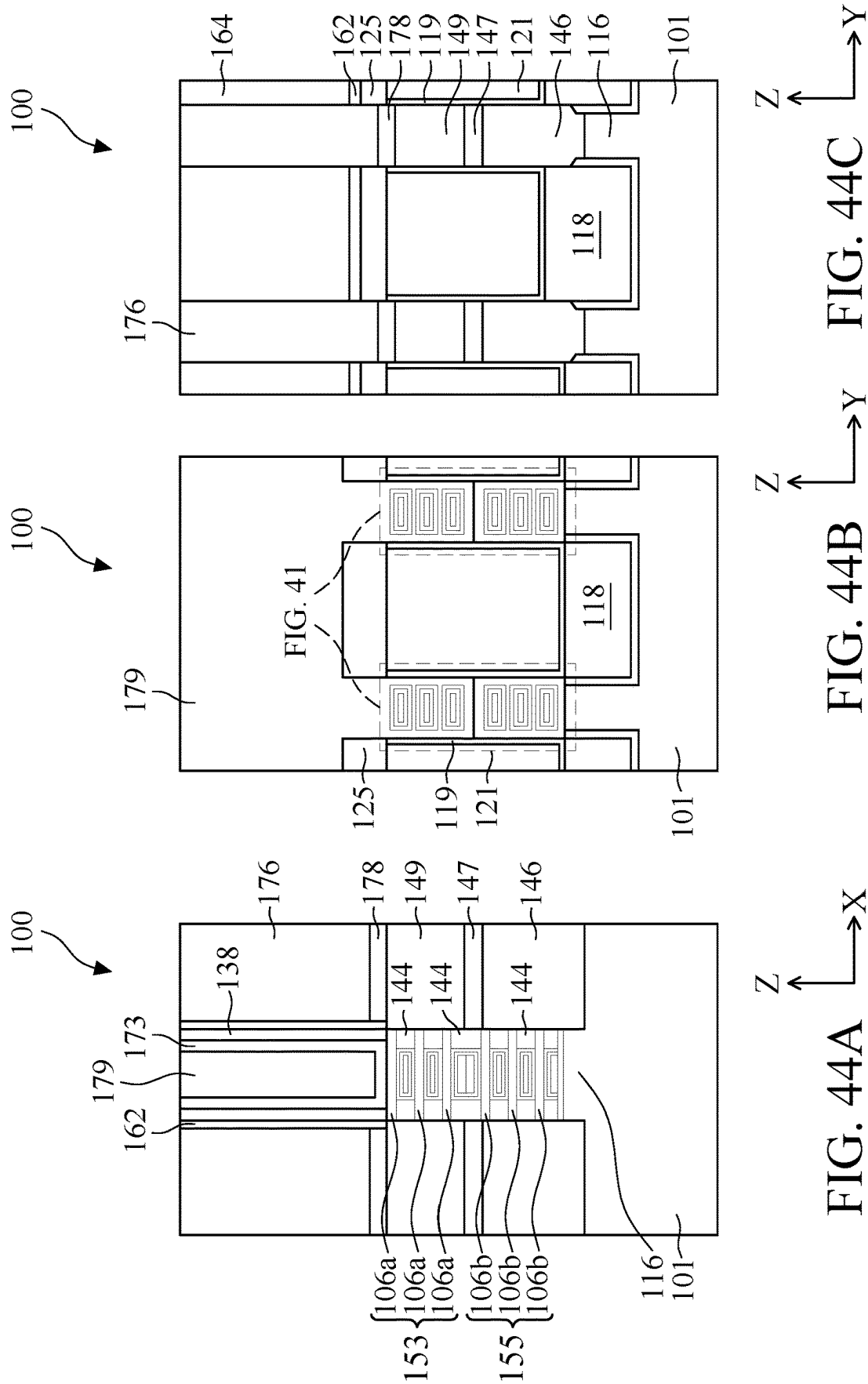


FIG. 44A

FIG. 44B

FIG. 44C

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SEMICONDUCTOR DEVICE HAVING NANOSHEET TRANSISTOR AND METHODS OF FABRICATION THEREOF

CROSS-REFERENCE TO RELATED APPLICATIONS

This application is a continuation application of U.S. patent application Ser. No. 17/883,971 filed Aug. 9, 2022, which is a continuation application of U.S. patent application Ser. No. 17/105,108 filed Nov. 25, 2020, which are incorporated by reference in their entirety.

BACKGROUND

The semiconductor integrated circuit (IC) industry has experienced exponential growth. Technological advances in IC materials and design have produced generations of ICs where each generation has smaller and more complex circuits than the previous generation. In the course of IC evolution, functional density (i.e., the number of interconnected devices per chip area) has generally increased while geometry size (i.e., the smallest component (or line) that can be created using a fabrication process) has decreased. This scaling down process generally provides benefits by increasing production efficiency and lowering associated costs. Such scaling down presents new challenge.

In pursuit of higher device density, higher performance, and lower costs, challenges from both fabrication and design issues have resulted in the development of three-dimensional designs, such as a multi-gate field effect transistor (FET), including a nanosheet FET. In a nanosheet FET, all side surfaces of the channel are surrounded by the gate electrode, which allows for fuller depletion in the channel and results in less short-channel effects and better gate control. As transistor dimensions are continually scaled down, further improvements of the nanosheet FET are needed.

BRIEF DESCRIPTION OF THE DRAWINGS

Aspects of the present disclosure are best understood from the following detailed description when read with the accompanying figures. It is noted that, in accordance with the standard practice in the industry, various features are not drawn to scale. In fact, the dimensions of the various features may be arbitrarily increased or reduced for clarity of discussion.

FIGS. 1-12 are perspective views of various stages of manufacturing a semiconductor device structure in accordance with some embodiments.

FIGS. 13A, 13B, 13C to 22A, 22B, 22C are cross-sectional side views of various stages of manufacturing the semiconductor device structure taken along line A-A, line B-B, and line C-C of FIG. 12, respectively, in accordance with some embodiments.

FIGS. 23-30 are enlarged views of a region of FIG. 22B showing various stages of manufacturing the semiconductor device structure in accordance with some embodiments.

FIGS. 31A-31B illustrate a portion of the first semiconductor layers of FIG. 30 in accordance with alternative embodiments.

FIGS. 32-38 are enlarged views of a region of FIG. 22B showing various stages of manufacturing the semiconductor device structure in accordance with some embodiments.

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FIGS. 39A-39B illustrate a portion of the first semiconductor layers of FIG. 38 in accordance with alternative embodiments.

FIGS. 40-41 are enlarged views of a region of FIG. 22B showing various stages of manufacturing the semiconductor device structure in accordance with some embodiments.

FIG. 42A illustrates an alternative embodiment incorporating various embodiments shown in FIGS. 29A-29B, 31A-31B, 37A-37B, and 39A-39B.

FIGS. 42B-42C are enlarged views of regions of FIG. 42A in accordance with alternative embodiments.

FIGS. 43A-43C are cross-sectional side views of one of various stages of manufacturing the semiconductor device structure taken along line A-A, line B-B, and line C-C of FIG. 12, respectively, in accordance with some embodiments.

FIG. 43D is an enlarged view of a portion of the semiconductor device structure showing first and second nanosheet transistors, in accordance with some embodiments.

FIGS. 44A, 44B, 44C are cross-sectional side views of one of various stages of manufacturing the semiconductor device structure taken along line A-A, line B-B, and line C-C of FIG. 12, respectively, in accordance with some embodiments.

DETAILED DESCRIPTION

The following disclosure provides many different embodiments, or examples, for implementing different features of the provided subject matter. Specific examples of components and arrangements are described below to simplify the present disclosure. These are, of course, merely examples and are not intended to be limiting. For example, the formation of a first feature over or on a second feature in the description that follows may include embodiments in which the first and second features are formed in direct contact, and may also include embodiments in which additional features may be formed between the first and second features, such that the first and second features may not be in direct contact. In addition, the present disclosure may repeat reference numerals and/or letters in the various examples. This repetition is for the purpose of simplicity and clarity and does not in itself dictate a relationship between the various embodiments and/or configurations discussed.

Further, spatially relative terms, such as “beneath,” “below,” “lower,” “above,” “over,” “on,” “top,” “upper” and the like, may be used herein for ease of description to describe one element or feature’s relationship to another element(s) or feature(s) as illustrated in the figures. The spatially relative terms are intended to encompass different orientations of the device in use or operation in addition to the orientation depicted in the figures. The apparatus may be otherwise oriented (rotated 90 degrees or at other orientations) and the spatially relative descriptors used herein may likewise be interpreted accordingly.

Embodiments of the present disclosure provide a semiconductor device structure having complementary field effect transistors (CFETs) with multi-threshold voltage schemes. Each CFET is formed by vertically stacking a first nanosheet FET (e.g., an n-channel FET) on a second nanosheet FET (e.g., a p-channel FET), and each nanosheet FET includes one or more semiconductor layers from a stack of semiconductor layers. The one or more semiconductor layers form nanosheet channels of the p-channel and n-channel nanosheet FET. Each of the one or more semiconductor layers is surrounded by a gate electrode layer. According to

embodiments of the present disclosure, an interfacial layer and/or a high-K dielectric layer between the nanosheet channels and the gate electrode layer are selectively doped or intermixed with one or more p-dipole and n-dipole metals to provide different threshold voltage schemes for the p-channel and the n-channel nanosheet FETs, respectively. As a result, the CFETs in different regions of the semiconductor device structure can be operated at different threshold voltages, resulting in improved device reliability and performance. Various embodiments are discussed in more detail below.

While the embodiments of this disclosure are discussed with respect to nanosheet channel FETs, implementations of some aspects of the present disclosure may be used in other processes and/or in other devices, such as planar FETs, Fin-FETs, Horizontal Gate All Around (HGAA) FETs, Vertical Gate All Around (VGAA) FETs, and other suitable devices. A person having ordinary skill in the art will readily understand other modifications that may be made are contemplated within the scope of this disclosure. In cases where gate all around (GAA) transistor structures are adapted, the GAA transistor structures may be patterned by any suitable method. For example, the structures may be patterned using one or more photolithography processes, including double-patterning or multi-patterning processes. Generally, double-patterning or multi-patterning processes combine photolithography and self-aligned processes, allowing patterns to be created that have, for example, pitches smaller than what is otherwise obtainable using a single, direct photolithography process. For example, in one embodiment, a sacrificial layer is formed over a substrate and patterned using a photolithography process. Spacers are formed alongside the patterned sacrificial layer using a self-aligned process. The sacrificial layer is then removed, and the remaining spacers may then be used to pattern the GAA structure.

FIGS. 1-44C show exemplary processes for manufacturing a semiconductor device structure 100 according to embodiments of the present disclosure. It is understood that additional operations can be provided before, during, and after processes shown by FIGS. 1-44C, and some of the operations described below can be replaced or eliminated, for additional embodiments of the method. The order of the operations/processes is not limiting and may be interchangeable.

FIGS. 1-12 are perspective views of various stages of manufacturing a semiconductor device structure 100 in accordance with some embodiments. As shown in FIG. 1, a semiconductor device structure 100 includes a stack of semiconductor layers 104 formed over a substrate 101. The substrate 101 may be a semiconductor substrate. The substrate 101 may include a single crystalline semiconductor material such as, but not limited to silicon (Si), germanium (Ge), silicon germanium (SiGe), gallium arsenide (GaAs), indium antimonide (InSb), gallium phosphide (GaP), gallium antimonide (GaSb), indium aluminum arsenide (InAlAs), indium gallium arsenide (InGaAs), gallium antimony phosphide (GaSbP), gallium arsenic antimonide (GaAsSb) and indium phosphide (InP). In one embodiment, the substrate 101 is made of silicon. In some embodiments, the substrate 101 is a silicon-on-insulator (SOI) substrate having an insulating layer (not shown) disposed between two silicon layers for enhancement. In one aspect, the insulating layer is an oxygen-containing layer.

The substrate 101 may include one or more buffer layers (not shown) on the surface of the substrate 101. The buffer layers can serve to gradually change the lattice constant

from that of the substrate to that of the source/drain (S/D) regions to be grown on the substrate 101.

The buffer layers may be formed from epitaxially grown single crystalline semiconductor materials such as, but not limited to Si, Ge, germanium tin (GeSn), SiGe, GaAs, InSb, GaP, GaSb, InAlAs, InGaAs, GaSbP, GaAsSb, GaN, GaP, and InP. In one embodiment, the substrate 101 includes SiGe buffer layers epitaxially grown on the silicon substrate 101. The germanium concentration of the SiGe buffer layers may increase from 30 atomic percent germanium for the bottom-most buffer layer to 70 atomic percent germanium for the top-most buffer layer.

The substrate 101 may include various regions that have been doped with impurities (e.g., dopants having p-type or n-type conductivity). Depending on circuit design, the dopants may be, for example boron for an n-type field effect transistors (NFET) and phosphorus for a p-type field effect transistors (PFET).

The stack of semiconductor layers 104 includes semiconductor layers made of different materials to facilitate formation of nanosheet channels in a multi-gate device, such as nanosheet channel FETs. In some embodiments, the stack of semiconductor layers 104 includes first semiconductor layers 106 (e.g., 106a and 106b) and second semiconductor layers 108 (e.g., 108a and 108b). In some embodiments, the stack of semiconductor layers 104 includes alternating first and second semiconductor layers 106, 108. The first semiconductor layers 106 and the second semiconductor layers 108 are made of semiconductor materials having different etch selectivity and/or oxidation rates. For example, the first semiconductor layers 106 may be made of Si and the second semiconductor layers 108 may be made of SiGe. In some examples, the first semiconductor layers 106 may be made of SiGe and the second semiconductor layers 108 may be made of Si. Alternatively, in some embodiments, either of the semiconductor layers 106, 108 may be or include other materials such as Ge, SiC, GeAs, GaP, InP, InAs, InSb, GaSbP, AlInAs, AlGaAs, InGaAs, GaInP, GaInAsP, or any combinations thereof.

The first semiconductor layers 106 or portions thereof may form nanosheet channel(s) of the semiconductor device structure 100 in later fabrication stages. The term nanosheet is used herein to designate any material portion with nanoscale, or even microscale dimensions, and having an elongate shape, regardless of the cross-sectional shape of this portion. Thus, this term designates both circular and substantially circular cross-section elongate material portions, and beam or bar-shaped material portions including, for example, a cylindrical in shape or substantially rectangular cross-section. The nanosheet channel(s) of the semiconductor device structure 100 may be surrounded by a gate electrode. The semiconductor device structure 100 may include a nanosheet transistor. The nanosheet transistors may be referred to as nanowire transistors, gate-all-around (GAA) transistors, multi-bridge channel (MBC) transistors, or any transistors having the gate electrode surrounding the channels. The use of the first semiconductor layers 106 to define a channel or channels of the semiconductor device structure 100 is further discussed below.

The first and second semiconductor layers 106, 108 are formed by any suitable deposition process, such as epitaxy. By way of example, epitaxial growth of the layers of the stack of semiconductor layers 104 may be performed by a molecular beam epitaxy (MBE) process, a metalorganic chemical vapor deposition (MOCVD) process, and/or other suitable epitaxial growth processes.

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In some embodiments, the semiconductor device structure **100** includes a complementary FET (CFET) in which two or more nanosheet FETs are vertically stacked on top of one another. In such a case, the first semiconductor layers **106** can include channels for the two or more nanosheet FETs. In the embodiment shown in FIG. 1, for example, the first semiconductor layers **106a** may define the channels of a first FET, such as a n-type FET (N-FET), and the first semiconductor layers **106b** may define the channels of a second FET, such as a p-type FET (P-FET). The thickness of the first semiconductor layers **106** is chosen based on device performance considerations. In some embodiments, each first semiconductor layer **106** has a thickness ranging from about 3 nanometers (nm) to about 10 nm. The second semiconductor layers **108** (e.g., **108a-108c**) may eventually be removed and serve to define spaces for a gate stack to be formed therein. Likewise, each second semiconductor layer **108** (e.g., **108a** and **108b**) may have a thickness that is equal, less, or greater than the thickness of the first semiconductor layer **106**, depending on device performance considerations. In one aspect, each second semiconductor layer **108** (e.g., **108a** and **108b**) has a thickness that is equal to the thickness of the first semiconductor layer **106** (e.g., **106a** and **106b**).

In some embodiments, the second semiconductor layer **108c** disposed between the first semiconductor layer **106a** in the first FET (e.g., n-channel FET) and the first semiconductor layer **106b** in the second FET (e.g., p-channel FET) has a greater thickness than the thickness of the rest second semiconductor layers **108a** and **108b** to help define boundary of the first FET and the second FET at a later stage. In such cases, the thickness of the second semiconductor layer **108c** may be about 1.5 to about 3 times thicker than the first semiconductor layer **106** (e.g., **106a** and **106b**) or the second semiconductor layer **108** (e.g., **108a** and **108b**).

While six first semiconductor layers **106** and seven second semiconductor layers **108** are alternately arranged as illustrated in FIG. 1, it can be appreciated that any number of first and second semiconductor layers **106**, **108** can be formed in the stack of semiconductor layers **104**, depending on the predetermined number of nanosheet channels needed for each FET of the semiconductor device structure **100**.

FIG. 2 is a perspective view of one of the various stages of manufacturing the semiconductor device structure **100**, in accordance with some embodiments. As shown in FIG. 2, fin structures **112** are formed from the stack of semiconductor layers **104**. Each fin structure **112** has an upper portion including the semiconductor layers **106**, **108**, a well portion **116** formed from the substrate **101**, and a portion of a mask structure **110**. The mask structure **110** is formed over the stack of semiconductor layers **104** prior to forming the fin structures **112**. The mask structure **110** may include an oxygen-containing layer **110a** and a nitrogen-containing layer **110b**. The oxygen-containing layer **110a** may be a pad oxide layer, such as a SiO₂ layer. The nitrogen-containing layer **110b** may be a pad nitride layer, such as Si₃N₄. The mask structure **110** may be formed by any suitable deposition process, such as chemical vapor deposition (CVD) process.

The fin structures **112** may be fabricated using suitable processes including double-patterning or multi-patterning processes. Generally, double-patterning or multi-patterning processes combine photolithography and self-aligned processes, allowing patterns to be created that have, for example, pitches smaller than what is otherwise obtainable using a single, direct photolithography process. For example, in one embodiment, a sacrificial layer is formed over a substrate and patterned using a photolithography

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process. Spacers are formed alongside the patterned sacrificial layer using a self-aligned process. The sacrificial layer is then removed, and the remaining spacers, or mandrels, may then be used to pattern the fin structures **112** by etching the stack of semiconductor layers **104** and the substrate **101**. The etch process can include dry etch, wet etch, reactive ion etch (RIE), and/or other suitable processes. While two fin structures **112** are shown, the number of the fin structures is not limited to two.

In some embodiments, the fin structures **112** may be fabricated using suitable processes including photolithography and etch processes. The photolithography process may include forming a photoresist layer (not shown) over the mask structure **110**, exposing the resist to a pattern, performing post-exposure bake processes, and developing the resist to form a patterned resist. In some embodiments, patterning the resist to form the patterned resist may be performed using an electron beam (e-beam) lithography process. The patterned resist may then be used to protect regions of the substrate **101**, and layers formed thereupon, while an etch process forms trenches **114** in unprotected regions through the mask structure **110**, the stack of semiconductor layers **104**, and into the substrate **101**, thereby leaving the extending fin structures **112**. The trenches **114** may be etched using a dry etch (e.g., RIE), a wet etch, and/or combination thereof.

In FIG. 3, a liner **115** is formed over the substrate **101** and the fin structures **112**. The liner **115** may be formed of a semiconductor material, such as Si. In some embodiments, the liner **115** is made of the same material as the substrate **101**. The liner **115** may be a conformal layer formed by any suitable process, such as an atomic layer deposition (ALD) process. The term "conformal" may be used herein for ease of description upon a layer having substantial same thickness over various regions.

In FIG. 4, after the fin structures **112** are formed, an insulating material **118** is formed on the substrate **101**. The insulating material **118** fills the trenches **114** between neighboring fin structures **112** until the fin structures **112** are embedded in the insulating material **118**. Then, a planarization operation, such as a chemical mechanical polishing (CMP) method and/or an etch-back method, is performed to expose the top of the fin structures **112**. The insulating material **118** may be made of silicon oxide, silicon nitride, silicon oxynitride (SiON), SiOCN, SiCN, fluorine-doped silicate glass (FSG), a low-K dielectric material, or any suitable dielectric material. The insulating material **118** may be formed by any suitable method, such as low-pressure chemical vapor deposition (LPCVD), plasma enhanced CVD (PECVD) or flowable CVD (FCVD).

In FIG. 5, the insulating material **118** is recessed to form an isolation region **120**. The recess of the insulating material **118** exposes portions of the fin structures **112**. The recess of the insulating material **118** reveals the trenches **114** between the neighboring fin structures **112**. The isolation region **120** may be formed using a suitable process, such as a dry etching process, a wet etching process, or a combination thereof. A top surface of the insulating material **118** may be level with or below a surface of the second semiconductor layers **108** (e.g., **108b**) in contact with the well portion **116**.

In FIG. 6, a cladding layer **117** is formed on the exposed surface of the liner **115** (FIG. 5). In some embodiments, the liner **115** may be diffused into the cladding layer **117** during the formation of the cladding layer **117**, resulting in the cladding layer **117** in contact with the stack of semiconductor layers **104**. The cladding layer **117** may be or include a semiconductor material, which allows the cladding layer **117**

to grow on semiconductor materials but not on dielectric materials. For example, the cladding layer 117 may be SiGe and is grown on the Si of the liner 115 but not on the dielectric material of the insulating material 118. In some embodiments, the cladding layer 117 may be formed by first forming a semiconductor layer on the liner 115 and the insulating material 118. An etch process is then performed to remove portions of the semiconductor layer formed on the insulating material 118. In some embodiments, the cladding layer 117 and the second semiconductor layers 108 include the same material having the same etch selectivity. For example, the cladding layer 117 and the second semiconductor layers 108a, 108b, 108c may be or include SiGe. The cladding layer 117 and the second semiconductor layers 108a, 108b, 108c may be removed subsequently to create space for the gate electrode layer.

In FIG. 7, a liner 119 is formed on the cladding layer 117 and the top surface of the insulating material 118. The liner 119 may include a low-k dielectric material (e.g., a material having a k value lower than 7), such as SiO₂, SiN, SiCN, SiOC, or SiOCN. The liner 119 may be formed by a conformal process, such as an ALD process. The liner 119 may have a thickness ranging from about 1 nm to about 6 nm. The liner 119 may function as a shell to protect a flowable oxide material to be formed in the trenches 114 (FIG. 6) during subsequent removal of the cladding layer 117. Thus, if the thickness of the liner 119 is less than about 1 nm, the flowable oxide material may not be sufficiently protected. On the other hand, if the thickness of the liner 119 is greater than about 6 nm, the trenches 114 (FIG. 6) may be filled.

A dielectric material 121 is formed in the trenches 114 (FIG. 6) and on the liner 119, as shown in FIG. 7. The dielectric material 121 may be an oxygen-containing material, such as an oxide, formed by FCVD. The oxygen-containing material may have a K value less than about 7, for example less than about 3. A planarization process, such as a CMP process, may be performed to remove portions of the liner 119 and the dielectric material 121 formed over the fin structures 112. The portion of the cladding layer 117 disposed on the nitrogen-containing layer is exposed after the planarization process.

In FIG. 8, the liner 119 and the dielectric material 121 are recessed to the level of the topmost second semiconductor layer 108a. For example, in some embodiments, after the recess process, the dielectric material 121 may include a top surface 121a that is substantially level with a top surface 108a-1 of the topmost second semiconductor layer 108a. The top surface 108a-1 of the topmost second semiconductor layer 108a may be in contact with the mask structure 110, such as in contact with the oxygen-containing layer 110a. Likewise, the liner 119 may be recessed to the same level as the dielectric material 121. The recess of the liner 119 and the dielectric material 121 may be performed by any suitable process, such as dry etch, wet etch, or a combination thereof. In some embodiments, a first etch process may be performed to recess the dielectric material 121, followed by a second etch process to recess the liner 119. The etch processes may be selective etch processes that do not remove the semiconductor material of the cladding layer 117. As a result of the recess process, trenches 123 are formed between the fin structures 112.

In FIG. 9, a dielectric material 125 is formed in the trenches 123 (FIG. 8) and on the dielectric material 121 and the liner 119. The dielectric material 125 may include SiO, SiN, SiC, SiCN, SiON, SiOCN, AlO, AlN, AlON, ZrO, ZrN, ZrAlO, HfO, or other suitable dielectric material. In some

embodiments, the dielectric material 125 includes a high-k dielectric material (e.g., a material having a k value greater than 7). The dielectric material 125 may be formed by any suitable process, such as a CVD, PECVD, FCVD, or ALD process. A planarization process, such as a CMP process, is performed until the nitrogen-containing layer 110b of the mask structure 110 is exposed. The planarization process removes portions of the dielectric material 125 and the cladding layer 117 disposed over the mask structure 110. The liner 119, the dielectric material 121, and the dielectric material 125 together may be referred to as a dielectric feature 127. The dielectric feature 127 serves as a dielectric fin that separates adjacent source/drain (S/D) epitaxial features and adjacent gate electrode layers.

In FIG. 10, the cladding layers 117 are recessed, and the mask structures 110 are removed. The recess of the cladding layers 117 may be performed by any suitable process, such as dry etch, wet etch, or a combination thereof. The recess process may be controlled so that the remaining cladding layers 117 are substantially at the same level as the top surface 108a-1 of the topmost second semiconductor layer 108a in the stack of semiconductor layers 102. The etch process may be a selective etch process that does not remove the dielectric material 125. The removal of the mask structures 110 may be performed by any suitable process, such as dry etch, wet etch, or a combination thereof. The removal of the mask structure 110 exposes the top surfaces 180a-1 of the topmost second semiconductor layers 108a in the stacks of semiconductor layers 104.

In FIG. 11, one or more sacrificial gate structures 130 (only one is shown) are formed over the semiconductor device structure 100. The sacrificial gate structures 130 are formed over a portion of the fin structures 112. Each sacrificial gate structure 130 may include a sacrificial gate dielectric layer 132, a sacrificial gate electrode layer 134, and a mask layer 136. The sacrificial gate dielectric layer 132, the sacrificial gate electrode layer 134, and the mask layer 136 may be formed by sequentially depositing blanket layers of the sacrificial gate dielectric layer 132, the sacrificial gate electrode layer 134, and the mask layer 136, followed by pattern and etch processes. For example, the pattern process includes a lithography process (e.g., photolithography or e-beam lithography) which may further include photoresist coating (e.g., spin-on coating), soft baking, mask aligning, exposure, post-exposure baking, photoresist developing, rinsing, drying (e.g., spin-drying and/or hard baking), other suitable lithography techniques, and/or combinations thereof. In some embodiments, the etch process may include dry etch (e.g., RIE), wet etch, other etch methods, and/or combinations thereof.

By patterning the sacrificial gate structure 130, the stacks of semiconductor layers 104 of the fin structures 112 are partially exposed on opposite sides of the sacrificial gate structure 130. The portions of the fin structures 112 that are covered by the sacrificial gate electrode layer 134 of the sacrificial gate structure 130 serve as channel regions for the semiconductor device structure 100. The fin structures 112 that are partially exposed on opposite sides of the sacrificial gate structure 130 define source/drain (S/D) regions for the semiconductor device structure 100. While one sacrificial gate structure 130 is shown, two or more sacrificial gate structures 130 may be arranged along the X direction in some embodiments.

Next, gate spacers 138 are formed on sidewalls of the sacrificial gate structures 130. The gate spacers 138 may be formed by first depositing a conformal layer that is subsequently etched back to form sidewall gate spacers 138. For

example, a spacer material layer can be disposed conformally on the exposed surfaces of the semiconductor device structure **100**. The conformal spacer material layer may be formed by an ALD process. Subsequently, anisotropic etch is performed on the spacer material layer using, for example, RIE. During the anisotropic etch process, most of the spacer material layer is removed from horizontal surfaces, such as the tops of the fin structures **112**, the cladding layer **117**, the dielectric material **125**, leaving the gate spacers **138** on the vertical surfaces, such as the sidewalls of sacrificial gate structures **130**. The gate spacer **138** may be made of a dielectric material such as silicon oxide, silicon nitride, silicon carbide, silicon oxynitride, SiCN, silicon oxycarbide, SiOCN, and/or combinations thereof.

In FIG. **12**, exposed portions of the fin structures **112**, exposed portions of the cladding layers **117**, and exposed portions of the dielectric material **125** not covered by the sacrificial gate structures **130** and the gate spacers **138** are selectively recessed by using one or more suitable etch processes, such as dry etch, wet etch, or a combination thereof. In some embodiments, exposed portions of the stacks of semiconductor layers **104** of the fin structures **112** are removed, exposing portions of the well portions **116**. As shown in FIG. **12**, the exposed portions of the fin structures **112** are recessed to a level at or below the top surface **118a** of the insulating material **118**. The recess processes may include an etch process that recesses the exposed portions of the fin structures **112** and the exposed portions of the cladding layers **117**.

FIGS. **13A**, **13B**, and **13C** are cross-sectional side views of the semiconductor device structure **100** taken along line A-A, line B-B, and line C-C of FIG. **12**, respectively, in accordance with some embodiments.

FIGS. **14A**, **14B**, and **14C** are cross-sectional side views of one of various stages of manufacturing the semiconductor device structure **100** taken along line A-A, line B-B, and line C-C of FIG. **12**, respectively, in accordance with some embodiments. As shown in FIG. **14A**, edge portions of each second semiconductor layer **108** (e.g., **108a**, **108b**, **108c**) of the stack of semiconductor layers **104** are removed horizontally along the X direction. The removal of the edge portions of the second semiconductor layers **108** forms cavities. In some embodiments, the portions of the second semiconductor layers **108** are removed by a selective wet etching process. In cases where the second semiconductor layers **108** are made of SiGe and the first semiconductor layers **106** are made of silicon, the second semiconductor layer **108** can be selectively etched using a wet etchant such as, but not limited to, ammonium hydroxide (NH₄OH), tetramethylammonium hydroxide (TMAH), ethylenediamine pyrocatechol (EDP), or potassium hydroxide (KOH) solutions.

After removing edge portions of each second semiconductor layers **108**, a dielectric layer (or so-called inner spacer) is deposited in the cavities to form dielectric spacers **144**. The dielectric spacers **144** may be made of a low-K dielectric material, such as SiON, SiCN, SiOC, SiOCN, or SiN. The dielectric spacers **144** may be formed by first forming a conformal dielectric layer using a conformal deposition process, such as ALD, followed by an anisotropic etching to remove portions of the conformal dielectric layer other than the dielectric spacers **144**. The dielectric spacers **144** are protected by the first semiconductor layers **106** during the anisotropic etching process. The remaining second semiconductor layers **108** (e.g., **108a**, **108b**, **108c**) are capped between the dielectric spacers **144** along the X direction.

FIGS. **15A**, **15B**, and **15C** are cross-sectional side views of one of various stages of manufacturing the semiconductor device structure **100** taken along line A-A, line B-B, and line C-C of FIG. **12**, respectively, in accordance with some embodiments. As shown in FIGS. **15A** and **15C**, epitaxial S/D features **146** are formed on the well portion **116** of the fin structures **112**. The epitaxial S/D features **146** may include or be made of one or more layers of Si, SiP, SiC and SiCP for n-channel FETs or Si, SiGe, Ge for p-channel FETs. For p-channel FETs, p-type dopants, such as boron (B), may also be included in the epitaxial S/D features **146**. In some embodiments, the epitaxial S/D feature **146** uses one or more layers of Si, SiGe, and Ge for a p-channel FET. The epitaxial S/D features **146** may be formed by an epitaxial growth method using CVD, ALD or MBE. The epitaxial S/D features **146** may grow both vertically and horizontally to form facets, which may correspond to crystalline planes of the material used for the substrate **101**. The epitaxial S/D features **146** are in contact with the first semiconductor layers **106a**, **106b** and dielectric spacers **144**, as shown in FIG. **15A**. The epitaxial S/D epitaxial features **146** may be the S/D regions. In this disclosure, a source region and a drain region are interchangeably used, and the structures thereof are substantially the same.

In FIGS. **16A** and **16C**, the epitaxial S/D features **146** are recessed by removing a portion of each epitaxial S/D feature **146**. The recess of the epitaxial S/D features **146** may be performed by any suitable process, such as dry etch or wet etch that selectively removes a portion of each epitaxial S/D feature **146** but not the gate spacer **138**, the dielectric material **125**, and the liner **119**. After the removal process, the epitaxial S/D features **146** are in contact with the first semiconductor layers **106b** and the dielectric spacers **144**, as shown in FIG. **16A**. In some embodiments, the semiconductor device structure **100** includes a nanosheet p-channel FET having a source epitaxial feature/terminal **146** and a drain epitaxial feature/terminal **146** both in contact with one or more first semiconductor layers **106b**, or one or more channels.

In FIGS. **17A** and **17C**, a dielectric material **147** is formed over the epitaxial S/D features **146**. The dielectric material **147** may include the same material as the insulating material **118** and may be formed by the same method as the insulating material **118**. In some embodiments, the dielectric material **147** includes an oxide that is formed by FCVD. The dielectric material **147** may be recessed to a level below the level of the first semiconductor layers **106a**, as shown in FIG. **17A**. The recess of the dielectric material **147** may be performed by any suitable process, such as dry etch or wet etch that selectively removes a portion of the dielectric material **147** but not the gate spacer **138**, the first semiconductor layer **106a**, and the dielectric spacers **144**.

In FIGS. **18A** and **18C**, epitaxial S/D features **149** are formed on the dielectric material **147**. The epitaxial S/D feature **149** may include one or more layers of Si, SiP, SiC and SiCP for an n-channel FET or Si, SiGe, Ge for a p-channel FET. In some embodiments, the epitaxial S/D feature **149** uses one or more layers of Si, SiP, SiC and SiCP for a n-channel FET. The epitaxial S/D features **149** may be formed from the first semiconductor layers **106a** (FIG. **17A**). The epitaxial S/D features **149** may grow both vertically and horizontally to form facets, which may correspond to crystalline planes of the material used for the first semiconductor layers **106a**. The epitaxial S/D features **149** may be formed by an epitaxial growth method using CVD, ALD or MBE. Likewise, the epitaxial S/D features **149** may be the S/D regions.

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As shown in FIGS. 18A and 18C, the source regions of the n-channel FETs and p-channel FETs may be vertically stacked and aligned, the drain regions of the n-channel FETs and the p-channel FETs may be vertically stacked and aligned, and the source of the n-channel FET and the source of the p-channel FET may be separated by the dielectric material 147. Vertical stacking of the n-channel FETs and p-channel FETs can increase the density of the FETs while reducing the cell active area footprint for the semiconductor devices, such as SRAMs.

In FIGS. 19A and 19C, a contact etch stop layer (CESL) 162 is conformally formed on the exposed surfaces of the semiconductor device structure 100. The CESL 162 covers the epitaxial S/D features 146, the gate spacers 138, the dielectric material 125, and the exposed surface of the stack of semiconductor layers 104. The CESL 162 may include an oxygen-containing material or a nitrogen-containing material, such as silicon nitride, silicon carbon nitride, silicon oxynitride, carbon nitride, silicon oxide, silicon carbon oxide, or the like, or a combination thereof, and may be formed by CVD, PECVD, ALD, or any suitable deposition technique. Next, an interlayer dielectric (ILD) layer 164 is formed on the CESL 162 over the semiconductor device structure 100. The materials for the ILD layer 164 may include tetraethylorthosilicate (TEOS) oxide, un-doped silicate glass, or doped silicon oxide such as borophosphosilicate glass (BPSG), fused silica glass (FSG), phosphosilicate glass (PSG), boron doped silicon glass (BSG), and/or other suitable dielectric materials comprising Si, O, C, and/or H. The ILD layer 164 may be deposited by a PECVD process or other suitable deposition technique. In some embodiments, after formation of the ILD layer 164, the semiconductor device structure 100 may be subject to a thermal process to anneal the ILD layer 164.

In FIGS. 20A, 20B, and 20C, after the ILD layer 164 is formed, a planarization operation, such as CMP, is performed on the semiconductor device structure 100 to remove portions of the ILD layer 164, the CESL 162 and the mask layer 136 until the sacrificial gate electrode layer 134 is exposed.

In FIGS. 21A and 21B, the sacrificial gate structure 130 is removed. The removal of the sacrificial gate structure 130 forms a trench 166 in the regions where the sacrificial gate electrode layer 134 and the sacrificial gate dielectric layer 132 were removed. The trench 166 exposes portions of the cladding layer 117 and the top of the second semiconductor layer 108a. The ILD layer 164 protects the epitaxial S/D features 146 during the removal of the sacrificial gate structure 130. The sacrificial gate structure 130 can be removed using plasma dry etching and/or wet etching. The sacrificial gate electrode layer 134 may be first removed by any suitable process, such as dry etch, wet etch, or a combination thereof, followed by the removal of the sacrificial gate dielectric layer 132, which may also be performed by any suitable process, such as dry etch, wet etch, or a combination thereof. In some embodiments, a wet etchant such as a tetramethylammonium hydroxide (TMAH) solution can be used to selectively remove the sacrificial gate electrode layer 134 but not the gate spacers 138, the dielectric material 125, and the CESL 162. In some embodiments, the gate spacers 138 may be recessed by the etchant used to remove the sacrificial gate electrode layer 134 and/or the sacrificial gate dielectric layer 132.

In FIGS. 22A and 22B, the cladding layers 117 and the second semiconductor layers 108 (e.g., 108a, 108b, 108c) are removed. The removal of the cladding layers 117 and the second semiconductor layers 108 exposes the dielectric

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spacers 144 and the first semiconductor layers 106 (e.g., 106a, 106b). The removal process may be any suitable etch processes, such as dry etch, wet etch, or a combination thereof. The etch process may be a selective etch process that removes the cladding layers 117 (FIG. 21B) and the second semiconductor layers 108 but not the gate spacers 138, the CESL 162, the dielectric material 125, and the first semiconductor layers 106. As a result, openings 151 are formed around the first semiconductor layers 106, as shown in FIG. 22B. That is, the portion of the first semiconductor layers 106 not covered by the dielectric spacers 144 is exposed to the openings 151.

FIGS. 23-28, 29A-29B, 30, 32-36, 37A-37B, 38, 40-41, and 42A are enlarged views of a region 220 of FIG. 22B showing various stages of manufacturing the semiconductor device structure 100 in accordance with some embodiments. For the sake of clarity, the dielectric material 125 is omitted in FIGS. 23-28, 29A-29B, 30, 32-36, 37A-37B, 38, 40-41, and 42A. In the embodiment shown in FIG. 23, each first semiconductor layer 106b may be a nanosheet channel of a first nanosheet transistor 155, such as a p-channel FET, and each first semiconductor layer 106a may be a nanosheet channel of a second nanosheet transistor 153, such as a n-channel FET. The second nanosheet transistors (e.g., 106a) are disposed over and aligned with the first nanosheet transistors (e.g., 106b) along the Z-direction. Depending on the application, nanosheet transistors having higher thermal budget (e.g., p-channel FETs) may be arranged below nanosheet transistors having lower thermal budget (e.g., n-channel FETs).

The top surface 133 of the topmost first semiconductor layer 106a is below the top surface 135 of the liner 119 by a height "H" due to the removal of the topmost second semiconductor layer 108a (FIG. 21B). Each first semiconductor layer 106a, 106b has a first height "H1" of about 2 nm to about 15 nm, for example about 3 nm to about 10 nm. In some embodiments, one or more first semiconductor layers 106a may have a first height "H1" different from that of one or more first semiconductor layers 106a, and one or more first semiconductor layers 106b may have the first height "H1" different from that of one or more first semiconductor layers 106b. In some embodiments, one or more first semiconductor layers 106a, 106b may have a first height while one or more first semiconductor layers 106a, 106b may have a second height different from the first height. The first height "H1" as used in this disclosure may also refer to nanosheet channel height or a thickness of the first semiconductor layers 106a, 106b.

The opening 151 between the first semiconductor layer 106b of the first nanosheet transistor 155 and the first semiconductor layer 106a of the second nanosheet transistor 153 has a second height "H2" that is greater than the first height "H1". The ratio of the second height "H2" to the first height "H1" may be in a range of about 1.5 to 3. The second height "H2" provides additional space to compensate for possible over-etch or under-etch that may occur during recess of the hardmask at a later stage (e.g., FIG. 28) and thus can help define boundary between the first nanosheet transistor 155 and the second nanosheet transistor 153. Therefore, if the ratio of the second height "H2" to the first height "H1" is less than about 1.5, the hardmask might be recessed to a level that is within the first nanosheet transistor 155 or the second nanosheet transistor 153 due to under-etching or over-etching. On the other hand, if the ratio of the second height "H2" to the first height "H1" is greater than about 3, the manufacturing cost is increased without significant advantage.

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In FIG. 24, the semiconductor device structure 100 is subjected to a pre-clean process to remove residues or unwanted films from exposed surfaces of the first semiconductor layers 106b of the first nanosheet transistor 155 and the first semiconductor layer 106a of the second nanosheet transistor 153. The pre-clean process may be any suitable wet cleaning process such as an APM process, which includes at least water (H₂O), ammonium hydroxide (NH₄OH), and hydrogen peroxide (H₂O₂), a HPM process, which includes at least H₂O, H₂O₂, and hydrogen chloride (HCl), a SPM process (also known as piranha clean), which includes at least H₂O₂ and sulfuric acid (H₂SO₄), or any combination thereof.

Next, an interfacial layer (IL) 148 is formed to surround the exposed surfaces of the first semiconductor layers 106a, 106b, as shown in FIG. 24. In some embodiments, the IL 148 may also form on the well portion 116 of the substrate 101. The IL 148 may include or be made of an oxygen-containing material or a silicon-containing material, such as silicon oxide, silicon oxynitride, oxynitride, hafnium silicate, etc. The IL 148 may be formed by CVD, ALD or any suitable conformal deposition technique. In one embodiment, the IL 148 is formed using ALD. The thickness of the IL 148 is chosen based on device performance considerations. In some embodiments, the IL 148 has a thickness ranging from about 0.5 nm to about 2 nm.

In FIG. 25, a first dipole layer 150 is formed on the IL 148 and the exposed surfaces of the semiconductor device structure 100. In some embodiments, the first dipole layer 150 forms on the liner 119, the insulating material 118, the liner 115, and the IL 148 that is in contact with the well portion 116 of the substrate 101. Depending on the conductivity type of the first nanosheet transistor 155, the first dipole layer 150 may be configured to include positive polarity or negative polarity. The first dipole layer 150 serves to enhance or modify threshold voltage for the first nanosheet transistor 155. In cases where the first nanosheet transistor 155 is a p-channel FET, the first dipole layer 150 can be a positive polarity dipole (p-dipole) layer formed from a material inherently including a positive polarity. Such materials may include, but are not limited to, aluminum oxide (Al₂O₃), titanium dioxide (TiO₂), germanium oxide (GeO₂), or the like. The first dipole layer 150 can be formed by ALD, CVD, or any suitable conformal deposition technique in order to ensure uniform thickness of the first dipole layer 150. The thickness of the first dipole layer 150 is chosen based on device performance considerations. In some embodiments, the first dipole layer 150 has a thickness ranging from about 0.05 nm to about 2 nm.

In some alternative embodiments, which can be combined with any embodiment(s) of this disclosure, the first dipole layer 150 may be formed to surround the exposed surfaces of the first semiconductor layers 106a, 106b, followed by formation of the IL 148 on the first dipole layer 150.

In FIG. 26, a mask layer 152 is formed on the first dipole layer 150 and in each opening 151 (FIG. 25). The mask layer 152 fills the opening 151 to a level so that at least the first dipole layer 150 within the second nanosheet transistor 153 is submerged in the mask layer 152. The mask layer 152 protects the first dipole layer 150 within the first nanosheet transistor 155, such as a p-channel FET, during the subsequent mask layer etch-back process (FIG. 27). The mask layer 152 may be any suitable masking material, such as a photoresist layer, a BARC (bottom anti-reflective coating) layer, a SOG (spin-on-glass) layer, or a SOC (spin-on-carbon) layer, and may be deposited by spin coating or any suitable deposition technique.

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In FIG. 27, the mask layer 152 is recessed to expose the first dipole layer 150 within the second nanosheet transistor 153. The recess of the mask layer 152 may be performed using a suitable etch-back process, such as dry etch, wet etch, or reactive ion etch (RIE). The etch-back process uses an etchant that selectively removes the mask layer 152 but not the first dipole layer 150. The recessed mask layer 152 has a top surface 157 that may be at a level between the topmost first semiconductor layer 106b of the first nanosheet transistor 155 and the bottom-most first semiconductor layer 106a of the second nanosheet transistor 153. The top surface 157 defines the boundary between the first nanosheet transistor 155 and the second nanosheet transistor 153.

In FIG. 28, the first dipole layer 150 that is not covered by the mask layer 152 is removed. The removal process removes the first dipole layer 150 to expose the IL 148 (formed on the first semiconductor layers 106a) within the second nanosheet transistor 153. The removal process also exposes the top surface 157 of the mask layer 152 and a portion of the first dipole layer 150 surrounding the mask layer 152. The removal process may be any suitable processes, such as dry etch, wet etch, or a combination thereof. In some embodiments, the first dipole layer 150 is removed by a wet etch using NH₄OH, dilute hydrofluoric acid (HF) (e.g., 2% HF), buffered HF (e.g., 30-50% ammonium fluoride (NH₄F) and 5-10% HF), or any combination thereof.

Next, after removal of the first dipole layer 150 within the second nanosheet transistor 153, a second dipole layer 154 is formed on the exposed IL 148, the top surface 157 of the mask layer 152, and the exposed liner 119. The second dipole layer 154 surrounds the first semiconductor layers 106a within the second nanosheet transistor 153. Likewise, the second dipole layer 154 may be configured to include positive polarity or negative polarity, depending on the conductivity type of the second nanosheet transistor 153. The second dipole layer 154 serves to enhance or modify threshold voltage for the second nanosheet transistor 153. In cases where the second nanosheet transistor 153 is a n-channel FET, the second dipole layer 154 can be a negative polarity dipole (n-dipole) layer formed from a material inherently including a negative polarity. Such materials may include, but are not limited to, lanthanum oxide (La₂O₃), magnesium oxide (MgO), yttrium oxide (Y₂O₃), gadolinium oxide (Gd₂O₃), or the like. The second dipole layer 154 can be formed by ALD, CVD, or any suitable conformal deposition technique in order to ensure uniform thickness of the second dipole layer 154. The thickness of the first dipole layer 150 is chosen based on device performance considerations. In some embodiments, the first dipole layer 150 has a thickness ranging from about 0.05 nm to about 2 nm.

In FIG. 29A, the mask layer 152 is removed. The removal process may use ash process or dry/wet etch process identical or similar to the process used for removal of the mask layer 152 as discussed in FIG. 27. The removal process selectively removes the mask layer 152 but not the first dipole layer 150 and the second dipole layer 154. The second dipole layer 154 on the top surface 157 of the mask layer 152 (FIG. 28) is removed along with the mask layer 152. The openings 151 within the first nanosheet transistor 155 are revealed upon removal of the mask layer 152, exposing the first semiconductor layers 106b being surrounded by the first dipole layer 150.

FIG. 29A illustrates one embodiment where the first semiconductor layers 106b within the first nanosheet transistor 155 are surrounded by the first dipole layer 150, and the first semiconductor layers 106a within the second nanosheet transistor 153 are surrounded by the second

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dipole layer **154**. In some embodiments, which can be combined with one or more embodiments (e.g., FIGS. **37A**, **37B**, **39A**, **39B**) discussed in this disclosure, one or more second nanosheet transistors **153** in the semiconductor device structure **100** are further proceeded to have the second dipole layer **154** removed from the first semiconductor layers **106a** and from the liner **119**, such as an alternative embodiment shown in FIG. **29B**. The removal of the second dipole layer **154** can be done after removing the mask layer **152** by a suitable selective etch process that removes the second dipole layer **154** but not the first dipole layer **150**. By configuring one or more second nanosheet transistors **153** to include the second dipole layer **154** over the first semiconductor layers **106a** and one or more second nanosheet transistors **153** to not include the second dipole layer **154** over the first semiconductor layers **106a**, the semiconductor device structure **100** can provide CFETs having multi-threshold voltage schemes according to the application needs.

In FIG. **30**, the semiconductor device structure **100** is subjected to a thermal treatment **131**. The second dipole layer **154** and the first dipole layer **150** are diffused into and mixed with the IL **148** as a result of the thermal treatment **131**. In some embodiments, the interface between the IL **148** and the first dipole layer **150** and the interface between the IL **148** and the second dipole layer **154** may not be observed after the thermal treatment. In some embodiments, the diffusion of the second dipole layer **154** and the IL **148** forms a first intermixed layer **156** surrounding and in contact with the first semiconductor layers **106a**.

Likewise, the diffusion/intermixing of the first dipole layer **150** and the IL **148** forms a second intermixed layer **158** surrounding and in contact with the first semiconductor layers **106b**. The first dipole layer **150** also diffuses into the IL **148** in contact with the well portion **116** of the substrate **101** and forms a bottom intermixed layer **159**. The first intermixed layer **156**, the second intermixed layers **158**, and the bottom intermixed layer **159** can be considered as modified IL layers. The term “first/second intermixed layer” or “modified IL layer” as used in this disclosure denotes a reaction product of the IL **148** and the first/second dipole layer **150/154**, which can be a compound, a composition or a mixture, depending on the thermal treatment used. In some embodiments, the first/second intermixed layer **156/158** or the modified IL layer can be an IL doped with materials from the second dipole layer **154** or the first dipole layer **150**.

In some alternative embodiments, which can be combined with any embodiment(s) discussed in this disclosure, the thermal treatment may cause portions of the second dipole layer **154** to diffuse into the IL **148** and form an intermixed layer **156a** sandwiched between the IL **148** and the second dipole layer **154**, as an alternative embodiment shown in FIG. **31A**.

In some alternative embodiments, which can be combined with any embodiment(s) discussed in this disclosure, the thermal treatment may cause portions of the first dipole layer **150** to diffuse into the IL **148** and form an intermixed layer **158a** sandwiched between the IL **148** and the first dipole layer **150**, as an alternative embodiment shown in FIG. **31B**.

The thermal treatment **131** can be any type of anneal, such as rapid thermal anneal, a spike anneal, a soak anneal, a laser anneal, a furnace anneal, etc. The thermal treatment may be performed for about 0.5 seconds to about 60 seconds, such as about 10 seconds to about 30 seconds, and at a temperature range of about 450° C. to about 1200° C. The thermal treatment may be performed in an atmosphere of gas, such as a nitrogen-containing gas, an oxygen-containing gas, a

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hydrogen-containing gas, an argon-containing gas, a helium-containing gas, or any combinations thereof. Exemplary gas may include, but are not limited to, N₂, NH₃, O₂, N₂O, Ar, He, H, etc.

After the thermal treatment, the semiconductor device structure **100** is subjected to a wet treatment. The wet treatment removes the first and second dipole layers **150**, **154** formed on the liner **119**, the liner **115** and the insulating material **118**, as shown in FIG. **30**. The wet treatment may be a wet etch such as a SPM process, an APM process, or a HPM process discussed above.

In FIG. **32**, a high-K (HK) dielectric layer **160** is formed on the exposed surfaces of the semiconductor device structure **100**. In some embodiments, the HK dielectric layer **160** is formed to wrap around the first intermixed layer **156** (or the second dipole layer **154**) and the second intermixed layer **158** (or the first dipole layer **150**). The HK dielectric layer **160** also forms on the liner **119**, the insulating material **118**, the liner **115**, and the bottom intermixed layer **159**. The HK dielectric layer **160** may include or made of hafnium oxide (HfO₂), hafnium silicate (HfSiO), hafnium silicon oxynitride (HfSiON), hafnium aluminum oxide (HfAlO), hafnium lanthanum oxide (HfLaO), hafnium zirconium oxide (HfZrO), hafnium tantalum oxide (HfTaO), hafnium titanium oxide (HfTiO), lanthanum oxide (LaO), aluminum oxide (AlO), aluminum silicon oxide (AlSiO), zirconium oxide (ZrO), titanium oxide (TiO), tantalum oxide (Ta₂O₅), yttrium oxide (Y₂O₃), silicon oxynitride (SiON), or other suitable high-k materials. The HK dielectric layer **160** may be a conformal layer formed by a conformal process, such as an ALD process or a CVD process. The HK dielectric layer **160** may have a thickness ranging from about 0.5 nm to about 3 nm.

In FIG. **33**, a third dipole layer **163** is formed on the exposed surfaces of the semiconductor device structure **100**. In some embodiments, the third dipole layer **163** is formed on the HK dielectric layer **160**. Depending on the conductivity type of the first nanosheet transistor **155**, the third dipole layer **163** may be configured to include positive polarity or negative polarity. The third dipole layer **163** serves to enhance or modify threshold voltage for the first nanosheet transistor **155**. In cases where the first nanosheet transistor **155** is a p-channel FET, the third dipole layer **163** can be a positive polarity dipole (p-dipole) layer formed using ALD, CVD, or any suitable conformal deposition technique. In some embodiments, the third dipole layer **163** can be formed of the same or different material than the first dipole layer **150**. The third dipole layer **163** may have the same or different thickness than the first dipole layer **150**, depending on the application.

In FIG. **34**, a mask layer **165** is formed on the third dipole layer **163** and in each opening **151** to cover the entire first semiconductor layers **106b** and the third dipole layer **163** formed therearound within the first nanosheet transistor **155**. The mask layer **165** may be formed by first forming a mask layer filling the openings **151**, followed by an etch back process to recess the mask layer **165** to a level between the topmost first semiconductor layer **106b** of the first nanosheet transistor **155** and the bottom-most first semiconductor layer **106a** of the second nanosheet transistor **153**. The mask layer **165** may use the same material as the mask layer **152** and formed by spin coating or any suitable deposition technique. The recess of the mask layer **165** exposes the third dipole layer **163** within the second nanosheet transistor **153**, as shown in FIG. **34**.

In FIG. **35**, the third dipole layer **163** that is not covered by the mask layer **165** is removed. The removal process

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removes the third dipole layer **163** to expose the HK dielectric layer **160** (formed on the first intermixed layer **156**) within the second nanosheet transistor **153**. The removal process also exposes the top surface **167** of the mask layer **165** and a portion of the third dipole layer **163** surrounding the mask layer **165**. The removal process may be any suitable processes, such as the removal process discussed above in FIG. **28** or the pre-clean process discussed above in FIG. **24**.

In FIG. **36**, after removal of the third dipole layer **163** within the second nanosheet transistor **153**, a fourth dipole layer **169** is formed on the exposed HK dielectric layer **160**. In some embodiments, the fourth dipole layer **169** wraps around and in contact with the HK dielectric layer **160** around the first semiconductor layers **106a**. The fourth dipole layer **169** extends along the liner **119** and the top surface **167** of the mask layer **165** and in contact with the HK dielectric layer **160**, the third dipole layer **163**, and the mask layer **165**. The fourth dipole layer may be configured to include positive polarity or negative polarity, depending on the conductivity type of the second nanosheet transistor **153**. The fourth dipole layer **169** serves to enhance or modify threshold voltage for the second nanosheet transistor **153**. In cases where the second nanosheet transistor **153** is a n-channel FET, the fourth dipole layer **169** can be a negative polarity dipole (n-dipole) layer formed using ALD, CVD, or any suitable conformal deposition technique. In some embodiments, the fourth dipole layer **169** can be formed of the same or different material from the second dipole layer **154**. The fourth dipole layer **169** may have the same or different thickness than the second dipole layer **154**, depending on the application.

In FIG. **37A**, the mask layer **165** is removed. The removal process may use ash process or dry/wet etch process identical or similar to the process used for removal of the mask layer **152** as discussed in FIG. **29A**. The removal process selectively removes the mask layer **165** but not the fourth dipole layer **169** and the third dipole layer **163**. The fourth dipole layer **169** on the top surface **167** of the mask layer **165** (FIG. **36**) is removed along with the mask layer **165**. The openings **151** within the first nanosheet transistor **155** are revealed upon removal of the mask layer **165**, exposing the first semiconductor layers **106b** being surrounded by the third dipole layer **163** and the first semiconductor layers **106a** being surrounded by the fourth dipole layer **169**.

In some embodiments, which can be combined with one or more embodiments (e.g., FIGS. **29A**, **29B**, **31A**, and **31B**) discussed in this disclosure, one or more second nanosheet transistors **153** in the semiconductor device structure **100** are further proceeded to have the fourth dipole layer **169** removed from the first semiconductor layers **106a** and from the HK dielectric layer **160**, such as an alternative embodiment shown in FIG. **37B**. The removal of the fourth dipole layer **169** can be done after removing the mask layer **165** by a suitable selective etch process that removes the fourth dipole layer **169** but not the third dipole layer **163**. By configuring one or more second nanosheet transistors **153** to include the fourth dipole layer **169** over the first semiconductor layers **106a** and one or more second nanosheet transistors **153** to not include the fourth dipole layer **169** over the first semiconductor layers **106a**, the semiconductor device structure **100** can provide CFETs having multi-threshold voltage schemes according to the application needs.

In FIG. **38**, the semiconductor device structure **100** is subjected to a thermal treatment **175**. The thermal treatment **175** may be the same as the thermal treatment **131** discussed

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above with respect to FIG. **30**. The third dipole layer **163** and the fourth dipole layer **169** are diffused or driven into the HK dielectric layer **160** as a result of the thermal treatment **175**. In some embodiments, the third dipole layer **163** and the fourth dipole layer **169** mix with the HK dielectric layer **160** and form a third intermixed layer **171** and a fourth intermixed layer **173**, respectively. In some embodiments, the interface between the HK dielectric layer **160** and the third dipole layer **163** and the interface between the HK dielectric layer **160** and the fourth dipole layer **169** may not be observed after the thermal treatment. The third intermixed layer **171** wraps around and in contact with the second intermixed layer **158**. The third dipole layer **163** extending over the liner **119**, the insulating material **118**, the liner **115**, and the bottom intermixed layer **159** also diffuses into the HK dielectric layer **160** and forms a fifth intermixed layer **181** in contact with the liner **119**, the insulating material **118**, the liner **115**, and the bottom intermixed layer **159**. Likewise, the fourth intermixed layer **173** wraps around and in contact with the first intermixed layer **156**. The fourth dipole layer **169** extending over the liner **119** also diffuses into the HK dielectric layer **160** and forms the sixth intermixed layer **183** in contact with the liner **119**.

The third intermixed layer **171**, the fourth intermixed layers **173**, the fifth intermixed layer **181**, and the sixth intermixed layer **183** can be considered as modified HK dielectric layers. The term “third/fourth/fifth/sixth intermixed layer” or “modified HK dielectric layer” as used in this disclosure denotes a reaction product of the HK dielectric layer **160** and the third/fourth dipole layer **163/169**, which can be a compound, a composition or a mixture, depending on the thermal treatment used. In some embodiments, the third/fourth/fifth/sixth intermixed layer **171/173/181/183** or the modified HK dielectric layer can be a HK dielectric layer doped with materials from the third dipole layer **163** or the fourth dipole layer **169**.

In some embodiments, after the thermal treatment **175**, the third dipole layer **163** and the fourth dipole layer **169** that are not driven into the HK dielectric layer **160** may be optionally removed by a dry etch (e.g., RIE), a wet etch (e.g., SPM process, APM process, or HPM process), ozone-deionized Water (O₃-DI), and/or combination thereof.

FIG. **38** illustrates an embodiment where the first semiconductor layers **106b** within the first nanosheet transistor **155** are sequentially surrounded by the second intermixed layer **158** (e.g., the reaction product of the IL **148** and the first dipole layer **150**) and the third intermixed layer **171** (e.g., the reaction product of the HK dielectric layer **160** and the third dipole layer **163**), and the first semiconductor layers **106a** within the second nanosheet transistor **153** are sequentially surrounded by the first intermixed layer **156** (e.g., the reaction product of the IL **148** and the second dipole layer **154**) and the fourth intermixed layer **173** (e.g., the reaction product of the HK dielectric layer **160** and the fourth dipole layer **169**).

In some alternative embodiments, which can be combined with any embodiment(s) discussed in this disclosure, the thermal treatment may cause portions of the fourth dipole layer **169** to diffuse into the HK dielectric layer **160** and form an intermixed layer **173a** sandwiched between the HK dielectric layer **160** and the fourth dipole layer **169**, as an alternative embodiment shown in FIG. **39A**.

In some alternative embodiments, which can be combined with any embodiment(s) discussed in this disclosure, the thermal treatment may cause portions of the third dipole layer **163** to diffuse into the HK dielectric layer **160** and form an intermixed layer **171a** sandwiched between the HK

dielectric layer **160** and the third dipole layer **163**, as an alternative embodiment shown in FIG. **39B**.

In FIG. **40**, a first gate electrode layer **172** is formed in each opening **151** (FIG. **38**) and on the third intermixed layer **171** and the fourth intermixed layer **173**. The first gate electrode layer **172** is formed on the third intermixed layer **171** and the fourth intermixed layer **173** to surround a portion of each first semiconductor layer **106b**, **106a**, respectively. The first gate electrode layer **172** includes one or more layers of conductive material, such as polysilicon, aluminum, copper, titanium, tantalum, tungsten, cobalt, molybdenum, tantalum nitride, nickel silicide, cobalt silicide, TiN, WN, WCN, TiAl, TiTaN, TiAlN, TaN, TaCN, TaC, TaSiN, metal alloys, other suitable materials, and/or combinations thereof. The first gate electrode layers **172** may be formed by PVD, CVD, ALD, electro-plating, or other suitable method. In some embodiments, the first gate electrode layer **172** includes a p-type gate electrode layer such as TiN, TaN, TiTaN, TiAlN, WCN, W, Ni, Co, or other suitable material.

In FIG. **41**, an etch-back process is performed on the first gate electrode layer **172** to recess the first gate electrode layer **172**. The etch-back process uses an etchant that selectively removes the first gate electrode layer **172** but not the fourth intermixed layer **173**. The first gate electrode layer **172** is recessed so that a top surface **177** of the first gate electrode layer **172** is at a level between the topmost first semiconductor layer **106b** of the first nanosheet transistor **155** and the bottom-most first semiconductor layer **106a** of the second nanosheet transistor **153**. The top surface **177** of the first gate electrode layer **172** generally defines the boundary between the first nanosheet transistor **155** and the second nanosheet transistor **153**.

Next, a second gate electrode layer **179** is formed in the opening **151** revealed as the result of the recess of the first gate electrode layer **172** and on the fourth intermixed layer **173**. The second gate electrode layer **179** is formed on the fourth intermixed layer **173** to surround a portion of each first semiconductor layer **106a**. The second gate electrode layer **179** includes one or more layers of conductive material, such as polysilicon, aluminum, copper, titanium, tantalum, tungsten, cobalt, molybdenum, tantalum nitride, nickel silicide, cobalt silicide, TiN, WN, WCN, TiAl, TiTaN, TiAlN, TaN, TaCN, TaC, TaSiN, metal alloys, other suitable materials, and/or combinations thereof. The second gate electrode layers **179** may be formed by PVD, CVD, ALD, electro-plating, or other suitable method. In some embodiments, the second gate electrode layer **179** includes an n-type gate electrode layer such as TiAlC, TaAlC, TiSiAlC, TiC, TaSiAlC, or other suitable material. Therefore, the second gate electrode layer **179** serves as a gate electrode layer for an n-channel FET, and the first gate electrode layer **172** serves as a gate electrode layer for a p-channel FET. Depending on the application, the first gate electrode layer **172** and the second gate electrode layer **179** may include the same material.

FIG. **42A** illustrates an alternative embodiment incorporating various embodiments shown in FIGS. **29A-29B**, **31A-31B**, **37A-37B**, and **39A-39B**, in which the IL **148** of the first nanosheet transistor **155** is modified to selectively include the first dipole layer **150** (e.g., a first p-dipole layer), such as an alternative embodiment shown in FIG. **42C**; the IL **148** of the second nanosheet transistor **153** is modified to selectively include the second dipole layer **154** (e.g., a first n-dipole layer), such as an alternative embodiment shown in FIG. **42B**; the HK dielectric layer **160** is modified to selectively include the third dipole layer **163** (e.g., a second

p-dipole layer), such as the alternative embodiment shown in FIG. **42C**; and the HK dielectric layer **160** is modified to selectively include the fourth dipole layer **169** (e.g., a second n-dipole layer), such as the alternative embodiment shown in FIG. **42B**. By using these embodiments, CFETs having multi-threshold voltage schemes can be provided at different regions of the semiconductor device structure **100**. Particularly, these embodiments enable a zero thickness dipole patterning for multiple threshold voltage tuning by diffusing and/or intermixing the n-dipole or p-dipole layers with the IL **148** and the HK dielectric layer **160** to form modified IL **148** and HK dielectric layer **160** with minimum thickness increase of the modified IL **148** (e.g., first intermixed layer **156**/intermixed layer **156a** and second intermixed layer **158**/intermixed layer **158a**) and the modified HK dielectric layer **160** (e.g., third intermixed layer **171**/intermixed layer **171a** and the fourth intermixed layer **173**/intermixed layer **173a**).

FIGS. **43A**, **43B**, and **43C** illustrate cross-sectional side views of the semiconductor device structure **100** after the first and second gate electrode layers **172** and **179** are formed. FIG. **43D** is an enlarged view of a portion of the semiconductor device structure **100** showing the first and second nanosheet transistors **155**, **153**.

In FIGS. **44A** and **44C**, source/drain (S/D) contacts **176** are formed in the ILD layer **164**. Prior to forming the S/D contacts **176**, contact openings are formed in the ILD layer **164** to expose the epitaxial S/D feature **149**. Suitable photolithographic and etching techniques are used to form the contact openings through various layers, including the ILD layer **164** and the CESL **162** to expose the epitaxial S/D features **149**.

After the formation of the contact openings, a silicide layer **178** is formed on the epitaxial S/D features **149**. The silicide layer **178** conductively couples the epitaxial S/D features **149** to the subsequently formed S/D contacts **176**. The silicide layer **178** may be formed by depositing a metal source layer over the epitaxial S/D features **149** and performing a rapid thermal annealing process. During the rapid anneal process, the portion of the metal source layer over the epitaxial S/D features **149** reacts with silicon in the epitaxial S/D features **149** to form the silicide layer **178**. Unreacted portion of the metal source layer is then removed. The material of the silicide layer **178** is chosen based on the conductivity type of the second nanosheet transistor **153**. For n-channel FETs, the silicide layer **178** may be made of a material including one or more of TiSi, CrSi, TaSi, MoSi, ZrSi, HfSi, ScSi, Ysi, HoSi, TbSi, GdSi, LuSi, DySi, ErSi, YbSi, or combinations thereof. For p-channel FETs, the silicide layer **178** may be made of a material including one or more of NiSi, CoSi, MnSi, Wsi, FeSi, RhSi, PdSi, RuSi, PtSi, IrSi, OsSi, or combinations thereof. In some embodiments, the silicide layer **178** is made of a metal or metal alloy silicide, and the metal includes a noble metal, a refractory metal, a rare earth metal, alloys thereof, or combinations thereof.

Next, a conductive material is formed in the contact openings and form the S/D contacts **176**. The conductive material may be made of a material including one or more of Ru, Mo, Co, Ni, W, Ti, Ta, Cu, Al, TiN and TaN. While not shown, a barrier layer (e.g., TiN, TaN, or the like) may be formed on sidewalls of the contact openings prior to forming the S/D contacts **176**. Then, a planarization process, such as CMP, is performed to remove excess deposition of the contact material and expose the top surface of the second gate electrode layer **179**.

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It is understood that the semiconductor device structure **100** may undergo further complementary metal oxide semiconductor (CMOS) and/or back-end-of-line (BEOL) processes to form various features such as transistors, contacts/vias, interconnect metal layers, dielectric layers, passivation layers, etc. The semiconductor device structure **100** may also include backside contacts (not shown) on the backside of the substrate **101** by flipping over the semiconductor device structure **100**, removing the substrate **101**, and selectively connecting source or drain feature/terminal of the epitaxial S/D features **146** or **149** to a backside power rail (e.g., positive voltage VDD or negative voltage VSS) through the backside contacts. Depending on the application, the source or drain feature/terminal of the epitaxial S/D features **146** or **149** and the first and second gate electrode layers **172**, **179** may be connected to a frontside power source.

Embodiments of the present disclosure provide a semiconductor device structure including CFETs each having a first nanosheet transistor and a second nanosheet disposed over the first nanosheet transistor. The first nanosheet transistor may be a p-channel FET and the second nanosheet transistor may be an n-channel FET. Each of the nanosheet channels of the p-channel FET may be surrounded by a modified IL and a modified HK dielectric layer. Each of the modified IL and the modified HK may or may not include a p-dipole layer. Likewise, each of the nanosheet channels of the n-channel FET may be surrounded by the modified IL and the modified HK dielectric layer. Each of the modified IL and the modified HK may or may not include an n-dipole layer. As a result, the CFETs in different regions of the semiconductor device structure can be operated at different threshold voltages, resulting in improved device reliability and performance. Embodiments of the present disclosure also allow precise gate patterning of the CFETs even if the n-channel FET and the p-channel FET are vertically stacked at different levels of the CFET.

An embodiment is a semiconductor device structure. The structure includes a first gate electrode layer having at least three surfaces surrounded by a first intermixed layer, wherein the first intermixed layer comprises a first material and a second material. The structure also includes a second gate electrode layer disposed below and in contact with the first gate electrode layer, the second gate electrode layer having at least three surfaces surrounded by a second intermixed layer, wherein the second intermixed layer comprises the first material and a fifth material, wherein the first gate electrode layer and the second gate electrode layer are disposed between two adjacent dielectric spacers.

Another embodiment is a semiconductor device structure. The structure includes a first semiconductor layer, a second semiconductor layer disposed below the first semiconductor layer, a first dielectric spacer and a second dielectric spacer disposed between the first and second semiconductor layers, a first gate electrode layer in contact with a first intermixed layer, wherein the first intermixed layer comprises a first material and a second material. The structure also includes a second gate electrode layer disposed below the first gate electrode layer, the second gate electrode layer being in contact with a second intermixed layer, wherein the second intermixed layer comprises the first material and a fifth material, and wherein the first and second dielectric spacers are in contact with a dielectric material layer.

A further embodiment is a method for forming a semiconductor device structure. The method includes forming a plurality of first semiconductor layers vertically stacked, forming a first source/drain feature and a second source/

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drain feature disposed over the first source/drain feature, wherein the first and second source/drain features are in contact with a sidewall of each of the plurality of first semiconductor layers. The method also includes surrounding a first group of the plurality of first semiconductor layers by a first dipole layer, wherein the first dipole layer has a first polarity. The method also includes surrounding a second group of the plurality of first semiconductor layers by a second dipole layer, wherein the second dipole layer has a second polarity opposite of the first polarity. The method further includes subjecting the first and second dipole layers to a first thermal treatment.

The foregoing outlines features of several embodiments so that those skilled in the art may better understand the aspects of the present disclosure. Those skilled in the art should appreciate that they may readily use the present disclosure as a basis for designing or modifying other processes and structures for carrying out the same purposes and/or achieving the same advantages of the embodiments introduced herein. Those skilled in the art should also realize that such equivalent constructions do not depart from the spirit and scope of the present disclosure, and that they may make various changes, substitutions, and alterations herein without departing from the spirit and scope of the present disclosure.

The invention claimed is:

1. A semiconductor device structure, comprising:

- a first gate electrode layer having at least three surfaces surrounded by a first intermixed layer, wherein the first intermixed layer comprises a first material and a second material;
 - a high-K (HK) dielectric layer having at least three surfaces in contact with the first intermixed layer;
 - a second gate electrode layer disposed below and in contact with the first gate electrode layer, the second gate electrode layer having at least three surfaces surrounded by a second intermixed layer, wherein the second intermixed layer comprises the first material and a fifth material; and
 - a third intermixed layer having at least three surfaces in contact with the second intermixed layer, the third intermixed layer comprising a third material having a first polarity,
- wherein the first gate electrode layer and the second gate electrode layer are disposed between two adjacent dielectric spacers.

2. The semiconductor device structure of claim 1, wherein the second material is a dipole material having the first polarity, and the fifth material is a dipole material having a second polarity opposite of the first polarity.

3. The semiconductor device structure of claim 1, wherein the HK dielectric layer is in contact with the two adjacent dielectric spacers.

4. The semiconductor device structure of claim 3, wherein a portion of the HK dielectric layer is disposed between and in contact with the first intermixed layer and the two adjacent dielectric spacers.

5. The semiconductor device structure of claim 1, wherein the third intermixed layer comprises a third material and a sixth material.

6. The semiconductor device structure of claim 1, wherein the HK dielectric layer is in contact with the third intermixed layer.

7. The semiconductor device structure of claim 6, wherein a portion of the HK dielectric layer and a portion of the third intermixed layer are in fully contact with the two adjacent dielectric spacers.

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8. The semiconductor device structure of claim 6, further comprising:

one or more first semiconductor layers, each first semiconductor layer of the one or more first semiconductor layers is surrounded by the first intermixed layer.

9. The semiconductor device structure of claim 8, wherein the one or more first semiconductor layers further comprises a fourth intermixed layer surrounding and in contact with the first intermixed layer, wherein the fourth intermixed layer comprises the third material and a fourth material, and the fourth material is a dipole material having the first polarity.

10. The semiconductor device structure of claim 1, wherein the first material comprises an oxygen-containing material or a silicon-containing material, and the third material comprises a high-k (HK) dielectric material.

11. The semiconductor device structure of claim 1, wherein the second gate electrode layer is chemically different from the first gate electrode layer.

12. A semiconductor device structure, comprising:

a first semiconductor layer;

a second semiconductor layer disposed below the first semiconductor layer;

a first dielectric spacer and a second dielectric spacer disposed between the first and second semiconductor layers;

a first gate electrode layer in contact with a first intermixed layer, wherein the first intermixed layer comprises a first material and a second material;

a high-K (HK) dielectric layer having at least three surfaces in contact with the first intermixed layer;

a second gate electrode layer disposed below the first gate electrode layer, the second gate electrode layer being in contact with a second intermixed layer, wherein the second intermixed layer comprises the first material and a fifth material, and wherein the first and second dielectric spacers are in contact with a dielectric material layer; and

a third intermixed layer having at least three surfaces in contact with the second intermixed layer, the third intermixed layer comprising a third material having a first polarity.

13. The semiconductor device structure of claim 12, wherein the second material is a dipole material having the

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first polarity, and the fifth material is a dipole material having a second polarity opposite of the first polarity.

14. The semiconductor device structure of claim 13, wherein the first material comprises an oxygen-containing material or a silicon-containing material.

15. The semiconductor device structure of claim 12, wherein the dielectric material layer is disposed between and in contact with a first epitaxial source/drain feature and a second epitaxial source/drain feature.

16. A method for forming a semiconductor device structure, comprising:

forming a plurality of first semiconductor layers vertically stacked;

forming a first source/drain feature and a second source/drain feature disposed over the first source/drain feature, wherein the first and second source/drain features are in contact with a sidewall of each of the plurality of first semiconductor layers;

surrounding a first group of the plurality of first semiconductor layers by a first dipole layer, wherein the first dipole layer has a first polarity;

surrounding a second group of the plurality of first semiconductor layers by a second dipole layer, wherein the second dipole layer has a second polarity opposite of the first polarity; and

subjecting the first and second dipole layers to a first thermal treatment.

17. The method of claim 16, further comprising:

forming a high-K dielectric layer to surround the first dipole layer and the second dipole layer, respectively; and

forming a third dipole layer on the high-K dielectric layer surrounding the first group of the plurality of first semiconductor layers, wherein the third dipole layer comprises a third dipole material having the first polarity.

18. The method of claim 17, further comprising:

forming a fourth dipole layer on the high-K dielectric layer surrounding the second group of the plurality of first semiconductor layers, wherein the fourth dipole layer has the second polarity; and

subjecting the third and fourth dipole layers to a second thermal treatment.

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